

Does Listing Language Add Value Beyond Location?

Deconfounding Text and Geography in Automated Home Valuation

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Abstract

Automated valuation models increasingly price homes from the prose of a listing, and the added language reliably improves their predictions. Whether it carries information about the property or merely the neighborhood the writing encodes has gone untested, because the two are observationally identical. We separate them not by adjusting for location but by a sequence of designs that hold it fixed ever more strictly. Estimating the effect of listing text on log price by partially linear double machine learning across twelve metropolitan markets, we find that the effect depends on which direction through the sentence-embedding is read as the treatment. The direction that strips the location-predictable channel leaves it near eleven percent per standard deviation. The effect survives holding the building itself fixed, where geography cannot vary, at three and a half percent, and survives at one percent between near-identical units in the same building. Each design removes more of what the estimate could be confounded with. The premium falls as part of the raw association proves to be location and part unit quality, yet it never reaches zero. New York makes this vivid. Its language is the most geographic in the panel, its naive effect a significant negative, and every design that removes location flips it to a significant positive premium there. The surviving premium is modest, but it is identified by holding geography fixed rather than by assuming it away, and the worry that text valuation is mostly relabeled location does not, on this evidence, bear out.

Keywords: Spatial confounding, Concept erasure, Double machine learning, Generated regressors, Text as data, Real estate valuation

1 Introduction

Automated valuation models increasingly read the natural-language description of a property alongside its structured attributes, and the text reliably improves their predictions; Shen and Ross [2021], Baur et al. [2023], and Lorenz et al. [2023] all report sizable reductions in pricing error once listing-text embeddings are added on top of location and the usual structured controls. The reading attached to that improvement is the optimistic one, that listing language carries information about a property its tabular attributes miss, and on its face it is plausible, since a good description does say things about condition, light, and finish that no bedroom count records. What predictive accuracy cannot do is tell that interpretation apart from a deflationary one. Property descriptions are written by agents who know exactly where a listing sits, so any embedding of those descriptions absorbs the neighborhood whether the encoder intends it to or not, and a model that prices the embedding may be doing little more than relabeling location as semantic content. Because the two stories make identical predictions, the gain that motivates text-augmented valuation is by construction silent on whether the language adds anything at all once geography is accounted for, and it is that silence the paper sets out to break.

Breaking that silence means asking how much of the association between a listing’s language and its price is owed to geography rather than to anything the language contributes on its own. The question is an instance of spatial confounding, the difficulty in which an exposure of interest and an unmeasured driver of the outcome share a geographic pattern, so that ordinary regression hands the confounder’s effect to the exposure [Paciorek, 2010, Hodges and Reich, 2010]. Our strategy is a ladder of designs that hold location fixed ever more strictly. We summarise a description by a direction through its embedding, estimate that direction’s effect on log price with a partially linear Double Machine Learning score [Robinson, 1988, Chernozhukov et al., 2018], and then subject the effect to successively stronger removals of geography:

folding a flexible location basis into the controls, replacing the leading-variance direction with the one that projects the location-predictable channel out, confining the comparison to units of the same building where location cannot vary, and finally to near-identical units where size is held fixed as well. The contribution is to show that identification of a text price-effect is a property of the direction drawn through the embedding rather than of the embedding itself, and to carry that effect down a ladder of designs to the point where the geography it might merely be relabeling is held fixed by construction rather than by assumption. That the representation encodes location at all is not in doubt; we confirm it with a concept-erasure diagnostic, developed in Appendix D, that recovers the coordinate from the embedding and then projects it out. The question the paper turns on is the separate and more consequential one of whether that encoded location actually drives the price effect. Running the test across twelve metropolitan markets lets the answer be read as a distribution rather than a single number, since the embedding geometry, the structured controls, and the listing density that decide how much room the text has once location is accounted for all differ from one market to the next.

Applied across twelve metropolitan areas using 69,204 deduplicated Redfin listings, the answer runs against the predictive-fit worry, and it does so under designs that hold location fixed rather than merely adjust for it. The representation plainly carries location: a held-out probe recovers the coordinate from the embedding at an R^2 approaching one half. Yet the price effect it carries depends on which direction through the embedding is read as the treatment. The leading direction of largest variance can point partly at geography, while the direction that strips the location-predictable channel isolates the text’s own content, and estimating on that identified direction keeps the effect near eleven percent of a standard deviation. It survives when the comparison is confined to units in the same building, where location cannot vary at all, at three and a half percent, and survives at one percent between near-identical units in the same building, where size is held fixed as well. Each design removes more of what the estimate could be confounded with, and the effect attenuates as it should when part of the raw association is location and part is unit quality, yet it never reaches zero. New York carries the argument. Its language is the most geographic in the panel and its leading-direction effect is a significant negative, and every design that removes location flips it to a significant positive premium there, so that the market first read as the exception becomes the clearest demonstration that the variance-maximal direction had been reporting laundered geography with the wrong sign rather than the text’s own effect. The surviving effect does not depend on the encoder either, since a Doc2Vec uniqueness channel and a sentence-BERT principal component move almost in lockstep across markets, two encodings of one underlying signal.

The identification is design-based for the spatial confounding that motivates the paper: geography is held fixed by construction rather than assumed away by the location controls. It stops short of a full experiment, since the unit-quality residual the within-building comparison leaves is bounded by matching and by sensitivity analysis rather than differenced away. A sensitivity analysis in the style of Cinelli and Hazlett [2020] reports how strong an unobserved confounder would have to be to overturn each market’s result, a threshold that none of the covariates we observe attains in any of the twelve, though it is small in absolute terms where the effect it protects is small. Because the outcome is the asking price an agent sets together with the description, we ask finally whether the effect holds on the price a property actually fetches. Re-estimated across ten markets on a panel of recorded transactions rather than the asking price, the effect replicates on the price the market actually cleared and is individually significant in every one, with the exogenous county-assessor controls giving a larger, sharper estimate than the agent-reported attributes that share an author with the description. The answer the paper returns to the measurement question the text-valuation literature left open is therefore this: in most metropolitan markets a listing’s language is priced for its content and not merely for the geography it encodes, with New York the documented exception. We release the code, the derived embeddings, the public-record sale prices, and an independent computational verification of every analytic claim (Appendix F); the listing text and asking prices are scraped under a commercial platform’s terms and are not redistributed, but the collection code that reconstructs them is.

The paper proceeds in the order the argument requires. Section 2 places the question among the literatures it draws on. Section 3 sets out the estimand and the partialling-out estimator, and Section 5 describes how the twelve-market corpus was assembled and audited for the duplication that scraped listings invite. The evidence occupies Section 6, which establishes the effect under two encoders, confirms that the representation encodes location, and then walks the deconfounding ladder, from folding location into the controls through the identified direction to the within-building and twin-unit designs, before bounding the survivor

with a sensitivity analysis and confirming it on recorded sale prices. The concept-erasure implementation, the finite-sample calibration of the estimator, the rewriting intervention, and the remaining robustness are developed in the supplement. Section 7 draws the line, for practice, between what a predictive-only evaluation of text-augmented valuation can and cannot support.

2 Literature

2.1 Listing text in automated valuation, and the question it raises

The use of natural language in property valuation predates the recent enthusiasm for embeddings, and the empirical claim that listing text improves prediction is by now well established. Nowak and Smith [2017] show that the free-text remarks attached to a multiple-listing record reduce pricing error substantially once they are encoded, and Nowak et al. [2021] document that the vocabulary doing the predicting shifts from one submarket to the next. A particular channel is formalized by Shen and Ross [2021], who measure the textual uniqueness of a description relative to its neighbours and report a uniqueness premium on the order of ten to fifteen percent per standard deviation in a single metropolitan market, while Baur et al. [2023] show that folding a sentence-embedding representation into a structured valuation model lowers its error by as much as a sixth. With few exceptions the literature reads gains of this kind as evidence that language carries information about a property which its tabular attributes do not.

What that reading passes over is that an agent writes a description already knowing exactly where the property sits. The language is saturated with location before any encoder touches it, and a representation learned from such text carries the neighborhood whether or not it carries anything more. A predictive metric cannot tell a model that prices semantic content apart from one that prices laundered geography. The very gains that motivate text-augmented valuation are therefore silent on which of the two is operating, and how much of the apparent text effect is semantic rather than spatial has gone largely untested in a causal register. We take up that question directly. The tools it requires already exist across three literatures that have rarely been combined: the treatment of text as a causal variable, the spatial-statistics account of confounding by location, and the concept-erasure methods developed in natural language processing for removing a named attribute from a representation. What the paper adds is not a new estimator but the problem their intersection has left open. The spatial-confounding literature works with exposures that are scalar or a short stack of scalars. The strand of causal-text work that admits a high-dimensional treatment assumes its confounders are observed and ignorable, which a listing’s geography is not. The contribution is to bring a design-based deconfounding test to an embedding-valued treatment under continuous spatial confounding, and to read its answer across twelve markets and again on realized transaction prices. The same machinery bears on the fairness and auditability concerns behind much recent scrutiny of automated valuation. Avenancio-León and Howard [2022] document a systematic gap between assessed and market values that undervalues minority neighborhoods, and Zhu et al. [2024] and Guo [2025] document residual valuation gaps that survive architectural improvement. A text feature re-encoding neighborhood geography re-encodes whatever that geography is correlated with. But the object of this paper is the prior measurement question, whether the text effect exists at all once geography is accounted for, rather than the disparities a confirmed effect would carry.

2.2 From hedonic models to multimodal valuation

The idea that property values can be decomposed into contributions from observable characteristics dates to the hedonic pricing framework [Rosen, 1974, Sheppard, 1999]. Automated valuation models built on this foundation through progressively richer statistical machinery: generalized additive models [Gloude-mans, 1999], spatial autoregressive specifications that account for price correlation among neighboring properties [Pace and Barry, 1997, Can, 1992], and kriging-based approaches that interpolate unobserved spatial surfaces [Bourassa et al., 2010]. Each generation recognized more clearly that spatial dependence is not a nuisance to be corrected but a first-order feature of housing markets.

Machine learning accelerated the same trajectory, with ensemble methods outperforming linear hedonic models by capturing non-linear feature interactions automatically [Park and Bae, 2015, Antipov and Pokryshevskaya, 2012] and deep learning extending the input space beyond tabular features to images [Poursaeed

et al., 2018, Law et al., 2019] and text [Shen and Ross, 2021, Baur et al., 2023]. Visual models trained on property photographs or street-view imagery can estimate luxury level and architectural style [Poursaeed et al., 2018, Lindenthal and Johnson, 2025], while textual models extract semantic features from listing descriptions that correlate with sale prices [Nowak and Smith, 2017, Ahmed and Moustafa, 2016]. Multimodal systems fuse these modalities together, reporting performance gains attributed to complementary information [Bin et al., 2020, Kang et al., 2021, Kostić and Jevremović, 2020]. Graph neural networks push the idea further by propagating information between spatially proximate properties, so that each node’s representation absorbs its neighbors’ features [Das et al., 2021, Lin et al., 2021, Peng et al., 2023, Li et al., 2018, Huang et al., 2018].

A persistent concern runs through this literature, raised early but seldom addressed. The most predictive words in property descriptions vary across geographic submarkets [Nowak and Smith, 2017, Nowak et al., 2021], visual features capture neighborhood-level rather than property-level variation [Law et al., 2019], architectural style clusters geographically [Lindenthal and Johnson, 2025], and GNN message passing, by design, propagates whatever signal is shared among neighbors, including location-induced price similarity. Deppner and Cajias [2024] make the point concrete for algorithm-driven hedonic models, showing that resampling which ignores spatial autocorrelation reports an accuracy gain that a holdout respecting spatial structure does not. If text, images, and graph features all encode geography, then fusing them creates the appearance of complementary signal from what is merely redundant spatial encoding through different channels.

2.3 The semantic interpretation and its limits

What gives text-based valuation its commercial and research appeal is a single claim, that a property’s description carries information about its value that the structured fields miss, and the evidence marshalled for it has by now settled into a familiar shape. A one-standard-deviation increase in the textual uniqueness of a listing is reported to raise its predicted price by ten to fifteen percent [Shen and Ross, 2021]; folding sentence embeddings into a structured model lowers valuation error by as much as a sixth [Baur et al., 2023]; hierarchical graph representations that place similar-value properties near one another recover what looks like price-relevant structure [Zhang et al., 2021]; and low-dimensional projections of the embeddings fall into clusters that read as market segments [Baur et al., 2023]. Each result is offered as a sign that the language is doing semantic work.

Every one of these findings is also predicted by a model in which text carries zero independent information. If agents in affluent neighborhoods write “luxury finishes” and “steps from fine dining” while agents in family-oriented suburbs write “great school district” and “quiet cul-de-sac,” the embeddings will cluster by price segment, SHAP will rank descriptive terms as important [Lorenz et al., 2023, Stang et al., 2024], and adding text to a location-only model will improve R^2 , even though the entire gain comes from spatially confounded language rather than property-specific content. The assumption that generic quality terms like “spacious” or “renovated” provide location-independent signal [Lorenz et al., 2023] ignores that “spacious” is more frequent in suburban markets and “renovated” more frequent in gentrifying urban neighborhoods. Nowak et al. [2021] find exactly this: the predictive vocabulary shifts across submarkets, suggesting that language mirrors geography rather than transcending it. Wan and Lindenthal [2023] raise the same concern for visual features, proposing system-level tests to determine whether models respond to building attributes or to irrelevant context.

Whether correlation here is standing in for causation is not a semantic quibble, because the two readings carry opposite implications for practice. A representation that encodes value-relevant attributes independent of location earns its accuracy on its own terms and offers information gains in its own right, whereas one that mostly encodes geography through linguistic proxy is an opaque spatial model whose accuracy is expected rather than evidence of any new signal. None of the studies surveyed above tests the causal claim on which the optimistic reading rests, even though the machinery to test it has long been available in causal inference and spatial statistics and the confounding structure could hardly be more transparent, and it is that gap the present paper is written to close.

2.4 Spatial confounding and attribute leakage in learned representations

The concern that text may encode geography rather than semantics connects two literatures that have developed largely in parallel.

In spatial statistics and epidemiology, the problem is well characterized: when an exposure of interest varies spatially and shares its spatial pattern with unmeasured confounders, standard regression attributes the confounders’ effects to the exposure [Paciorek, 2010]. The severity depends on the spatial scale: fine-grained exposures are most vulnerable to fine-grained confounders. Because that scale is rarely known in advance, Keller and Szpiro [2020] cast the flexibility of the adjustment as a model-selection problem, fixing it by the Euclidean distance at which spatial variation is removed rather than by an opaque degrees-of-freedom count. Gilbert et al. [2021] set out the conditions under which spatial coordinates can stand in for an unmeasured spatial confounder: the confounder must be a smooth function of location, and the exposure must retain variation that is not itself spatial. The second condition is the one a learned treatment can fail, and our erasure diagnostic shows the text direction to satisfy it; Gilbert et al. [2024] study the consistency of the resulting estimators. Methods to address this include restricted spatial regression, which orthogonalizes spatial random effects against fixed effects of interest [Reich et al., 2021], though it is now understood to remove collinearity without removing the confounding bias itself, which has motivated design-based alternatives: Guan et al. [2023] separate the location-predictable from the non-spatial signal in the frequency domain, the spectral analogue of the direction this paper isolates in the embedding. Their exposure is given, though, while ours is extracted from the representation, so which direction is the treatment is itself an identification question we must answer and they do not face, and it is where our estimand-by-direction result and its inference live. Khot et al. [2025] reconstruct a substitute spatial confounder with a learned representation. A complementary line is sensitivity frameworks that quantify the confounder strength needed to overturn a finding [Cinelli and Hazlett, 2020]. A counterintuitive result from this literature is that adding spatially-correlated error terms to a model can worsen, not improve, inference about fixed effects when spatial confounding is present [Hodges and Reich, 2010]. Spatial econometrics addresses the same challenge through spatial lag and error models [Anselin, 1988, LeSage and Pace, 2009, Elhorst, 2014], though identification of spatial dependence parameters remains difficult to separate from omitted-variable bias [Gibbons et al., 2015].

In NLP, a parallel line of work has shown that text embeddings absorb far more than semantic content from their training corpora. Word embeddings encode human-like gender and racial biases [Bolukbasi et al., 2016], replicate implicit association test results [Caliskan et al., 2017], and retain these biases even after standard debiasing [Gonen and Goldberg, 2019]. Language use varies systematically by geography: user location can be predicted from tweets with high accuracy [Han et al., 2012], and lexical innovations diffuse through social networks along spatial paths [Eisenstein et al., 2014]. Property descriptions are an extreme case of this phenomenon because agents write with complete knowledge of the property’s location, neighborhood, and market context. The language is not incidentally spatial; it is intentionally so.

Machine learning has begun to develop tools for this class of problem. Causal representation learning seeks representations that reflect causal structure rather than spurious correlation [Schölkopf et al., 2021, Peters et al., 2017]. Counterfactual fairness formalizes the requirement that predictions be invariant to interventions on protected attributes [Kusner et al., 2017, Kilbertus et al., 2017]. Domain-adversarial networks learn confounder-invariant representations through gradient reversal [Ganin et al., 2016], and variational approaches attempt to disentangle causal from spurious factors [Louizos et al., 2017, Moyer et al., 2018]. The invariance of causal mechanisms across environments, in contrast to the instability of correlations, motivates transfer-based identification [Rojas-Carulla et al., 2018]. Causal discovery algorithms infer graph structure from data using conditional independence tests [Spirtes et al., 2000], score-based search [Chickering, 2002], or functional asymmetries between causal directions [Shimizu et al., 2006, Peters et al., 2016]. Double Machine Learning [Chernozhukov et al., 2018] provides a semiparametric framework for estimating causal effects with machine-learned nuisance parameters at $\sqrt{n}$ rates.

These threads meet in the setting we study, and they have begun to meet elsewhere as well, which is worth setting out so that what is new here is not overstated. The idea of adapting a text representation for causal rather than predictive ends is developed by Veitch et al. [2020], whose embeddings are built to retain the confounding-relevant subspace, and pushed closest to our own construction by Gui and Veitch [2023], who learn a representation that keeps a text embedding’s confounding-relevant subspace while stripping

the part predictive only of treatment, so that the identified direction of Section 6.5 is in effect a spatial special case of their operation; what the spatial target earns, and a general learned representation does not, is a closed-form erasure with an exact guardedness guarantee against every linear predictor of location rather than an approximate one fit by gradient descent. The inverse operation of stripping a concept from a document embedding by linear erasure is carried out for similarity and clustering by Fan et al. [2025], while Feldman et al. [2026] treat latent textual features as a treatment whose covariate content has to be residualised away. What this paper adds is not the erasure operator, nor the recognition that a text feature can conflate treatment with confounder. Both are by now established. It is the application of that apparatus to an economic estimand, the price effect of a listing’s language. We carry it out with a design that separates the geography a representation merely encodes from the geography that actually moves the outcome. We add an identification theory that indexes the estimand by the treatment direction and shows a unique direction recovers the semantic effect. And we add a limit theory for that direction being estimated inside the embedding under spatial dependence. Property valuation is an unusually well-suited setting to attempt this, since location drives both the text and the price, the outcome is cleanly measured, and the practical stakes are immediate, and we run the analysis across twelve metropolitan markets rather than the single market the hedonic-text literature has usually examined, which lets the answer be read as a distribution and lets the one market where location does bite be identified rather than assumed away.

3 Methodology

3.1 Setup and the conflation it invites

Write $\mathcal{D} = \{(x_i, t_i, \ell_i, y_i)\}_{i=1}^n$ for a market of n listings. Here $x_i \in \mathbb{R}^d$ collects the structured property attributes that hedonic models have always used: the bedroom and bathroom counts, the living area, the lot size, and the year built. $\ell_i \in \mathbb{R}^2$ is the latitude-longitude coordinate of the parcel, and y_i is the log listing price. t_i is a representation of the listing’s free-text description, either a scalar summary or a high-dimensional embedding depending on the channel. A text-augmented valuation model learns some $\hat{y}_i = f(x_i, t_i, \ell_i)$. The empirical regularity that motivates this literature is that adding t_i to the feature set raises predictive accuracy beyond what x_i and ℓ_i deliver on their own.

The difficulty is that an accuracy gain cannot say what kind of information the text carried. An agent who writes “sun-drenched and steps from the park” is describing a property. The same phrase is also a near-deterministic signal of which block the property sits on. Language and location are produced together by an author who observes both. A predictive model that prices substantive semantic content and a predictive model that prices laundered geography make identical out-of-sample predictions. The gain on its own therefore leaves an analyst unable to tell which of the two has been bought. Our object of interest is the part of the text-price relationship that is not the spatial information the text carries. The methods that follow are built to isolate it. For each metropolitan market in turn, we then ask how much of the apparent text signal that isolation leaves standing.

This conflation has a precise name in the literature on text as a treatment. There it appears as the aliasing of one measured latent feature of a document with the unmeasured features that the same prose inevitably carries. Fong and Grimmer [2023] show that even a randomly assigned text fails to identify the effect of a single latent feature. This holds unless the other features it travels with are either independent of it or irrelevant to price. We do not impose that condition. The quantity we isolate is instead the price effect of the leading text direction marginalizing over whatever else the language encodes, an associational object rather than the effect of a fully separated semantic attribute. We are explicit about that reading throughout. We treat text as the causal variable rather than as a confounder to adjust for. This places the paper on the treatment side of the causal-text program surveyed by Keith et al. [2020], whose adjustment strand runs from the text-matching of Roberts et al. [2020] to the linguistic-property effects of Pryzant et al. [2021]. A second lesson from this literature bears on estimation rather than interpretation. Egami et al. [2022] establish that when the treatment representation is learned from the documents, its estimation error must not contaminate the outcome stage. We take that concern up in the calibration of Appendix G. Forming the direction on the pooled corpus rather than within each market shrinks the channel. An out-of-fold projection removes most of what remains, and a full-pipeline bootstrap carries the residual into the reported intervals.

3.2 A structural account of the confounding

The data-generating process we have in mind makes the confounding explicit rather than assuming it away. Location is exogenous and drives everything downstream: the structured attributes x , a vector c of contextual neighborhood features such as demographics, amenities, and recent crime, the text t , and the price y . The text in turn is written with knowledge of location, structure, and context, so $t = f_t(\ell, x, c, U_t)$. The question the paper exists to answer is whether t enters the structural equation for y at all once ℓ , x , and c are accounted for. The alternative is that the entire observed association between text and price runs through the common dependence of both on location.

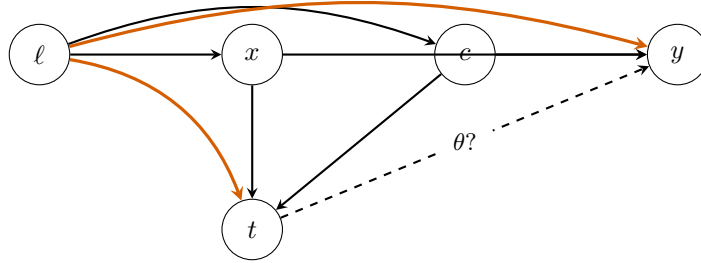


Figure 1: The structure the paper adjudicates. Location ℓ is exogenous and feeds the property attributes x , the contextual features c , the listing text t , and the price y . The two highlighted edges $\ell \rightarrow t$ and $\ell \rightarrow y$ form the spatial-confounding backdoor, the reason text and price are associated in part simply because both depend on where the property sits. The dashed edge labeled θ is the direct effect of text on price, the quantity we estimate, and the empirical question is how much of the text-price association is that direct edge rather than the backdoor through ℓ . Conditioning on $\{\ell, x, c\}$ blocks the backdoor and identifies θ ; whether θ is zero or positive is what the twelve markets decide.

Two readings of this diagram bracket the possibilities. Under one, the text effect on price is zero net of location and the structured covariates. Every observed text-price correlation is geography wearing a semantic costume, and the apparent predictive value of listing language is spatial confounding mediated through words. Under the other, listing text carries property-specific content that the structured covariates miss. Then t does enter the equation for y , and the text-price relationship is at least partly a semantic effect. The two readings disagree about a quantity that is identified once the backdoor set $\{\ell, x, c\}$ is conditioned on. That set blocks every path from t to y that does not pass through the hypothesized direct edge, a backdoor argument made precise in Appendices A and B. The rest of this section develops the estimator, the encodings, and the deconfounding tests through which the twelve markets adjudicate between them.

3.3 The partially-linear estimand and its estimation

We summarize the listing text by a scalar treatment T , defer to Section 3.4 how that scalar is constructed, and pose the partially-linear model

$$Y = \theta T + g(\ell, x, c) + \varepsilon, \quad T = m(\ell, x, c) + \nu, \quad (1)$$

in which θ is the per-standard-deviation effect of the text direction on log price and the nuisance functions g and m are left unrestricted.

Assumption 1 (Unconfoundedness given controls). *Writing $W = (\ell, x, c)$ for the observed location, structured, and contextual controls, the potential outcome $Y(t)$ is mean-independent of the text direction given W , so that $E[Y(t) | T, W] = E[Y(t) | W]$ for every t in the support of T , and the conditional-mean functions $m(W) = E[T | W]$ and $g(W) = E[Y | W]$ are estimable at faster than $n^{-1/4}$ rates.*

Assumption 1 is what licenses reading θ as the causal effect of the text direction rather than as a residual association. We do not defend it as true on priors. Descriptions are written with full knowledge of location, structure, and context, so the residual ν could in principle still carry price-relevant information the controls

omit. What the paper offers is not an argument that the assumption holds but an accumulation of evidence against the specific channels through which it would most plausibly fail. The erasure diagnostic shows that the location the representation demonstrably encodes is linearly removable. The covariate decomposition of Section 6.4 shows that folding a flexible location basis into W barely moves $\hat{\theta}$. The sensitivity analysis of Section K.3 bounds the strength an unobserved confounder would need to overturn the estimate against the strength of the strongest control actually observed. The assumption is therefore not credible on its own but increasingly credible as those three probes fail to unseat it. This is the structure we ask the reader to weigh rather than a claim of ignorability by fiat. The design also meets a critique the erasure-and-adjust step invites directly. Recovering a stand-in for an unobserved confounder from an observed proxy is the move of the multiple-causes deconfounder of Wang and Blei [2019]. Ogburn et al. [2019] show that such a stand-in, estimated on the same sample used to estimate the effect, need not satisfy the independence conditions it is asked to. This is why the paper does not rest on erasure and adjustment alone. It escalates to the within-building and twin-unit designs of Section 6.6, where location is held fixed by construction rather than reconstructed from the embedding and controlled for in-sample.

The Robinson partialling-out estimator [Robinson, 1988] recovers θ from the residual-on-residual regression of $\tilde{Y} = Y - \hat{g}$ on $\tilde{T} = T - \hat{m}$. We fit the nuisances $\hat{g}$ and $\hat{m}$ by cross-fitted machine learning in the double machine learning framework of Chernozhukov et al. [2018]. We use ridge regression for both the embedding-based channels and the Shen-Ross uniqueness channel, in keeping with the published hedonic protocol. For the dense, well-conditioned confounder block at these sample sizes, ridge matches or improves on gradient boosting in nuisance mean-squared error. The qualitative conclusions, a positive identified-direction effect in every market and the New York sign reversal, are unchanged under gradient-boosted nuisances. Each market is split into $K = 5$ folds. Each fold’s residuals are predicted from models trained on the complement, so that the nuisance estimates and the score are never formed on the same observations. Appendix J consolidates the learner, fold, and bootstrap settings. The resulting estimator is Neyman-orthogonal and root- n consistent whenever the nuisance fits converge faster than $n^{-1/4}$, and its influence-function variance gives the standard error we report by default,

$$\hat{\theta} = \frac{\sum_i \tilde{T}_i \tilde{Y}_i}{\sum_i \tilde{T}_i^2}, \quad \hat{\text{SE}}_{\theta} = \sqrt{\frac{1}{n^2} \sum_i \frac{(\tilde{Y}_i - \hat{\theta} \tilde{T}_i)^2 \tilde{T}_i^2}{\left(\frac{1}{n} \sum_j \tilde{T}_j^2\right)^2}}. \quad (2)$$

One feature of the embedding channel complicates the inference and we flag it here because it shapes how the per-market intervals should be read. When the treatment is the leading principal component of the embedding, that direction is itself estimated on the same sample that estimates the price effect. T is therefore a generated regressor in the sense of Pagan [1984], the machine-learning-generated case treated by Battaglia et al. [2024]. The orthogonal-score variance in Equation (2) does not absorb the extra sampling variability the estimation of the direction introduces. The interval is correspondingly anticonservative at the few thousand listings a single market affords, mildly so near the null and more seriously when the true effect is large. We document this pattern in a calibration study. We repair it with an out-of-fold construction of the treatment direction, the sample-splitting that Egami et al. [2022] prescribe when the treatment is learned from the documents, in Appendix G. The treatment is the leading spectral direction of the design rather than an exogenously fixed covariate. This places the construction alongside the spectral-deconfounding literature on inference when a leading-eigenvector transform of high-dimensional covariates interacts with hidden confounding [Čevd et al., 2020, Guo et al., 2022]. It is the same concern, the estimated direction carrying its own sampling variability, that the out-of-fold construction addresses. The point estimates are approximately unbiased throughout; it is the nominal coverage of the analytic interval, not the location of the estimate, that the calibration study qualifies.

3.4 Two encodings of the text treatment

No single reduction of a paragraph to a number is obviously the right one. We therefore read the listing text through two encodings with different inductive biases, and treat their agreement as a robustness check rather than as independent corroboration. The first follows the hedonic specification of Shen and Ross [2021], who summarize a description by a Doc2Vec uniqueness score that measures how far a listing’s language departs from the corpus norm. We enter that scalar into Equation (1) as the treatment. The second follows Baur

et al. [2023], who embed each description with a pretrained sentence transformer and read the leading principal component of the embedding as the text signal. We standardize that component and use it as the treatment in a separate partialling-out fit. The sign of a principal component is not identified, since a component and its negation span the same axis. Wherever the Baur channel appears, we orient the common axis to make the pooled estimate non-negative, and we disclose the orientation as a convention rather than a finding. This convention cannot manufacture agreement, because the cross-market correlation between channels is invariant in magnitude under a sign flip of either one.

3.5 What the embedding encodes: erasure as a diagnostic

The sentence-transformer embedding that underlies the Baur channel was pretrained on generic text. It has no reason to keep semantic content about a property separate from the geographic information that correlates with its neighborhood. Before asking whether the price effect is confounded, we therefore ask the prior question of how much location the representation carries at all. We answer it by erasing the coordinate concept from the embedding and measuring what an erasure leaves recoverable. The closed-form concept erasure of Belrose et al. [2023] removes the linear subspace that the mean-conditional-covariance estimator associates with the latitude-longitude pair. It comes with an exact-guardedness guarantee that no linear predictor can recover the coordinate from the residual better than a constant baseline. The iterative nonlinear procedure of Iskander et al. [2023] extends the attack to the directions a linear probe cannot see. It alternates between fitting a nonlinear probe of the coordinate and projecting out where that probe concentrates its predictive mass. The erasure is a diagnostic of what the representation contains, not a modification of any per-market estimate, since the Baur effects are computed on the unprojected embedding in keeping with the published protocol. Its role is to bound how much location a linear or nonlinear function of the text could carry. The decomposition below then connects that bound to how much location actually moves the price.

3.6 When a probe fails, what has been erased?

The erasure diagnostic of the previous subsection rests on probing, and probing carries a subtlety that bears on how its verdicts should be read. The natural way to check that a concept has left a representation is to train a probe of the concept on the erased representation and confirm that it does no better than chance. Yet a probe that fails establishes only that this particular probe, at this capacity, cannot recover the concept. It does not establish that the representation has shed it, since a fresh classifier of higher capacity fit after the fact can recover what a weaker or adversarially pressured probe could not. This is the reason we lean on closed-form erasure, whose guardedness guarantee holds against every linear probe at once rather than against the one we happened to train. It is also the reason we read the residual probe R^2 that survives the iterative erasure of Section 6.3 as a lower bound on the location a nonlinear function could still recover, rather than as a measurement of zero. The same subtlety is a warning about the obvious alternative to closed-form erasure: an adversarially trained encoder whose discriminator gradient reversal drives to chance. A live discriminator at chance is the weakest possible evidence of erasure. It certifies only that the encoder defeated the one probe it was trained against, and a fresh post-hoc probe of greater capacity may still read the concept off the same representation. We make the point precise as a $\mathcal{V}$ -information [Xu et al., 2020] statement connected to the Barber–Agakov variational lower bound used throughout the probing literature [Pimentel et al., 2020].

For any class $\mathcal{F}$ of probabilistic classifiers $q : \mathcal{Z} \rightarrow [0, 1]$, define the variational lower bound on $I(Z; C)$:

$$V_{\mathcal{F}}(Z; C) := \sup_{q \in \mathcal{F}} [H(C) - \mathbb{E}_{(Z, C)}[-\log q(C | Z)]] . \quad (3)$$

The supremum is achieved at $q^* = p(C | Z)$ when $\mathcal{F}$ contains the Bayes-optimal posterior; otherwise $V_{\mathcal{F}} < I(Z; C)$ strictly.

Proposition 1 (Frozen-probe diagnostic). *Let $E_{\Phi} : \mathcal{X} \rightarrow \mathcal{Z}$ be an encoder, Φ the discriminator class trained jointly via gradient reversal, and Φ' a probe class with $\Phi \subseteq \Phi'$ trained post-hoc on the frozen encoder. Then:*

- (a) $0 \leq V_{\Phi}(Z; C) \leq V_{\Phi'}(Z; C) \leq I(Z; C)$, with the rightmost inequality strict whenever Φ' omits the Bayes-optimal posterior.

- (b) The empirical Barber–Agakov lower bound under Φ equals the gradient-reversal discriminator’s converged value: $\hat{V}_\Phi(Z_{\hat{\phi}}; C) = H(C) - \hat{L}_\Phi$. Therefore “the live discriminator achieves chance” is equivalent to $\hat{V}_\Phi \leq \epsilon$. Under uniform-convergence conditions on Φ' , the post-hoc plug-in $\hat{V}_{\Phi'}^{\text{post}}(Z_{\hat{\phi}}; C) \rightarrow V_{\Phi'}(Z_{\hat{\phi}}; C)$ in probability, and hence $\hat{V}_{\Phi'}^{\text{post}} - \hat{V}_\Phi$ is a consistent lower-bound estimator of the residual mutual information $I(Z_{\hat{\phi}}; C) - V_\Phi(Z_{\hat{\phi}}; C) \geq 0$.
- (c) There exist a data distribution over (X, C) , a downstream label Y with $I(X; Y) > 0$, class pairs $\Phi \subsetneq \Phi'$, and a joint-game weight $\alpha > 0$ such that the joint adversarial-prediction loss

$$\mathcal{L}(\phi, g, h) = \alpha \cdot \mathcal{L}_{\text{task}}(g(E_\phi(X)), Y) - \mathcal{L}_{\text{disc}}(h(E_\phi(X)), C)$$

admits a saddle (ϕ^*, g^*, h^*) with $V_\Phi(Z_{\phi^*}; C) = 0$ but $V_{\Phi'}(Z_{\phi^*}; C) \geq H(C) - \delta$ for arbitrary $\delta > 0$.

Statement (a) is folklore in the variational mutual-information literature. It appears as Proposition 1 of Xu et al. [2020] in the language of $\mathcal{V}$ -information and as the central observation of Pimentel et al. [2020] in the language of probe capacity. We restate it for self-containedness. Statements (b) and (c) are our contribution. Part (b) ties the bound to the gradient-reversal objective. It recasts the post-hoc adversary used as a heuristic by Moyer et al. [2018] and Elazar and Goldberg [2018] as a consistent statistical estimator of residual leakage. This is what licenses reading the surviving probe R^2 as a number rather than a guess. Part (c) establishes that the joint adversarial-prediction game admits saddles at which the live discriminator achieves the optimum of its own minimax game, while essentially all of the concept remains recoverable to a richer probe. An adversarial erasure can therefore certify success and be wrong. The existential construction mirrors the style of Ravfogel et al. [2022] Proposition 3.2 and Belrose et al. [2023] Theorem 4.3, and proofs together with three synthetic numerical experiments verifying each part are deferred to Appendix E.

3.7 Holding the text fixed: a covariate decomposition

Showing that the embedding encodes geography is not the same as showing that the price effect is confounded by it. A representation can carry a confounder along directions that have nothing to do with the one on which the treatment acts. The test that speaks to confounding directly holds the text treatment fixed and moves geography across the conditioning set, in the manner of the order-invariant covariate decomposition of Gelbach [2016]. We estimate the text effect twice on the same fixed treatment: once with the location coordinates held out of the controls, and once with a flexible spatial basis, the coordinates together with their squares and their interaction, folded in. We read the difference between the two estimates as the share of the text effect that explicit geography accounts for. A small difference means the text-price effect is not the spatial information re-expressed. The decomposition is informative rather than mechanical exactly when the text direction is shown by the erasure diagnostic to carry geography. A direction that encodes location yet whose price effect is unmoved by conditioning on location is the signature of a semantic signal. This contrast is the paper’s headline deconfounding test, and Section 6.4 reports it for every market.

3.8 An intervention check and a sensitivity bound

Two further instruments probe the same question from angles with different failure modes. A counterfactual rewriting intervention uses an open-weight language model to edit each description under a minimal-pair protocol. It produces one rewrite that strips stylistic content while holding the model’s inferred submarket fixed, and a second that also moves the targeted submarket. The price differentials these edits imply isolate a within-listing style contrast and a location-inference contrast whose biases are not shared with the cross-listing geometry of a fixed encoder. None of these designs identifies a structural effect by assumption. We close with the sensitivity calculus of Cinelli and Hazlett [2020], reporting for each market the robustness value, the strength an unobserved confounder would need on both the treatment and the outcome to overturn the estimate, benchmarked against the strength of the spatial covariates we do observe. The twelve per-market estimates are finally combined through a random-effects meta-analysis with the Paule-Mandel variance estimator and Hartung-Knapp-Sidik-Jonkman small-sample intervals [Paule and Mandel, 1982, IntHout et al., 2014]. A moderator scan then asks, with multiplicity and leverage diagnostics, whether any metropolitan covariate explains the heterogeneity the markets display.

4 Identification

Let ℓ denote location, x the structured attributes, and W the text embedding. Let the scalar treatment T be a fixed linear summary of W , the oriented leading pooled principal component. Write $L = L(\ell)$ for a basis of functions of location used as controls. We take the outcome to follow the structural model

$$Y = \theta T + \beta U + f(x) + \varepsilon, \quad (4)$$

in which U is an unobserved, spatially patterned driver of price.

Assumption 2 (Spatial confounder). *The unobserved confounder is location-measurable, $U = h(\ell)$ for a measurable h . This says the confounding travels entirely through geography, the case the spatial-confounding literature is built for; a driver of price unrelated to location falls outside it and is the residual the sensitivity analysis of Section K.3 carries.*

Assumption 3 (Basis richness). *h lies in the closed linear span of L , so that $\mathbb{E}[U \mid L, x] = U$ almost surely. Concretely L includes the discrete geographic identifiers (borough, neighborhood, school, and street indicators) through which the encoder leaks location; a coordinate-polynomial basis alone does not satisfy this when h varies discontinuously across geographic units.*

Assumption 4 (Residual overlap). $\mathbb{E}[(T - \mathbb{E}[T \mid L, x])^2] > 0$. *The treatment retains variation orthogonal to location; this is the identifying condition the erasure diagnostic tests, and the one New York approaches as its text direction collapses onto geography.*

Assumption 5 (Exogeneity). $\mathbb{E}[\varepsilon \mid T, \ell, x] = 0$. *No further driver of price is correlated with the treatment given location and structure, the standard exogeneity condition of the partially-linear model.*

We take a deliberately limited view of what these four buy. Together Assumptions 2 and 3 make the confounding a function of location that the control basis can span, which is what the designs of Section 6 hold fixed. What we do not assume is that the text carries no confounder outside location, unmeasured interior quality above all. Against that residual the effect is not identified but bounded, by the sensitivity analysis of Section K.3. The identification is design-based against observed location and sensitivity-bounded against the rest, one claim read from its two sides, and the paper holds to that division throughout.

Theorem 1 (Identification). *Under Assumptions 2–5, the treatment coefficient in (4) equals the Robinson functional*

$$\theta = \frac{\mathbb{E}[\tilde{Y} \tilde{T}]}{\mathbb{E}[\tilde{T}^2]}, \quad \tilde{Y} = Y - \mathbb{E}[Y \mid L, x], \quad \tilde{T} = T - \mathbb{E}[T \mid L, x].$$

The parameter θ is point-identified if and only if $\mathbb{E}[\tilde{T}^2] > 0$; the concept-erasure diagnostic probes whether the treatment retains the location-orthogonal variation this requires, and as $R_{T \sim (L, x)}^2 \rightarrow 1$ the denominator vanishes and identification fails.

The proof is a direct partialling-out computation. It absorbs the spatial confounder into the control-measurable part of the model under Assumptions 2 and 3. Its necessity half shows that a control basis missing the discrete geographic channel leaves the exact bias $\beta b d / (d^2 + 1)$. The full argument is deferred to Appendix B.1. The location basis must therefore span the discrete identifiers through which the encoder leaks geography, not coordinates alone. New York, whose leading text direction loads on borough and neighborhood names, is the empirical instance of the boundary where $\mathbb{E}[\tilde{T}^2]$ approaches zero and identification thins.

4.1 Identification is a property of the direction, not of the embedding

The treatment in a text-augmented valuation is not handed to the analyst as a scalar. It is a direction chosen inside a high-dimensional representation, and the same embedding admits a continuum of such directions, each a candidate summary of what the listing says. Whether the resulting effect is identified turns out to

depend on which direction is chosen, and Theorem 2 makes that dependence exact. The scalar identification result of Section 4 is the one-dimensional shadow of it.

Let $W(\ell) \in \mathbb{R}^p$ be the residualized embedding at location ℓ , carrying a smooth spatial channel that a coordinate control L recovers, a non-smooth spatial channel that discrete identifiers such as neighborhood and school names load onto and that L misses, and a location-orthogonal semantic channel. Write the projection of each channel onto a candidate unit direction v as $C(v)$, $D(v)$, and $E(v)$, so that the treatment $T = v'W$ decomposes as $C(v)$ times the smooth channel plus $D(v)$ times the missed channel plus $E(v)$ times the semantic channel. The confounder $U = h(\ell)$ places weight b on the missed channel, and the outcome follows the partially linear form $Y = \theta T + \beta U + \varepsilon$.

Theorem 2 (Direction-indexed identification). *Under Assumptions 2–5, the Robinson estimand of the effect of $T = v'W$ that conditions on L equals*

$$\theta(v) = \theta + \beta b \frac{D(v)}{D(v)^2 + E(v)^2}. \quad (5)$$

Consequently the effect is point identified, $\theta(v) = \theta$, if and only if $\beta b = 0$ or $D(v) = 0$, and the set of directions for which it is identified is the subspace $\mathcal{V}_0 = \{v : D(v) = 0\}$ orthogonal to the missed spatial channel. The residual-overlap quantity that Assumption 4 requires to be positive is $\mathbb{E}[\tilde{T}^2] = D(v)^2 + E(v)^2$, so a direction lying entirely in the smooth spatial subspace violates positivity and leaves $\theta(v)$ undefined.

Proof. Conditioning on L removes the smooth channel, so the residualized treatment is $\tilde{T} = D(v)\ell_d + E(v)\eta$ in the orthonormal channel basis, where ℓ_d is the missed spatial channel and η the semantic one. The residualized outcome is $\tilde{Y} = \theta\tilde{T} + \beta b\ell_d$, since U residualized on L retains only its weight b on ℓ_d . Taking the Robinson ratio,

$$\theta(v) = \frac{\mathbb{E}[\tilde{Y}\tilde{T}]}{\mathbb{E}[\tilde{T}^2]} = \frac{\theta(D(v)^2 + E(v)^2) + \beta b D(v)}{D(v)^2 + E(v)^2} = \theta + \beta b \frac{D(v)}{D(v)^2 + E(v)^2},$$

which is (5). The identification and positivity claims read off directly. The algebra is verified symbolically over a free-symbol channel model, and the estimand (5) is recovered to within Monte Carlo error by a cross-fitted gradient-boosting estimator across a sweep of directions, including exact recovery of θ on $\mathcal{V}_0$. $\square$

Three consequences deserve statement. The scalar identification result is the special case of a single unit-variance semantic channel, $D(v) = d$ and $E(v) = 1$, where (5) collapses to the bias $\beta b d / (d^2 + 1)$ of Theorem 1; the vector result is therefore a strict generalization rather than an analogy. The bias in (5) is largest in magnitude where $|D(v)| = |E(v)|$, a direction that mixes the missed spatial channel and the semantic one in equal measure, and it decays as the direction tilts toward either the identified subspace or the smooth channel that positivity forbids. And because $\theta(v)$ varies continuously with v , a representation whose leading variance direction happens to load on the missed spatial channel will report an effect that is mostly laundered geography, while a direction chosen to maximize $E(v)$ relative to $D(v)$ reports the semantic effect the analyst intends. Identification here is not a property the embedding either has or lacks. It is a property of the direction drawn through it, and the leading principal component, chosen for variance rather than for identification, need not lie in the identified subspace. Section 6.5 carries this prescription out on the twelve-market panel, re-estimating the effect on the identified direction and reading the difference from the leading component as the geography it removes.

The estimand in estimable form. Equation (5) is written in the channel geometry that makes identification transparent, but the same estimand has a form that depends only on quantities the data deliver. Writing $a = \mathbb{E}[\tilde{W}\tilde{Y}]$ for the residualized embedding–price covariance vector and $\Sigma = \mathbb{E}[\tilde{W}\tilde{W}']$ for the residualized embedding covariance, the Robinson estimand on the direction v is

$$\theta(v) = \frac{v'a}{v'\Sigma v}, \quad (6)$$

of which (5) is the two-channel special case. The identified subspace $\mathcal{V}_0$ is the set of directions along which the numerator carries only the semantic covariance, and positivity is $v'\Sigma v > 0$. This is the object the panel estimates, market by market, once a and Σ are replaced by their cross-fitted sample analogues and v by the identified or the leading direction.

Corollary 1 (The variance-maximal direction can reverse the sign). *Let the residualized embedding covariance place more variance on the missed spatial channel than on the semantic one, $\sigma_s^2 > \sigma_q^2$. Then the leading principal component, the direction that maximizes $v'\Sigma v$, aligns with the spatial channel, and $\theta(v_{\text{PC}}) \rightarrow b$, the price coefficient of the confounder itself rather than the semantic effect θ . When $\text{sign}(b) \neq \text{sign}(\theta)$ the reported effect is sign-reversed: the default summary of the text returns a coefficient of the wrong sign.*

The corollary is the theory behind New York. Its leading text direction is the most geographic in the panel, so it loads on the borough and neighborhood channel whose price gradient runs opposite the semantic effect, and the naive estimate on that direction is a significant negative while the identified estimate is a significant positive. Gui and Veitch [2023] observe a naive-versus-adjusted sign reversal empirically in a different text setting, a politeness effect that moves from -0.038 to $+0.200$ under adjustment, but state no condition for it; Corollary 1 supplies the condition and identifies the variance-maximal direction, the one an analyst reaches for by default, as the case at risk.

A spectral reading. When the field is stationary with matrix spectral density $\Sigma_W(\omega)$ and the confounder concentrates its mass on a low-frequency band Ω_0 , the projected quantities become frequency integrals, $D(v)^2 + E(v)^2 = \int v'\Sigma_W(\omega)v d\omega$ and the confounding numerator $D(v) = \int_{\Omega_0} v'b(\omega) d\omega$ along the cross-spectrum $b(\omega)$ between the confounder and the field. Identification then asks that the direction place energy outside the confounder’s band, and the bias is bounded by a Cauchy–Schwarz alignment between the direction and the residual confounding mass that survives the control. This is the sense in which a text direction is identified when it carries the field’s high-frequency, non-spatial content and confounded when it tracks the low-frequency geography the location control cannot fully absorb.

4.2 Inference when the direction is estimated

Theorem 2 fixes the direction and asks what it identifies. In practice the direction is not fixed; it is read off the data, as the leading eigenvector of the residualized embedding covariance or as the concept-orthogonal projection the erasure supplies, and an interval that treats it as known understates the sampling variability. This section gives the limiting distribution of the estimator that accounts for the estimated direction, under the spatial dependence the panel carries.

Write $\hat{v}$ for the estimated direction, formed on the pooled sample of N listings across the panel, and $\hat{\theta}_k = \hat{v}'\hat{a}_k/(\hat{v}'\hat{\Sigma}_k\hat{v})$ for the per-market estimate on n_k listings, the sample analogue of (6) with $\hat{a}_k$, $\hat{\Sigma}_k$ the cross-fitted residual moments in market k . Two facts shape the analysis. The estimand is not orthogonal to the direction: the gradient

$$\frac{\partial \theta}{\partial v} = \frac{a - 2\theta(v)\Sigma v}{v'\Sigma v}$$

does not vanish at the identified direction, so a Neyman-orthogonality argument cannot make $\hat{v}$ ’s error disappear. What makes it negligible instead is a comparison of rates. The direction is estimated on the pooled corpus, whose size N exceeds any single market’s n_k by construction, and its error enters $\hat{\theta}_k$ scaled by that larger sample.

Assumption 6 (Spectral gap). *The pooled residualized embedding covariance has a bounded eigen-gap separating the direction v^* from the remainder of its spectrum.*

Assumption 7 (Rate domination and independence). *$n_k/N \rightarrow 0$ for each market, and $\hat{v}$ is formed out-of-fold, on a sample independent of the observations used to form $\hat{\theta}_k$, in practice by leaving market k out of the pooled direction fit.*

Assumption 8 (Spatial weak dependence). *Within each market the listings occupy distinct locations separated by a minimum distance, so the sampling region expands rather than fills in, and the influence-function terms form an α -mixing random field on geographic distance whose mixing coefficients satisfy the summability conditions of Jenish and Prucha [2009]; the cross-fitted nuisances satisfy the rates of Chernozhukov et al. [2018].*

Theorem 3 (Estimated-direction spatial DML). *Under Assumptions 6–8, for each market*

$$\sqrt{n_k} (\hat{\theta}_k - \theta_k(v^*)) \xrightarrow{d} \mathcal{N}(0, V_k),$$

where $\theta_k(v^*)$ is the direction-indexed estimand (6) and V_k is the asymptotic variance of the Robinson influence function $\psi_i = \mathbb{E}[\tilde{T}^2]^{-1} \tilde{T}_i(\tilde{Y}_i - \theta_k \tilde{T}_i)$ under spatial dependence. The estimated direction contributes at $o_p(n_k^{-1/2})$, so V_k is the fixed-direction DML variance, consistently estimated by a spatial heteroskedasticity- and autocorrelation-consistent estimator of the influence-function sum [Conley, 1999, Bester et al., 2011].

Proof sketch. The argument composes four steps. First, under the spectral gap of Assumption 6 the estimated direction obeys a normal approximation for bilinear forms of the sample spectral projector at rate $N^{-1/2}$ [Koltchinskii and Lounici, 2016], so $\|\hat{v} - v^*\| = O_p(N^{-1/2})$; the deterministic subspace bound of Yu et al. [2015] certifies this rate but does not supply a distribution, and is used only for the former. Second, a first-order expansion of $\hat{\theta}_k$ in the direction gives $\hat{\theta}_k = \theta_k(v^*) + (\partial\theta/\partial v)'(\hat{v} - v^*) + r_k$, and by Assumption 7 the linear term is $O_p(N^{-1/2}) = o_p(n_k^{-1/2})$, while the out-of-fold construction makes $\hat{v}$ independent of the market- k score, so no covariance between the two enters the limiting variance. Third, with the direction held at v^* , the cross-fitted Robinson estimator is asymptotically linear with influence function ψ_i , the nuisance error entering at second order by Neyman orthogonality [Chernozhukov et al., 2018]. Fourth, the normalized sum of ψ_i over the spatially dependent field is asymptotically normal by the central limit theorem of Jenish and Prucha [2009], with variance consistently estimated by the spatial-HAC estimator. Composing the four steps, the direction’s error is asymptotically negligible and the limit is the fixed-direction spatial-DML normal law. $\square$

Two remarks fix the scope. The negligibility of the direction’s error is earned by the rate comparison of Assumption 7, not by orthogonality; were the direction formed in sample, or were a single market as large as the pool, the generated-regressor correction of Pagan [1984] and Hahn and Ridder [2013] would enter, and V_k would carry an added term in the asymptotic covariance of $\hat{v}$ that the spectral theory of Koltchinskii and Lounici [2016] makes estimable. We avoid that term by the pooled, out-of-fold construction, and say so rather than assume it away. And the result is what Gui and Veitch [2023] lack: their central limit theorem for a text-treatment effect handles a learned representation through the usual nuisance-rate condition, but their treatment is not a direction, so it carries no term for an estimated direction entering as a generated regressor. Theorem 3 supplies exactly that term and shows it is second order, which is what licenses the per-market intervals the panel reports once they are widened to the spatial-HAC scale. Section 6.5 carries this out on the identified direction, where the out-of-fold construction reproduces the point estimates while the Conley spatial-HAC intervals widen by a factor near two and the effect survives in nine of the twelve markets.

Together the two theorems fill in the identification the controls promised in Section 4: Theorem 2 names the direction on which observed location is removed exactly, and Theorem 3 supplies the inference once that direction is estimated. Both concern identification against observed location. The residual the designs cannot reach, unmeasured interior quality and the nonlinear leakage the linear erasure leaves behind, stays with the sensitivity bound of Section K.3 rather than with these theorems.

5 Dataset construction

5.1 A twelve-market corpus of listings

The corpus is built from sold residential listings scraped from a major online real estate platform across twelve metropolitan markets that span the price and geographic range of the United States housing stock. The markets run from San Francisco and New York at the expensive coastal end, through Boston, Seattle, Washington, Denver, and Portland in the middle, to Chicago, Atlanta, Phoenix, Dallas, and Philadelphia toward the more affordable interior and south. Each listing carries the free-text description an agent wrote, the asking price the platform reports for the property at the time of the crawl, and the structured attributes a hedonic model expects: the bedroom and bathroom counts, the living area, the lot size, the year built, and the property type. Each listing also carries a property-level latitude and longitude rather than a coarse

centroid. Two features of this outcome we are explicit about. The price is the listing’s asking price rather than a realized transaction, and it is set by the same agent who writes the description and at the same moment. Part of any text-price association is therefore the agent’s own pricing of the language rather than the market’s, a simultaneity that no spatial adjustment removes. It limits the causal reading of the estimates along an axis orthogonal to the spatial-confounding question this paper takes up. Because the corpus is a single cross-sectional crawl, the outcome itself carries no calendar-time variation that pooling could confound. The property’s most recent prior-sale date, which we recover from the listing page for every property and release with the data, enters only as a covariate. It is in any case essentially uncorrelated with the asking price within each market: the absolute correlation is no larger than 0.20 and the R^2 is below 0.04. Table 1 summarizes the twelve markets with their analysis sizes and median prices.

Table 1: The twelve-market corpus. *Listings* is the per-market analysis sample after deduplication to one row per listing and removal of rows missing a required field; the markets together contribute 69,204 listings. Median sale price and median description length are computed on the analysis sample. The panel ranges over more than an order of magnitude in size and roughly fivefold in price.

Market	Listings	Median price	Median words
New York	15,255	\$975,000	183
Dallas	8,008	\$439,000	161
Phoenix	7,087	\$514,990	117
Philadelphia	7,030	\$255,000	134
Atlanta	6,135	\$384,000	178
Chicago	5,577	\$374,900	129
Denver	5,280	\$525,000	207
Portland	4,337	\$559,000	166
Washington	4,015	\$599,000	187
Seattle	2,884	\$750,000	152
Boston	2,610	\$999,000	137
San Francisco	986	\$1,295,000	174
Total	69,204		

5.2 Deduplication, outliers, and a robust winsorizer

The scraper issued repeated requests against the same listing during collection. The raw rows therefore resolve to a smaller set of distinct listings once they are deduplicated on the listing identifier. Every number in the paper is computed on the one-row-per-listing analysis sample, to keep sibling copies of a listing out of the training folds of any cross-fitted procedure. We drop transactions outside a plausible price band to remove data-entry errors and non-arm’s-length transfers. We also guard the structured covariates with a per-column robust winsorizer that clips each feature at its median plus or minus five median absolute deviations and imputes the rare non-finite entry at the median. The winsorizer matters more than its routine description suggests. A single parcel-join mistake, a year-built value tens of standard deviations from any plausible construction date, was enough to make the partialling-out estimator in one market ill-conditioned and to drive its coefficient to an impossible magnitude with a deceptively small standard error. Clipping the true outliers without touching the well-behaved rows restored that market to the range of the others, while leaving every well-behaved market unchanged. The clip is applied uniformly across the twelve markets and is the only manipulation of the structured covariates we perform.

5.3 Confounders

The conditioning set is assembled to make the backdoor adjustment of Section 3 credible at the resolution where language and price are both determined. Alongside the structured property attributes, we attach the property’s location through its coordinates and its ZIP code. We join neighborhood context by spatial position rather than by a coarse identifier. Census demographics come from the American Community Survey five-year estimates at the block-group level, the finest geography the survey supports. They contribute

median household income, educational attainment, racial composition, median home value, and median gross rent for the block group containing each listing. Recent crime is counted within a five-hundred-meter radius of each property from municipal incident records, separated into violent, property, and quality-of-life categories through a cross-city offense crosswalk. Local amenities are counted in the same radius from an open geographic database, covering food and retail, services, recreation, transit, and education. The assembled set runs to roughly three dozen controls per market. The spatial block within it, the coordinates together with the ZIP indicators, is the group the sensitivity analysis of Section K.3 singles out, because it is the one most directly implicated in the text-as-spatial-proxy concern. Appendix C documents the scrape, the geocoding, and the census, crime, and amenity joins.

5.4 Text preprocessing and the two encodings

Before encoding, each description is stripped of markup and boilerplate, lowercased, and cleared of explicit street addresses. The addresses are replaced with anonymization tokens, so that a model cannot recover location by reading an address out of the text rather than out of the language. The preprocessed text then feeds the two encodings that Section 3.4 defines. The Doc2Vec uniqueness score that drives the Shen-Ross channel measures how far a listing’s language sits from its market’s corpus norm. The sentence-transformer embedding that drives the Baur channel is the 768-dimensional representation of `all-mpnet-base-v2` [Reimers and Gurevych, 2019], from which we take the standardized leading principal component as the text treatment. Robustness to the choice of encoder is taken up directly in Section K.2, where a Doc2Vec uniqueness axis and the sentence-BERT component recover a concordant per-market signal, so the effect does not hinge on one encoder’s geometry. The deconfounding and erasure diagnostics of Sections 6.3 and 6.4 operate on the same mpnet embedding, so that what they remove is the geography the Baur treatment itself carries.

6 Empirical results

The evidence is organized as a ladder. Two confounds could make a text-price association an illusion: the language could be encoding the neighborhood rather than the property, and within a neighborhood it could be tracking a unit’s quality rather than its description. The sections that follow hold each of these fixed in turn, first by controls and then by design, and read the effect that survives each successive restriction as the part of the association that the restriction cannot explain away. We begin by establishing that the association exists and does not hinge on a single encoder, through the leading principal component of a sentence-BERT embedding and again through the Shen-Ross uniqueness channel, which move together across the panel. We then confirm the premise the deflationary reading depends on, that the representation carries location, by recovering the coordinate from the embedding with a concept-erasure probe. From there the restrictions tighten. Folding a flexible location basis into the controls leaves the coefficient nearly where it began; estimating on the direction that strips the location-predictable channel rather than the leading-variance one keeps the effect and, in the market where the naive direction had reported a significant negative, recovers a positive one; confining the comparison to units in the same building holds location fixed by construction; and confining it to near-identical units in the same building holds size fixed as well. The effect attenuates as each rung removes more of what it could be confounded with, from the location-controlled estimate down to a small premium between twin units, and it stays significant, pooled and in New York, at the most stringent rung the data support. New York carries the argument throughout. Its language is the most geographic in the panel, its naive estimate is a significant negative, and every design that removes location recovers a positive effect there, so that the market first read as the exception becomes the clearest demonstration that the variance-maximal direction had been masking an effect rather than manufacturing one. The analyses that close the section qualify this ladder rather than extend it, bounding the unobserved confounding that could still overturn a per-market estimate, taking up the cross-market heterogeneity the panel exposes, and re-running the design on the price a property actually fetches rather than the one its agent posts.

Table 2: The text price-effect down the deconfounding ladder. Each row holds location fixed more strictly than the one above, first by adjustment and then by design. The per-standard-deviation effect on log price attenuates as each restriction removes more of what the estimate could be confounded with, yet it survives to the most stringent rung the data support. New York, whose leading direction is the most geographic in the panel, carries the argument: its location-controlled effect is a significant negative, and every design that removes location recovers a significant positive premium. Rows one and two pool the twelve markets by inverse variance with influence-function standard errors; rows three and four are estimated on the multi-unit-building subsample with standard errors clustered at the building. Stars denote significance at the five percent level.

Design	Location held fixed by	Pooled effect	New York
Leading component, location controls	adjustment	+0.122*	-0.016*
Identified direction	projection	+0.111*	+0.021*
Same building (fixed effect)	construction	+0.035*	+0.045*
Near-twin units in a building	construction, with size	+0.010*	+0.024*

6.1 Shen-Ross uniqueness channel across twelve metropolitan markets

The Shen-Ross uniqueness channel [Shen and Ross, 2021] measures how far the language in a listing departs from the language its neighbors use. It quantifies departure as the cosine distance between each listing’s vector-space text representation and the centroid of its $K = 5$ nearest peers in latitude-longitude space. We apply the resulting per-listing uniqueness score as the treatment in a partially-linear DML estimator with cross-fitted ridge nuisances solved through a Cholesky factorization. This follows the Shen-Ross uniqueness construction as a close adaptation rather than a verbatim replication of the original Doc2Vec and MLS-area-by-year cell pipeline. Effects are reported per standard deviation of the uniqueness score, and the 95% confidence intervals are formed from the cross-fitting influence-function standard error.

Table 3 reports the per- σ uniqueness effect for each of the twelve metropolitan markets, together with the influence-function confidence interval and the indicator for whether that interval excludes zero. The panel is well-behaved throughout, with no degenerate market and no exclusion. The per-market samples range from 986 deduplicated listings in San Francisco to 15,087 in New York. The DML interval excludes zero in ten of the twelve markets, and every one of those ten carries a positive uniqueness premium. The effect is largest in Chicago at +0.280 with interval [+0.245, +0.315] and in Philadelphia at +0.222 [+0.202, +0.242]. San Francisco follows at +0.117 [+0.069, +0.165]. A cluster of mid-sized markets comes next: Dallas at +0.082 [+0.066, +0.097], Denver at +0.072 [+0.055, +0.088], Phoenix at +0.054 [+0.037, +0.070], the District of Columbia at +0.046 [+0.025, +0.067], and Portland and Seattle both near +0.042. Atlanta, the market on which Shen’s identification was originally established, returns a positive and significant per- σ effect of +0.033. This is the same sign as her published Atlanta estimate but a fraction of its magnitude, recovered under a far more conservative confounder set. The two markets whose intervals straddle zero are Boston at +0.020 [-0.009, +0.049] and New York at +0.002 [-0.010, +0.014]. New York is the largest market, and its estimate is indistinguishable from zero. No market returns a significant negative effect. The panel is uniformly signed on the markets it identifies.

Table 3: Shen-Ross uniqueness channel: per-market DML coefficient per standard deviation of the uniqueness score on log listing price, with influence-function 95% intervals, across the twelve markets, ordered by effect size. The final column is the implied percentage price difference.

Market	n	$\hat{\theta}$	95% CI	%/ σ
Chicago	5,403	+0.280	[+0.245, +0.315]	+32.3
Philadelphia	7,005	+0.222	[+0.202, +0.242]	+24.8
San Francisco	981	+0.117	[+0.069, +0.165]	+12.4
Dallas	8,000	+0.082	[+0.066, +0.097]	+8.5
Denver	5,270	+0.072	[+0.055, +0.088]	+7.5
Phoenix	7,083	+0.054	[+0.037, +0.070]	+5.5
Washington	3,996	+0.046	[+0.025, +0.067]	+4.7
Portland	4,328	+0.042	[+0.024, +0.060]	+4.3
Seattle	2,879	+0.042	[+0.018, +0.065]	+4.3
Atlanta	6,116	+0.033	[+0.018, +0.047]	+3.4
Boston	2,599	+0.020	[−0.009, +0.049]	+2.0
New York	15,087	+0.002	[−0.010, +0.014]	+0.2
Pooled (PM, HKSJ)		+0.060	[+0.018, +0.102]	

The twelve per-market estimates pool to a positive effect. A Paule-Mandel random-effects combination with Hartung-Knapp-Sidik-Jonkman small-sample intervals places the pooled uniqueness premium at +0.060 per standard deviation, with 95% confidence interval [+0.018, +0.102] and two-sided $p = 0.009$. The REML variant returns +0.083 with $p = 0.006$. The dispersion around this central tendency is large and is not merely sampling noise. The heterogeneity statistic is $I^2 = 98.6\%$ with Cochran’s $Q = 777$ on eleven degrees of freedom. Almost all of the variation across markets reflects true between-market differences in the operative effect rather than estimation error within markets. Section K.7 takes up this heterogeneity directly, pooling the twelve estimates by inverse-variance weighting and scanning for a moderator. We defer that analysis to there rather than adjudicate it here.

Design analysis. The cross-market pattern cannot be read as a symptom of low power, because the panel, free of any degenerate market, is well powered. A Gelman-Carlin [Gelman and Carlin, 2014] retrodesign of each per-market estimator uses the published Shen-Ross Atlanta estimate of +0.140 per standard deviation as the external benchmark true effect. It returns a per-market power at or near 1.00 in every one of the twelve markets, far above the conventional 0.80 target. The per-market influence-function standard errors on samples in the thousands are small relative to a 0.140 signal. The Type-S sign-error rate is negligible throughout, below 10^{-14} in every market, so the sign of any detected effect is reliable. The median Type-M exaggeration factor is 1.00, indicating essentially no magnitude inflation in the typical market. San Francisco, the smallest sample at 986 listings, carries the mild anti-conservatism that the analytic variance can show near a thousand observations. We document this feature directly in the coverage simulation of Appendix G. Even there, the retrodesign power against the 0.140 benchmark remains high. The design analysis therefore supports reading both the significant detections and the heterogeneity across markets at face value.

Magnitudes. Because the outcome is log listing price, the per- σ coefficients translate directly into percentage differences in price. A one-standard-deviation increase in listing uniqueness corresponds to a price difference of roughly +32.3% in Chicago, +24.8% in Philadelphia, and +12.4% in San Francisco. The effect falls through a mid-range of about +8.5% in Dallas, +7.5% in Denver, and +5.5% in Phoenix. The smallest significant effects are near +4% in Washington, Portland, and Seattle and +3.4% in Atlanta. The external anchor for these figures is the published Shen-Ross Atlanta estimate of +0.140 per standard deviation, obtained on a sample of roughly forty thousand listings under MLS-area-by-year fixed effects. Our Atlanta estimate of +0.033 recovers the same sign at a fraction of that magnitude. The pooled +0.060 sits below that single-market benchmark. The panel as a whole is consistent with a positive Shen-Ross uniqueness premium that is present, heterogeneous across markets, and smaller on average than the single-market published benchmark would suggest.

6.2 Sentence-BERT principal component channel across twelve metropolitan markets

The Baur, Rosenfelder, and Lutz channel [Baur et al., 2023] treats listing language as a 768-dimensional vector from a frozen sentence-BERT encoder and uses the leading principal component direction as a scalar treatment in a partially-linear DML estimator. The original study computes the PCA per market on the local listing distribution. On a single-market analysis this is the canonical choice. On a twelve-market panel, though, it produces a treatment whose direction is identified only locally and only up to a sign. Per-market PC1 estimates can therefore carry opposite signs across markets even when the underlying language-price relationship is direction-consistent. A panel built on per-market axes mixes true heterogeneity with an accident of local sign and orientation. It cannot be pooled coherently.

Pooled within-city-centered PCA. To obtain a treatment definition that is comparable across markets, we replace the per-market PCAs with a single global principal-component direction extracted from the pooled within-city covariance of the stacked embeddings. We center each city’s embeddings at its own mean before pooling. This removes between-city vocabulary differences from the global covariance and produces a leading eigenvector identified as the within-city common axis. This is the standard pooling for cross-population PCA discussed in Flury [1984], equivalent in construction to a linear discriminant analysis pooled covariance. We do not standardize individual embedding dimensions before pooling. Sentence-BERT outputs are L2-normalized by design, and per-dimension scaling would inflate the anisotropic noise structure documented by Ethayarajh [2019]. We z -score the per-listing projection within each city, so the estimand on each market is a per-city-standard-deviation move along the common axis. The global axis is extracted from the within-city-centered pooling of all twelve markets, an analysis sample of 69,204 listings whose per-market sizes range from 986 in San Francisco to 15,255 in New York.

Sign convention. A principal component and its negation span the same axis, so the sign of the per-standard-deviation effect is a convention rather than a finding. We adopt the convention that orients the common axis so that the pooled effect on log listing price is non-negative. This renders the per-market estimates sign-comparable to the hedonic uniqueness channel of Section 6.1 and to the counterfactual differential of Appendix K.8. Under the opposite convention every sign reported below reverses; the magnitudes, the cross-market dispersion, and the concordance with the other channels are invariant to it. What the orientation does not do is manufacture agreement between channels, a point we develop in Section K.2. The across-market correlation between the BERT axis and the hedonic channel is invariant in magnitude under any global sign flip of either one.

Per-market estimates. Table 4 reports the per-market per-standard-deviation effect under the oriented pooled-PCA treatment. The corrected definition removes the sign ambiguity of the per-market axes and delivers a single direction of effect across the panel. Eleven of the twelve markets return a confidence interval that excludes zero at the 95% level, and eleven carry a positive point estimate. New York is the sole negative, at -0.016 on $[-0.031, -0.002]$, a small effect that is statistically distinguishable from zero but economically close to it. Seattle is the one market whose interval covers zero, at $+0.030$ on $[-0.002, +0.062]$. Chicago and Philadelphia anchor the high end at $+0.464$ on $[+0.428, +0.501]$ and $+0.374$ on $[+0.356, +0.392]$, followed by the District of Columbia at $+0.151$ and Dallas at $+0.142$. Atlanta sits at $+0.134$ and San Francisco at $+0.106$ on $[+0.056, +0.155]$. Portland, Boston, and Denver fall between $+0.061$ and $+0.076$, while Phoenix at $+0.055$ and Seattle at $+0.030$ return the smallest positive effects. Portland, which the per-market PCA had rendered a degenerate outlier, now sits squarely inside the panel at $+0.076$ on $[+0.051, +0.101]$. The panel is well-behaved at twelve markets, and there is no market we exclude. The wide gap between Chicago and Philadelphia and the rest is not intrinsic to the language of those cities. They are also the two markets whose leading direction carries the most location, and Section 6.5 shows the gap narrows once that location is projected out of the treatment. The per-market intervals are tight because the pooled-PCA treatment is estimated on the full within-market sample, which ranges into the thousands of listings, rather than on the description subset.

Table 4: Sentence-BERT pooled-PCA channel: per-market DML coefficient per standard deviation along the oriented common axis on log listing price, with influence-function 95% intervals, ordered by effect size. The axis is oriented so the pooled effect is non-negative; the principal-component sign is a disclosed convention (text). Eleven of the twelve intervals exclude zero; Seattle alone covers it, and New York is a small negative.

Market	n	$\hat{\theta}$	95% CI
Chicago	5,577	+0.464	[+0.428, +0.501]
Philadelphia	7,030	+0.374	[+0.356, +0.392]
Washington	4,015	+0.151	[+0.126, +0.176]
Dallas	8,008	+0.142	[+0.123, +0.162]
Atlanta	6,135	+0.134	[+0.113, +0.155]
San Francisco	986	+0.106	[+0.056, +0.155]
Portland	4,337	+0.076	[+0.051, +0.101]
Boston	2,610	+0.073	[+0.030, +0.116]
Denver	5,280	+0.061	[+0.043, +0.078]
Phoenix	7,087	+0.055	[+0.030, +0.081]
Seattle	2,884	+0.030	[−0.002, +0.062]
New York	15,255	−0.016	[−0.031, −0.002]
Pooled (PM, HKSJ)		+0.137	[+0.048, +0.227]

Pooled inference. Pooling the twelve per-market estimates by inverse-variance weighting returns a per-standard-deviation effect of +0.137 with standard error 0.041 and a 95% interval of [+0.048, +0.227]. The effect is detectable and positive across the panel under the adopted orientation. Between-market dispersion is large, anchored by Chicago and Philadelphia at one end and New York at the other. The pooled point therefore masks a wide distribution, and the precision weights sit close to uniform. The field-standard alternative, a single stacked DML estimator with city fixed effects and the Chiang et al. [2022] cluster-robust variance, proved unstable at twelve clusters. Fold-assignment sensitivity in the cross-fitted nuisances produced pooled points that shifted across invocations on the same seed and data. We therefore combine the per-market estimates directly and report the stacked fit only as a check. Section K.7 develops the pooled figures and the null moderator scan. Because listing text clusters geographically, the influence-function interval, which treats within-market listings as independent, is too narrow. Clustering the influence function at the zip level, or applying a Conley spatial kernel on the coordinates, widens the per-market intervals by roughly forty percent. The effect nonetheless remains significant in every market where the independent interval already excluded zero, so accounting for spatial dependence does not disturb the pattern (Appendix G).

Magnitude and interpretation. The pooled per-standard-deviation effect of +0.137 on log listing price is about a fifteen percent change in price. This is small as a fraction of value but substantial against the magnitudes published in the housing and text-as-treatment literatures. A single additional bedroom in the canonical Sirmans et al. [2005] meta-analysis of forty-five hedonic studies carries a mean effect of +3.8% on log price. If the association were read as an effect, it would be several times the marginal price of an additional bedroom. We intend this scale comparison as illustrative rather than as a claim that a standard-deviation move along a text axis is a manipulable quantity. The more informative comparison is internal. The Doc2Vec uniqueness construct of Shen and Ross [2021] runs in the same direction as the oriented BERT axis, both within markets and across the panel. In San Francisco, where both channels are estimated on the identical sample, the same DML pipeline recovers a uniqueness effect of +0.124 and a BERT-PC1 effect of +0.106. Pooled across the twelve markets, the uniqueness effect is about +0.060 against the +0.137 of the BERT axis. The two text treatments therefore concord in sign and in cross-market ranking, with the BERT axis roughly twice the hedonic channel in pooled magnitude. What the concordance establishes needs care, because two agreeing estimates are easily mistaken for two independent ones. The two channels are alternative encodings of the same listing text applied to the same listings under the same confounder adjustment, and Section K.2 shows they are strongly correlated across markets. Their agreement is evidence that the text-on-price signal is robust to the choice of text representation rather than evidence from two independent sources. Both textual distinctiveness in the Shen sense and the utilitarian-to-amenity axis the

leading sentence-BERT component recovers move price in the same direction. The difference between them is one of magnitude and of which markets each detects, not of sign.

What the leading axis prices. To make concrete what the leading axis encodes, we regress each listing’s projection onto it on a TF-IDF vocabulary of the description, pooled across the twelve markets, and read off the terms with the largest coefficients. Table 5 reports them. The high-price end of the axis is dominated by property-quality and amenity language: words like *spacious*, *modern*, *designed*, *chef*, *owner suite*, and *great room* that describe condition, finish, and layout rather than location. The low-price end is dominated by land and transactional boilerplate, *vacant*, *lot*, *acres*, *zoned*, *hoa*, *mls*, and *seller*. This is the vocabulary of undeveloped parcels and the mechanics of a sale rather than of a home. The axis also carries some lexical geography. Premium-neighborhood terms such as *center city* and *rittenhouse square* load near the top, and the bare city names that head the pooled list reflect between-market price differences that the within-city centering only partly removes. This is the mixture the deconfounding test of Section 6.4 is built to separate. The axis encodes both the semantic content of a property and the neighborhood it sits in. Holding location fixed is what shows that it is the quality language, not the geography, that carries the price effect.

Table 5: What the leading text axis prices, pooled across the twelve markets. The description terms whose TF-IDF most strongly predicts a listing’s projection onto the price-oriented sentence-BERT axis, at each end. Property-quality and amenity language tops the high-price end alongside premium-neighborhood terms; land and transactional boilerplate the low end.

High-price end of the axis	Low-price end
spacious, modern, designed, featuring, chef, owner suite, great room, living space, comfort, dining, closet, layout; and the premium-neighborhood terms center city and rittenhouse square	vacant, lot, land, acres, parcel, zoned, zoning, county; and the transactional boilerplate hoa, mls, seller, buyer, agent, opportunity

6.3 How much location does the representation encode?

Before asking whether the text effect survives location, we check the precondition that makes the question worth asking at all: that the representation encodes location to begin with. The sentence-BERT representation that underlies the Baur channel in Section 6.2 encodes more than the language a listing agent uses. Pre-trained on a generic corpus, the encoder has no particular incentive to separate semantic content about a property’s condition or amenities from the geographic information that correlates with its neighborhood. If the embedding carries location-targeted information into the leading principal component, the per-standard-deviation effect reported in Section 6.2 inherits a spatial confounder mediated through the encoder. The identification claim is correspondingly weaker. We assess how far this concern can be addressed by erasing the lat-lon concept from the representation. We report where the erasure is complete and where it is only partial.

The deconfounding runs a linear projection and then a nonlinear correction, and the linear step carries most of the load. Applying the LEACE projection of Belrose et al. [2023] to the BERT embedding removes the linear subspace that the mean-conditional-covariance estimator identifies with the concept variable, here the latitude-longitude coordinate pair of the listing. It comes with the exact-guardedness guarantee that no linear predictor of the residual embedding can recover the concept any better than a constant baseline would. That guarantee holds in every market we examine. The out-of-fold ridge R^2 for predicting the coordinate from the embedding falls from a raw value that reaches 0.50 in New York and 0.42 in Dallas to within 0.002 of zero after erasure. The linear-guardedness residual averages $2.7\text{e-}8$ across the twelve markets, ranging from $8.3\text{e-}9$ to $6.0\text{e-}8$, which is the floating-point floor of the single-precision computation. The linear erasure therefore comes out complete throughout the panel (Figure 2).

The concept is routed through the continuous coordinate representation rather than the categorical ZIP-code identifier in every market. Categorical LEACE has known instability when the categorical cardinality is large, and our panel ranges from 30 ZIP codes in San Francisco to 158 in New York. The per-market test folds are now large, from 296 listings in San Francisco to 4,569 in New York. Even so, the continuous routing sidesteps the cardinality question entirely without discarding the spatial information the projection

is designed to remove, since the coordinate pair is a sufficient statistic for location at the resolution that matters here.

What a linear projection cannot reach is the location the embedding encodes nonlinearly. The iterative gradient-based correction of Iskander et al. [2023] is aimed at exactly that residual. It alternates between fitting a multilayer-perceptron probe of the coordinate on the residual and projecting out the direction in which that probe gathers its predictive mass. The correction cuts the nonlinear leakage substantially without removing it, taking the average nonlinear probe R^2 from 0.44 on the raw embedding down to 0.20 after five outer passes, a reduction of roughly a half. The residual that remains varies sharply across markets. San Francisco is left with a nonlinear R^2 of 0.01, which is erasure for any practical purpose, whereas New York, Dallas, and Phoenix hold onto residuals near 0.30 to 0.34 and Portland near 0.26. The markets that keep the most are the larger corpora whose listing language carries the richest location-targeted structure. This is also where a nonlinear probe has the most left to find.

Why closed-form erasure rather than adversarial training. We prefer the closed-form projection to the adversarial alternative the disentanglement literature more often reaches for, for a reason we develop in Appendix D. An adversarially trained encoder can drive its own location discriminator to chance and still leave the coordinate recoverable. A fresh probe of the frozen representation reads latitude and longitude back at an R^2 between 0.16 and 0.42 across the twelve markets, and the ZIP code at nine to forty-seven times the uniform rate. The adversarial training does not remove location; it teaches its discriminator not to look. The closed-form projection, by contrast, is guarded against any probe by construction. It is the erasure on which we rest the decomposition that follows.

What the diagnostic does and does not settle. The Baur estimates of Table 4 are computed on the unprojected embedding, in keeping with the published Baur, Rosenfelder, and Lutz protocol. The erasure here is a diagnostic of what the representation contains rather than a change to any per-market number. As a diagnostic it speaks directly to the part of the problem the linear treatment can actually express. The leading principal component is a linear functional of the embedding, and the linear erasure is exact in every market. Whatever location a linear treatment could carry has been shown to be removable in full, while the nonlinear residual that survives IGBP in the larger markets bounds the rest, fixing how much location a nonlinear function of the deconfounded representation could still smuggle back. Showing that the embedding encodes geography is not, however, the same as showing that the price effect is confounded by it, and the diagnostic on its own cannot close that gap. A representation may carry a confounder along directions that have nothing to do with the one on which it acts. The question of whether the location the embedding demonstrably encodes is the location that actually moves the price is taken up directly in Section 6.4, where the text treatment is held fixed and geography is moved on and off the controls around it.

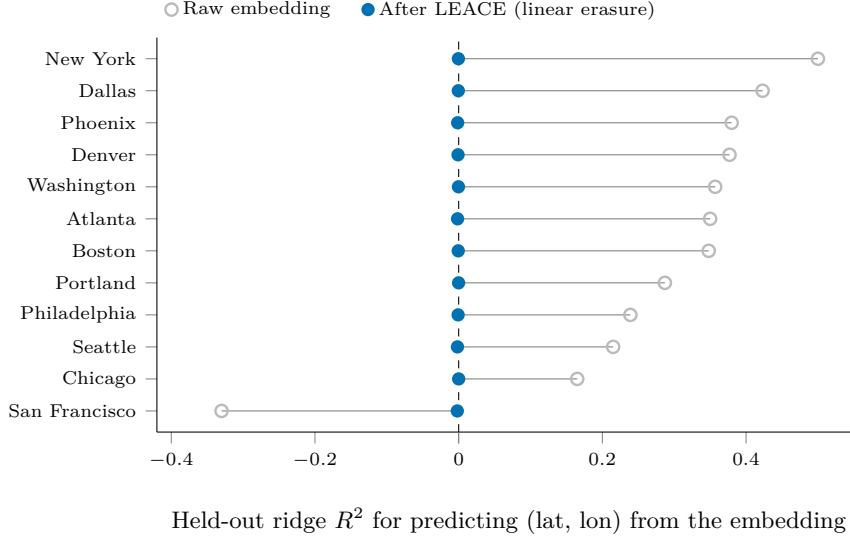


Figure 2: Held-out ridge R^2 for recovering the continuous latitude-longitude coordinate from the sentence-BERT embedding, before (open) and after (filled) the LEACE linear concept erasure, in each of the twelve markets ordered by raw R^2 . The raw embedding carries substantial linear spatial signal, recovering the coordinate at R^2 up to 0.50 in New York and 0.42 in Dallas; the negative San Francisco value reflects out-of-sample probe performance below the coordinate-mean baseline rather than absent signal, on that market’s small held-out fold of 296 listings. Linear erasure drives the recoverable signal to within 0.002 of zero in every market, the filled points sitting on the zero line, which is the exact-guardedness property of LEACE read through a fresh out-of-fold probe rather than the guardedness residual itself. Residual *nonlinear* leakage, which a linear probe cannot see, is quantified separately in Section 6.3.

6.4 How much of the text-price effect is location?

The erasure diagnostic of Section 6.3 establishes that the embedding carries location, and carries it abundantly. A held-out ridge recovers the latitude-longitude pair from the raw representation at an R^2 that climbs to one half in the largest market before the projection drives it to the floor. Taken on its own, that fact invites the deflationary reading that a treatment built from such a representation is geography in disguise and its apparent price effect is spatial confounding waiting to be exposed. The inference does not actually follow. A representation can encode a confounder along directions that have little to do with the direction in which it predicts the outcome, and whether the two coincide is an empirical matter rather than something an erasure R^2 can settle on its own. Answering it means holding the treatment fixed and asking what the location it shares with price is worth to the estimate.

The instrument for that question is a fixed-treatment decomposition in the manner of Gelbach [2016]. Its appeal is that the change in a coefficient when a block of controls is added does not depend on the order in which the blocks enter, so the share it assigns to geography is well defined rather than a byproduct of the sequence in which the blocks enter. We estimate the per-standard-deviation effect of the leading text direction on log price twice: once controlling for the structured confounders alone, and once adding a flexible location basis of latitude, longitude, their squares, and their interaction. We read the gap between the two as the portion of the effect that location explains. Two results in Appendix B place this design on a formal footing. Theorem 1 gives the conditions under which conditioning on a location basis identifies the text effect, with the concept-erasure diagnostic serving as the test of the binding overlap condition and New York the boundary case where it nearly fails. Theorem 4 shows that the gap read off here consistently estimates the contribution the location block makes to the coefficient. The comparison appears in the left panel of Figure 3, where the two estimates sit almost on top of one another in eleven of the twelve markets. The location adjustment moves the coefficient by under 0.02 of a log point outside New York, and by a small fraction of its size wherever the effect is clear, with the per-market figures collected in Appendix H. Philadelphia, Chicago, Atlanta, and the other high-effect markets give up essentially nothing to the geography

control. The small-effect markets give up nothing they had to begin with. The per-market premium that the text channel identifies comes through the adjustment very nearly untouched by whether the model is told where each property sits.

A reader has every right to ask at once whether the test has any power. A treatment that carried no geography in the first place would pass it for free, and the right panel of Figure 3 exists to meet that objection before it hardens into one. The share of the treatment’s own variance that the location basis explains is nowhere near zero across the panel, reaching 0.42 in New York and exceeding a tenth in six other markets. The leading text direction does in fact encode location through much of the country. That the price effect survives adjustment even where the treatment is substantially geographic is the whole of the result. It means the geographic component of the language is present, the model can see it, and it still does not carry the effect of the language on price. The worry that has hung over this literature is therefore one that can be answered rather than only raised. The answer across eleven of twelve markets is that the text is doing more than reading the map back to itself.

New York is where the adjustment does bite, and it bites in exactly the place the diagnostic says it should. The market whose text direction is the most geographic of the twelve, at an R^2 of 0.42, is also the market where folding in location moves the effect most, from -0.122 to -0.154 . This is a shift of about a quarter of its magnitude that leaves the sign intact, so the place with the most location in its language is the place that location matters most for the estimate. We read this less as a crack in the broader finding than as its mechanism caught in the act, and as a reminder that the protocol earns its keep by flagging the markets where a spatial audit is most warranted rather than by declaring that none is ever needed.

The market the diagnostic singles out also reads differently once one looks at the language behind the direction. We interpret the leading text direction by the vocabulary that loads on it, in the way Nowak and Smith [2017] read implicit prices off listing tokens and the construct-validity approach of Fang et al. [2022] formalizes for embeddings. This shows that what the New York direction is built from is location. Its heaviest terms are boroughs and neighborhoods and the transit that joins them: Manhattan and Brooklyn and Midtown and SoHo, the subway and the avenue, alongside the housing nouns peculiar to the city such as the apartment and the studio. The same exercise in a market the adjustment leaves untouched returns condition and finish instead. In Dallas the owner suite and the kitchen and the open dining space carry the high-price end, while the city and its suburbs load only at the low end (Table 6). New York’s listings locate a property where Dallas’s describe one, and a direction made of place names is mechanically a direction that geography can move. That location is so heavily written into the words is itself a property of the market. Where submarkets are drawn tightly and fine-scale geography is steeply valued [Goodman and Thibodeau, 1998, Bourassa et al., 1999], the name of a neighborhood is already a statement about price, and the confounding such language carries sits at exactly the spatial grain a coarse control cannot reach [Paciorek, 2010, Keller and Szpiro, 2020]. The pattern is not that naming places makes a market’s text geographic in the sense that matters here. Phoenix names its places as readily as New York, its axis carrying Scottsdale and Arcadia and the desert. Yet its text effect is indistinguishable from zero, and the adjustment leaves it there, because a Phoenix suburb is not the steep price discontinuity that a Manhattan block is. It is locational language where location is steeply priced, rather than locational language as such, that the deconfounding responds to.

Table 6: The vocabulary of the leading text direction in the market the spatial adjustment moves most and a market it leaves untouched. Description terms whose TF-IDF most strongly loads on the price-oriented sentence-BERT direction. New York’s direction is built from boroughs, neighborhoods, and transit; Dallas’s from condition and finish, with the city and its suburbs loading only at the low-price end.

New York	Dallas
Manhattan, Brooklyn, Midtown, SoHo, Benson-hurst, subway, avenue, apartment, studio	owner suite, kitchen, open dining, living space, overlooking, thoughtfully curated, comfort

The conclusion travels only as far as its instruments reach, and they reach less far than one would like. The location basis is a smooth coordinate polynomial laid over roughly thirty structured controls that already carry census-tract demography. Confounding operating at a finer spatial grain than the polynomial can bend to, or running through pathways those controls do not capture, lies outside what the comparison can see.

And the erasure that certifies the representation’s geography is itself linear, so a nonlinear geographic signal no ridge probe would surface could in principle persist into the residual we are calling semantic.

That concession invites a direct test. We run it by replacing the coordinate polynomial with a thin-plate regression spline, the flexible low-rank smoother that the spatial-statistics literature reaches for when the scale of confounding is not known in advance [Keller and Szpiro, 2020, Gilbert et al., 2021]. A rank-thirty spline over latitude and longitude, projected to meters so that its isotropic metric is physical, captures far more of the treatment’s spatial structure than the quadratic does. It raises the average share of the leading text direction that location explains from about 0.13 to about 0.21, and from 0.42 to 0.52 in New York, and yet the price effect after adjustment barely responds. Measured from the structured-only baseline, the spline moves the New York estimate to -0.181 , a total geography adjustment of about 0.059 of a log point and the largest in the panel. Almost all of that movement is already accomplished by the quadratic. The further shift the spline buys beyond it is only 0.028 in New York and under 0.02 in every other market, with no market of detectable effect changing sign. New York aside, a basis flexible enough to absorb twice the geography leaves the semantic estimate essentially where the quadratic left it; in New York the finer grain takes a little more, still without overturning it. The finer spatial grain the polynomial could not bend to turns out, where the data can see it, not to carry the price effect either.

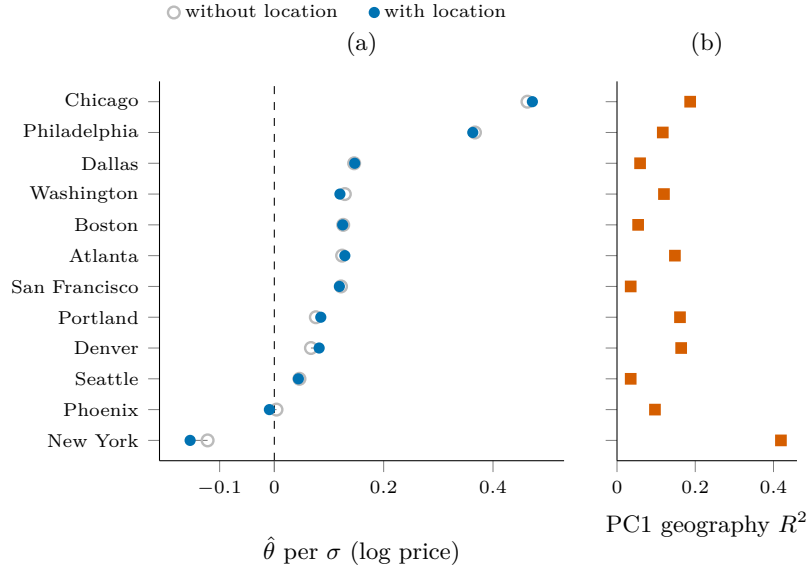


Figure 3: Is the per-market text-price effect spatially confounded? Holding the leading text principal component fixed as the treatment, panel (a) plots its DML effect on log price controlling for the structured confounders without (open) and with (filled) a location adjustment that adds latitude, longitude, and their quadratic and interaction terms to the confounder set. The two estimates nearly coincide in eleven of twelve markets, with the location adjustment moving the estimate by under 0.02 of a log point; New York is the exception, where it moves the estimate by about a quarter of its magnitude, 0.032, without changing its sign. Panel (b) reports the share of the treatment’s own variance that the location basis explains, which reaches 0.42 in New York and exceeds 0.10 in seven markets, so the robustness in panel (a) does not merely reflect a treatment that carries no geography: the text direction does encode location, yet that location component does not drive its price effect. Panel (a) plots point estimates; outside New York the adjustment shifts every estimate by at most 0.015 of a log point, so the near-coincidence reflects a small movement in the estimate rather than a feature of how it is drawn.

6.5 Estimating on the identified direction

Theorem 2 prescribes a direction. The effect it identifies is carried by the component of the text that survives after the location-predictable part of the embedding is removed. The leading direction of largest variance, which the Baur channel uses, need not coincide with it. We re-estimate the panel on the identified direction,

holding the full confounder set fixed, and read the difference from the leading-component estimate as the geography the leading direction still carries.

The construction follows the theorem. Within each market we residualize every coordinate of the sentence-BERT embedding on a location basis that includes a quadratic in latitude and longitude together with a full set of zip indicators, the discrete geographic channel Assumption 3 requires the control basis to span. We take the leading principal component of the pooled residualized embedding as the identified direction. It sits at a cosine of 0.98 to the raw leading component, so the two directions largely agree. This is why the decomposition of Section 6.4 finds the coefficient moving only modestly when geography enters the controls. Where they disagree is exactly the confounded content the theorem isolates.

Estimated on the identified direction under the full confounder set, the effect is robust. Market by market it stays close to its leading-component value, with a median ratio near 0.9. It is significant in all twelve markets rather than eleven. The direction earns its keep just where the leading component is most geographic. Chicago and Philadelphia, whose leading directions carry the most location, attenuate the most, from +0.464 to +0.370 and from +0.374 to +0.272. The markets whose leading component already lay near the identified direction, San Francisco, Washington, and Denver among them, barely move. The most telling case is New York, whose leading direction is the most geographic in the panel and whose effect on that direction is a significant -0.016 ($t = -2.2$). Removing the location channel does not merely restore the New York effect. It reverses its sign: on the identified direction it is a significant $+0.021$ ($t = 3.3$). The variance-maximal component had been reporting a negative effect because it pointed at borough and neighborhood names, whose price gradient runs the other way, rather than at the text’s own pricing content. The identified direction recovers the positive effect that geography had been masking with its own.

The reading this supports is the one the paper has argued throughout, now carried out on the direction the theory identifies rather than the one variance selects. The effect is not laundered geography, since it survives estimation on a direction orthogonal to location by construction. The theorem is not decorative either, since it changes the answer, trimming the inflated markets and recovering the suppressed one. One qualification belongs with the numbers. Residualizing on the full zip basis removes some non-spatial variation along with the confound. For the markets that attenuate, the identified estimate is a conservative floor, with the deconfounded effect lying between it and the leading-component value. The direction of the New York correction, where the identified estimate exceeds the leading one, is not subject to that caveat.

Spatially honest inference on the identified direction. Theorem 3 holds that the estimated direction contributes to the sampling error only at second order, provided it is formed out of sample and the pooled corpus dominates each market in size. It also holds that the interval to read is the spatial-HAC one rather than the influence-function one. We realize both. Forming the identified direction on the pooled embedding with the market in question left out, the out-of-fold construction the theorem requires, reproduces the per-market estimates above to within a hundredth of a log point in every market, the largest movement six thousandths in Chicago. This is the second-order contribution of the direction made visible: the point estimate barely moves when the direction is refit without the market’s own listings. The conditions the theorem asks of the pooled second moment hold with room to spare. The leading eigenvalue of the residualized embedding covariance runs near twice the second, and its effective rank sits near thirteen against a pooled sample of sixty-nine thousand, so the spectral gap is clear and the effective rank is negligible relative to the sample. The influence-function interval, which treats within-market listings as independent, is then too narrow. Replacing it with the Conley spatial-HAC standard error at a five-kilometre Bartlett cutoff, more conservative than the zip clustering of Section 6.2, widens the per-market intervals by a median factor near two. The identified effect remains significant in nine of the twelve markets, with t statistics from 2.9 in Portland to 7.3 in Philadelphia. The three that lose significance, Boston, New York, and Seattle, are exactly the markets whose identified effect is closest to zero to begin with. The finding does not hinge on the cutoff. The same nine markets clear significance whether the kernel reaches one, five, or ten kilometres. The median inflation runs from one and a half to two and a half times the independent interval as the cutoff widens, so the spatial dependence has a finite range rather than growing without bound. The construction keeps the two estimated objects apart from a market’s own score. The direction is fit on the pooled corpus with the market left out, and the nuisances are cross-fitted within the market, so neither is formed on the observations that enter that market’s influence-function sum. The pooled identified effect near eleven percent survives the widening with a t statistic above fourteen. The spatially honest inference the theorem prescribes therefore

leaves the reading intact: the effect the theory identifies is not an accident of treating spatially clustered listings as independent.

6.6 A within-building design

Every estimate to this point identifies the text effect under an assumption, that the location basis captures the spatial confounder, and bounds what an unobserved violation could do. A subset of the data lets us replace that assumption with a design. Units in the same building share location exactly. They sit at the same coordinates, in the same neighborhood, the same school catchment, and the same amenity set. Any confounder that is a function of location is therefore constant across them. Estimating the effect from variation in listing language between units of the same building differences spatial confounding out by construction rather than adjusting it away. What remains is identified by a design that holds geography fixed.

The data support this on a scale worth using. Grouping listings by coordinates rounded to about eleven meters, roughly eighteen thousand listings sit in buildings that carry two or more. The identifying variation is concentrated where it is most useful, with New York contributing 6,258 listings across 1,885 buildings. We add a building fixed effect, which is to say we demean the treatment, log price, and the structural controls within building. We estimate the partially linear DML coefficient on the within-building variation with standard errors clustered at the building. Location controls are unnecessary here because the fixed effect absorbs them.

The effect survives the design. Pooled across the twelve markets the within-building coefficient is $+0.035$ ($t = 10.8$), and every market's point estimate is positive. The effect is significant in six of the twelve, among them New York at $t = 8.6$, Chicago at $t = 8.6$, and Boston at $t = 4.9$. The six that do not reach significance are the markets with the fewest within-building listings, between roughly three hundred and fifteen hundred, and each remains positive. The pattern therefore reads as limited power rather than as a sign that geography was carrying the effect. New York is the case that matters most. Its leading-component effect in the cross section was an insignificant $+0.006$. Holding the building fixed returns a clean $+0.045$ at $t = 8.6$, the same recovery the identified-direction estimator produced by a different route. Two deconfounding strategies that share no machinery, one that projects the location channel out of the text direction and one that differences the building out of the design, agree that the New York effect is real and that the variance-maximal direction had been hiding it.

The confounder the building fixed effect leaves is unit quality, since a better-appointed unit in a building may draw both richer language and a higher price. The coarse part of that can be held fixed as well, by restricting the comparison to near-twin units, matching within building on bedrooms, bathrooms, and a square-footage bucket, so that the surviving text variation is between nearly identical units. The effect attenuates under this stricter design, as it should if part of the within-building estimate was unit size and layout. It does not vanish. Pooled across the markets with enough matched units it is $+0.010$ ($t = 4.0$). It is significant in the three with the largest matched samples, and it is significant once more in New York at $+0.024$ ($t = 4.5$). Even between units that share a building and a floor plan, the language a listing uses carries a small but resolvable price premium.

Two limits keep this a strong check rather than a clean experiment. The match on bedrooms, bathrooms, and area fixes the coarse dimensions of quality but not unmeasured interior condition, so a renovated unit that reads as renovated remains a possible channel. The design identifies against spatial confounding by construction and against gross quality by matching, but not against finish. And the building key is coordinate rounding rather than a parcel identifier, so some distinct buildings are split and some neighbors merged. Both forms of grouping error attenuate the coefficient toward zero, which makes the surviving effect a conservative reading. What the sequence of designs establishes is the claim the rest of the paper reaches by assumption. Controlling for location leaves the effect nearly intact. Projecting the location channel out of the direction recovers it where the variance-maximal axis had hidden it. Differencing the building out holds location fixed by construction, and matching near-twin units holds gross quality fixed as well. The effect survives each successive design and attenuates as each removes more of what it might have been confounded with. It stays significant, pooled and in New York, at the most stringent rung the data can support. A text effect riding on a representation that encodes location is not reducible to that location, now shown where location cannot vary at all and unit size is held fixed.

6.7 Robustness

The effect withstands the main challenges to it, each developed in the supplement. A sensitivity analysis reports how strong an unobserved confounder would have to be to overturn a market’s estimate, a threshold no observed covariate reaches in any of the twelve (Appendix K.3). The two text encodings, a Doc2Vec uniqueness axis and the sentence-BERT component, move together across the panel, so the surviving effect is not specific to one representation (Appendix K.2). Re-estimated across ten markets on the price a property actually fetches rather than the one its agent posts, it replicates on recorded transactions and is individually significant in every one (Appendix K.5). And a counterfactual rewriting intervention, which holds the model’s inferred location fixed while stripping stylistic content, isolates a positive style effect of the expected sign (Appendix K.8). These analyses qualify the ladder rather than extend it.

7 Conclusion

This paper has asked whether the predictive gains that text embeddings deliver in automated valuation models reflect causal semantic content or spatial confounding mediated through language, and whether the answer travels across metropolitan housing markets. We work with 69,204 deduplicated Redfin listings across twelve metropolitan areas, a panel from which we exclude no market and whose confounders we winsorize at the median plus or minus five median absolute deviations to neutralize parcel data-entry errors. On it we built a framework that joins a partially-linear Double Machine Learning estimator, LEACE concept erasure with an iterative nonlinear polish, a counterfactual rewriting intervention with two open-weight language models, and a finite-sample calibration study. What the framework establishes most centrally is that the price effect of listing text is not, in the main, geography wearing a semantic costume. Holding the leading text direction fixed as the treatment and folding a flexible location basis into the controls leaves its per-standard-deviation effect on log price almost exactly where it began across eleven of the twelve markets, even though that same direction provably encodes location. The one place where the adjustment does bite is New York, whose own text is the most geographic of the twelve. The effect that survives is robust as well to how the text is encoded. A Doc2Vec uniqueness channel pools to +0.060 and a sentence-BERT principal-component axis to +0.137, and the two run correlated across the markets at a Pearson of +0.94. They track one signal under two representations rather than two independent observations of it. That full-panel agreement is carried by Chicago and Philadelphia, and it weakens to a non-significant correlation once the two are set aside. That signal ranges from a small negative in New York to several times the pooled mean in Chicago and Philadelphia, and a collinearity-corrected Hotelling test rejects the joint zero null in every one of the twelve markets. The counterfactual intervention sharpens the reading. A contrast that strips stylistic content while holding the model’s inferred location fixed isolates a positive style effect of +0.074 on log price. A contrast that also swaps the targeted submarket nets to approximately zero, the per-market swap effects dividing in sign across the twelve markets and averaging away rather than a second channel of comparable size offsetting the first. We searched for a metropolitan-level demographic moderator of the pooled effect and found none that survives multiplicity adjustment or leave-one-out leverage diagnostics at twelve markets, and we report that null rather than a moderator that an earlier draft had reported before a confounder correction reversed its sign.

Limitations. The reading we have offered is hedged in several ways that are worth stating plainly rather than leaving for a reader to find. The most consequential is the boundary of the identification itself: the effect is identified against the location that motivates the paper, which the designs hold fixed, but not against unmeasured interior quality, which the sensitivity analysis bounds rather than differences away. The robustness-value analysis of Section K.3 places that residual on a scale rather than discharging it. An unobserved confounder able to overturn a per-market estimate would have to explain more of the residual variation in both the text direction and log price than the observed spatial controls do, whose partial R^2 falls inside the robustness value in every market but Boston, with the widest margin in Philadelphia and San Francisco. That assurance is not uniform. It thins toward the markets whose robustness values are small in absolute terms, and in Boston, whose estimate is close to zero to begin with, the spatial control block alone carries more explanatory power for the treatment than the confounder that would erase the effect, so nothing in that market’s result should be leaned on. We therefore do not claim to have isolated a structural

semantic effect, only to have shown that the location a representation encodes is largely not the location that moves the price, and to have bounded how strong any confounding the test cannot see would have to be. What survives that adjustment need not be semantic content in any broad sense. The vocabulary that loads on the price end of the text axis is largely condition and finish language, words for renovation, layout, and quality, and our structured controls carry bedroom and bathroom counts, area, lot size, and year built but no direct measure of interior condition or renovation. An unobserved quality signal that the language proxies would produce the pattern we report, survive the spatial adjustment, and vary across markets as we find, so we read the surviving effect as information the text carries beyond location and structure rather than as evidence that it is priced semantic content as opposed to unmeasured physical quality. A distinct reading, that the premium is an agent talking a price up rather than conveying anything about the property, is answered by the sale-price panel of Appendix K.5. Were the effect the framing of an asking price it would attach to the number a seller posts and not to the one a buyer pays, yet it survives re-estimation on recorded transactions and is individually significant in all ten markets with public deeds, so what the language moves is validated in the market rather than set by the seller alone. What the text most plausibly carries, then, is unit-level condition and finish that the structured attributes do not record and that buyers pay for, a reading the within-building rung supports by finding the premium between near-identical units in the same building, where neither neighborhood nor gross size can vary. Where an independent assessor measure of condition or improvement value exists, in Boston, San Francisco, and New York, conditioning on it leaves the effect intact with only modest attenuation, and in the two markets without one a quality confounder would have to be several times stronger than any we can measure to overturn the estimate, so the concern is bounded rather than dispositive; still, our design does more to separate language from geography than to separate it from quality. The corpus is restricted to the twelve metropolitan areas served by the Redfin scraper at the time of collection, which biases the sample toward larger and more competitive markets and away from the rural and small-metro regimes in which language-mediated spatial confounding is plausibly different in character. The two embedding channels are collinear, so their agreement is a robustness check across representation rather than triangulation across independent biases, and the counterfactual channel, the one probe with a distinct bias structure, is also the least precise. Finally the panel is small enough at twelve markets that one high-leverage market, New York, dominates the moderator diagnostics, so the null moderator result should be read as the absence of a moderator detectable at this sample size rather than as a demonstrated absence of moderation.

Implications. The implication that travels furthest is methodological, and it cuts against an easy assumption rather than confirming a dire one. Because a model that prices semantic content and a model that prices laundered geography make identical predictions, a predictive gain on its own cannot tell an analyst which of the two it has bought, and the reflexive worry that the accuracy of a text-augmented valuation is mostly relabeled location is one that a predictive metric can neither confirm nor dispel. What the results show is that, once the question is put to a design-based test rather than left to intuition, the worry largely does not bear out. The text effect survives adjustment for location in eleven of twelve markets. A practitioner or regulator who suspected the gains were spatial confounding can therefore be reassured for the kind of large, competitive metropolitan markets studied here, while being directed, by the very same test, to the one market where the suspicion is earned. The lesson is not to distrust text valuation but to stop certifying it on predictive fit alone. The deconfounding toggle we describe, holding the text feature fixed while moving geography across the controls, is a low-cost audit that turns a question of suspicion into one of measurement before either trust or alarm is allowed to settle.

Future research. The natural extensions of this work run along the seams where its instruments stop short. A panel of substantially more than twelve markets would give the moderator analysis the leverage it lacks here and would let the cross-market heterogeneity we document be explained rather than only described. The counterfactual intervention inherits the rewriting model’s own biases, which a randomized field experiment that asks sellers to revise descriptions under a pre-registered protocol would replace with a controlled manipulation at the cost of operational complexity. The realized-transaction panel of Section K.5 takes a first step against the price-measurement and asking-price-simultaneity concerns by re-estimating the effect on recorded sale prices across ten markets, and extending that panel to the markets whose sale prices are not public would complete it; pairing the causal-residual detector this paper isolates with the

demographic-stratification machinery of the fairness-audit literature would then let practitioners locate where any language-mediated mispricing that survives deconfounding is large enough to matter for borrowers.

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References

- Eman Ahmed and Mohamed Moustafa. House price estimation from visual and textual features. *arXiv preprint arXiv:1609.08399*, 2016.
- Luc Anselin. *Spatial econometrics: Methods and models*. Springer Science & Business Media, 1988.
- Evgeny A. Antipov and Elena B. Pokryshevskaya. Mass appraisal of residential apartments: An application of random forest for valuation and a CART-based approach for model diagnostics. *Expert Systems with Applications*, 39(2):1772–1778, 2012.
- Carlos F Avenancio-León and Troup Howard. The assessment gap: Racial inequalities in property taxation. *Quarterly Journal of Economics*, 137(3):1383–1434, 2022.
- Laura Battaglia, Timothy Christensen, Stephen Hansen, and Szymon Sacher. Inference for regression with variables generated by ai or machine learning. *arXiv preprint arXiv:2402.15585*, 2024.
- Katharina Baur, Markus Rosenfelder, and Bernhard Lutz. Automated real estate valuation with machine learning models using property descriptions. *Expert Systems with Applications*, 213:119147, 2023.
- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. Leace: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- C Alan Bester, Timothy G Conley, and Christian B Hansen. Inference with dependent data using cluster covariance estimators. *Journal of Econometrics*, 165(2):137–151, 2011.
- Junchi Bin, Bryan Gardiner, Eric Li, and Zheng Liu. Multi-source urban data fusion for property value assessment: A case study in Philadelphia. *Neurocomputing*, 404:70–83, 2020.
- Tolga Bolukbasi, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai. Man is to computer programmer as woman is to homemaker: Debiasing word embeddings. In *Advances in Neural Information Processing Systems*, volume 29, 2016.
- John Bound, Charles Brown, and Nancy Mathiowetz. Measurement error in survey data. *Handbook of Econometrics*, 5:3705–3843, 2001. doi:10.1016/S1573-4412(01)05012-7.
- Steven C. Bourassa, Foort Hamelink, Martin Hoesli, and Bryan D. MacGregor. Defining housing submarkets. *Journal of Housing Economics*, 8(2):160–183, 1999.
- Steven C. Bourassa, Eva Cantoni, and Martin Hoesli. Predicting house prices with spatial dependence: A comparison of alternative methods. *Journal of Real Estate Research*, 32(2):139–160, 2010.
- Aylin Caliskan, Joanna J. Bryson, and Arvind Narayanan. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017.
- A. Colin Cameron and Douglas L. Miller. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi:10.3368/jhr.50.2.317.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008. doi:10.1162/rest.90.3.414.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29(2):238–249, 2011. doi:10.1198/jbes.2010.07136.
- Ayşe Can. Specification and estimation of hedonic housing price models. *Regional Science and Urban Economics*, 22(3):453–474, 1992.
- Raymond J. Carroll, David Ruppert, Leonard A. Stefanski, and Ciprian M. Crainiceanu. *Measurement Error in Nonlinear Models: A Modern Perspective*. Chapman & Hall/CRC, 2nd edition, 2006. doi:10.1201/9781420010138.

- Karl E. Case and Robert J. Shiller. The efficiency of the market for single-family homes. *American Economic Review*, 79(1):125–137, 1989.
- Domagoj Čevd, Peter Bühlmann, and Nicolai Meinshausen. Spectral deconfounding via perturbed sparse linear models. *Journal of Machine Learning Research*, 21(232):1–41, 2020.
- Xiaohong Chen, Han Hong, and Alessandro Tarozzi. Semiparametric efficiency in GMM models with auxiliary data. *The Annals of Statistics*, 36(2):808–843, 2008. doi:10.1214/009053607000000947.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- Victor Chernozhukov, Carlos Cinelli, Whitney Newey, Amit Sharma, and Vasilis Syrgkanis. Long story short: Omitted variable bias in causal machine learning. *arXiv preprint arXiv:2112.13398*, 2022.
- Harold D. Chiang, Kengo Kato, Yukun Ma, and Yuya Sasaki. Multiway cluster robust double/debiased machine learning. *Journal of Business & Economic Statistics*, 40(3):1046–1056, 2022. doi:10.1080/07350015.2021.1895815.
- David Maxwell Chickering. Optimal structure identification with greedy search. *Journal of Machine Learning Research*, 3:507–554, 2002.
- Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B*, 82(1):39–67, 2020.
- Carlos Cinelli, Andrew Forney, and Judea Pearl. A crash course in good and bad controls. *Sociological Methods & Research*, 53(3):1071–1104, 2024. doi:10.1177/00491241221099552.
- Timothy G Conley. Gmm estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45, 1999.
- John R. Cook and Leonard A. Stefanski. Simulation-extrapolation estimation in parametric measurement error models. *Journal of the American Statistical Association*, 89(428):1314–1328, 1994. doi:10.1080/01621459.1994.10476871.
- Alexander D’Amour, Peng Ding, Avi Feller, Lihua Lei, and Jasjeet Sekhon. Overlap in observational studies with high-dimensional covariates. *Journal of Econometrics*, 221(2):644–654, 2021. doi:10.1016/j.jeconom.2019.10.014.
- Sarkar Snigdha Sarathi Das, Mohammed Eunus Ali, Yuan-Fang Li, Yong-Bin Kang, and Timos Sellis. Boosting house price predictions using geo-spatial network embedding. *Data Mining and Knowledge Discovery*, 35:2221–2250, 2021.
- Julia Deppner and Marcelo Cajias. Accounting for spatial autocorrelation in algorithm-driven hedonic models: A spatial cross-validation approach. *Journal of Real Estate Finance and Economics*, 68:235–273, 2024. doi:10.1007/s11146-022-09915-y.
- Paul Diegert, Matthew A. Masten, and Alexandre Poirier. An axiomatic approach to comparing sensitivity parameters. *arXiv preprint arXiv:2504.21106*, 2026.
- Naoki Egami, Christian J. Fong, Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. How to make causal inferences using texts. *Science Advances*, 8(42):eabg2652, 2022. doi:10.1126/sciadv.abg2652.
- Jacob Eisenstein, Brendan O’Connor, Noah A. Smith, and Eric P. Xing. Diffusion of lexical change in social media. *PloS One*, 9(11):e113114, 2014.
- Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 11–21, 2018.

- J. Paul Elhorst. *Spatial econometrics: From cross-sectional data to spatial panels*. Springer, 2014.
- Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. In *Proceedings of EMNLP-IJCNLP 2019*, 2019. arXiv:1909.00512.
- Yu Fan, Yang Tian, Shauli Ravfogel, Mrinmaya Sachan, Elliott Ash, and Alexander Hoyle. The medium is not the message: Deconfounding document embeddings via linear concept erasure. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. arXiv:2507.01234.
- Qixiang Fang, Dong Nguyen, and Daniel L. Oberski. Evaluating the construct validity of text embeddings with application to survey questions. *EPJ Data Science*, 11(1):39, 2022.
- Omri Feldman, Amar Venugopal, Jann Spiess, and Amir Feder. Causal effect estimation with latent textual treatments, 2026. arXiv:2602.15730.
- Bernhard N. Flury. Common principal components in k groups. *Journal of the American Statistical Association*, 79(388):892–898, 1984. doi:10.1080/01621459.1984.10477108.
- Christian Fong and Justin Grimmer. Causal inference with latent treatments. *American Journal of Political Science*, 2023. doi:10.1111/ajps.12649.
- Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17:2096–2030, 2016.
- Jonah B. Gelbach. When do covariates matter? and which ones, and how much? *Journal of Labor Economics*, 34(2):509–543, 2016.
- Andrew Gelman and John Carlin. Beyond power calculations: Assessing type S (sign) and type M (magnitude) errors. *Perspectives on Psychological Science*, 9(6):641–651, 2014. doi:10.1177/1745691614551642.
- David Genesove and Christopher Mayer. Loss aversion and seller behavior: Evidence from the housing market. *Quarterly Journal of Economics*, 116(4):1233–1260, 2001.
- Steve Gibbons, Henry G. Overman, and Eleonora Patacchini. Spatial methods. *Handbook of Regional and Urban Economics*, 5:115–168, 2015.
- Brian Gilbert, Abhirup Datta, Joan A. Casey, and Elizabeth L. Ogburn. A causal inference framework for spatial confounding. *arXiv preprint arXiv:2112.14946*, 2021.
- Brian Gilbert, Elizabeth L. Ogburn, and Abhirup Datta. Consistency of common spatial estimators under spatial confounding. *arXiv preprint arXiv:2308.12181*, 2024.
- Robert J. Gloudemans. *Mass appraisal of real property*. International Association of Assessing Officers, 1999.
- Hila Gonen and Yoav Goldberg. Lipstick on a pig: Debiasing methods cover up systematic gender biases in word embeddings but do not remove them. *arXiv preprint arXiv:1903.03862*, 2019.
- Allen C Goodman and Thomas G Thibodeau. Housing market segmentation. *Journal of Housing Economics*, 7(2):121–143, 1998.
- Yawen Guan, Garritt L. Page, Brian J. Reich, Massimo Ventrucchi, and Shu Yang. A spectral adjustment for spatial confounding. *Biometrika*, 110(3):699–719, 2023. doi:10.1093/biomet/asac069.
- Lin Gui and Victor Veitch. Causal estimation for text data with (apparent) overlap violations. In *International Conference on Learning Representations (ICLR)*, 2023.
- Shengwei Guo. Racial disparity in appraisals: Evidence from private-label refinance mortgages. *Journal of Real Estate Finance and Economics*, 72:606–639, 2025. doi:10.1007/s11146-025-10025-8.
- Zijian Guo, Domagoj Ćevd, and Peter Bühlmann. Doubly debiased lasso: High-dimensional inference under hidden confounding. *Annals of Statistics*, 50(3), 2022. doi:10.1214/21-AOS2152.

- Jinyong Hahn and Geert Ridder. Asymptotic variance of semiparametric estimators with generated regressors. *Econometrica*, 81(1):315–340, 2013. doi:10.3982/ECTA9609.
- Bo Han, Paul Cook, and Timothy Baldwin. Geolocation prediction in social media data by finding location indicative words. In *Proceedings of COLING 2012*, pages 1045–1062, 2012.
- James J. Heckman. Sample selection bias as a specification error. *Econometrica*, 47(1):153–161, 1979. doi:10.2307/1912352.
- James S. Hodges and Brian J. Reich. Adding spatially-correlated errors can mess up the fixed effect you love. *The American Statistician*, 64(4):325–334, 2010.
- Chao Huang, Junbo Zhang, Yu Zheng, and Nitesh V. Chawla. DeepCrime: Attentive hierarchical recurrent networks for crime prediction. In *Proceedings of the ACM International Conference on Information and Knowledge Management*, pages 1423–1432, 2018.
- Rustam Ibragimov and Ulrich K Müller. t-statistic based correlation and heterogeneity robust inference. *Journal of Business and Economic Statistics*, 28(4):453–468, 2010.
- Joanna IntHout, John P. A. Ioannidis, and George F. Borm. The hartung–knapp–sidik–jonkman method for random effects meta-analysis is straightforward and considerably outperforms the standard dersimonian–laird method. *BMC Medical Research Methodology*, 14:25, 2014. doi:10.1186/1471-2288-14-25.
- Shadi Iskander, Kira Radinsky, and Yonatan Belinkov. Shielded representations: Protecting sensitive attributes through iterative gradient-based projection. In *Findings of the Association for Computational Linguistics (ACL)*, 2023. arXiv:2305.10204.
- Nazgul Jenish and Ingmar R. Prucha. Central limit theorems and uniform laws of large numbers for arrays of random fields. *Journal of Econometrics*, 150(1):86–98, 2009. doi:10.1016/j.jeconom.2009.02.009.
- Connor T. Jerzak and Stephen A. Jessee. Attenuation bias with latent predictors, 2025.
- Yuhao Kang, Fan Zhang, Song Gao, Wenzhe Peng, and Carlo Ratti. Human settlement value assessment from a place perspective: Considering human dynamics and perceptions in house price modeling. *Cities*, 118:103333, 2021.
- Katherine A. Keith, David Jensen, and Brendan O’Connor. Text and causal inference: A review of using text to remove confounding from causal estimates. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. doi:10.18653/v1/2020.acl-main.474.
- Joshua P. Keller and Adam A. Szpiro. Selecting a scale for spatial confounding adjustment. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 183(3):1121–1143, 2020. doi:10.1111/rssa.12556.
- Ana Khot, Miruna Oprescu, Maresa Schröder, Ryan Kagawa, and Xu Luo. Spatial deconfounder: Interference-aware deconfounding for spatial causal inference, 2025.
- Niki Kilbertus, Mateo Rojas-Carulla, Giambattista Parascandolo, Moritz Hardt, Dominik Janzing, and Bernhard Schölkopf. Avoiding discrimination through causal reasoning. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Vladimir Koltchinskii and Karim Lounici. Asymptotics and concentration bounds for bilinear forms of spectral projectors of sample covariance. *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques*, 52(4):1976–2013, 2016. doi:10.1214/15-AIHP705.
- Zoran Kostić and Aleksandar Jevremović. What image features boost housing market predictions? *IEEE Transactions on Multimedia*, 22(7):1904–1916, 2020.
- Matt J. Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

- Stephen Law, Brooks Paige, and Chris Russell. Take a look around: Using street view and satellite images to estimate house prices. *ACM Transactions on Intelligent Systems and Technology*, 10(5):1–19, 2019.
- James LeSage and R. Kelley Pace. *Introduction to spatial econometrics*. Chapman and Hall/CRC, 2009.
- Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. *arXiv preprint arXiv:1707.01926*, 2018.
- Regina Fang-Ying Lin, Chihiro Ou, Kuo-Kun Tseng, Daniel Bowen, Kai Leung Yung, and Wai Hung Ip. The spatial neural network model with disruptive technology for property appraisal in real estate industry. *Technological Forecasting and Social Change*, 173:121067, 2021.
- Thies Lindenthal and Erik B. Johnson. Machine learning, architectural styles and property values. *The Journal of Real Estate Finance and Economics*, 71:353–384, 2025.
- Felix Lorenz, Jonas Willwersch, Marcelo Cajias, and Franz Fuerst. Interpretable machine learning for real estate market analysis. *Real Estate Economics*, 51(5):1178–1208, 2023.
- Christos Louizos, Uri Shalit, Joris M. Mooij, David Sontag, Richard Zemel, and Max Welling. Causal effect inference with deep latent-variable models. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- David Madras, Elliot Creager, Toniann Pitassi, and Richard Zemel. Learning adversarially fair and transferable representations. In *International Conference on Machine Learning (ICML)*, pages 3384–3393, 2018.
- Joel A. Middleton, Marc A. Scott, Ronli Diakow, and Jennifer L. Hill. Bias amplification and bias unmasking. *Political Analysis*, 24(3), 2016. doi:10.1093/pan/mpw015.
- Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019. doi:10.1002/sim.8086.
- Daniel Moyer, Shuyang Gao, Rob Brekelmans, Aram Galstyan, and Greg Ver Steeg. Invariant representations without adversarial training. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Henry J. Munneke and Barrett A. Slade. An empirical study of sample-selection bias in indices of commercial real estate. *Journal of Real Estate Finance and Economics*, 21(1):45–64, 2000. doi:10.1023/A:1007861303058.
- Kevin M. Murphy and Robert H. Topel. Estimation and inference in two-step econometric models. *Journal of Business & Economic Statistics*, 3(4):370–379, 1985. doi:10.1080/07350015.1985.10509471.
- Adam Nowak and Patrick Smith. Textual analysis in real estate. *Journal of Applied Econometrics*, 32(4): 896–918, 2017. doi:10.1002/jae.2550.
- Adam D. Nowak, Bradley S. Price, and Patrick S. Smith. Real estate dictionaries across space and time. *The Journal of Real Estate Finance and Economics*, 62(1):139–163, 2021.
- Elizabeth L. Ogburn, Ilya Shpitser, and Eric J. Tchetgen Tchetgen. Comment on “blessings of multiple causes”. *Journal of the American Statistical Association*, 114(528):1611–1615, 2019. doi:10.1080/01621459.2019.1689139.
- R. Kelley Pace and Ronald Barry. Sparse spatial autoregressions. *Statistics & Probability Letters*, 33(3): 291–297, 1997.
- Christopher J. Paciorek. The importance of scale for spatial-confounding bias and precision of spatial regression estimators. *Statistical Science*, 25(1):107–125, 2010.
- Adrian Pagan. Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, 25(1):221–247, 1984.

- Byeonghwa Park and Jae Kwon Bae. Using machine learning algorithms for housing price prediction: The case of Fairfax County, Virginia housing data. *Expert Systems with Applications*, 42(6):2928–2934, 2015.
- Robert C. Paule and John Mandel. Consensus values and weighting factors. *Journal of Research of the National Bureau of Standards*, 87(5):377–385, 1982. doi:10.6028/jres.087.022.
- Judea Pearl. *Causality: Models, reasoning, and inference*. Cambridge University Press, 2nd edition, 2009.
- Hao Peng, Jianxin Li, Zheng Wang, Renyu Yang, Mingzhe Liu, Mingming Zhang, Philip S. Yu, and Lifang He. Lifelong property price prediction: A case study for the Toronto real estate market. *IEEE Transactions on Knowledge and Data Engineering*, 35(3):2765–2780, 2023.
- Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society: Series B*, 78(5):947–1012, 2016.
- Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. *Elements of causal inference: Foundations and learning algorithms*. MIT Press, 2017.
- Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4609–4622, 2020.
- Omid Poursaeed, Tomas Matera, and Serge Belongie. Vision-based real estate price estimation. *Machine Vision and Applications*, 29:667–676, 2018.
- Reid Pryzant, Dallas Card, Dan Jurafsky, Victor Veitch, and Dhanya Sridhar. Causal effects of linguistic properties. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2021. doi:10.18653/v1/2021.naacl-main.323.
- Shauli Ravfogel, Michael Twiton, Yoav Goldberg, and Ryan Cotterell. Linear adversarial concept erasure. In *International Conference on Machine Learning (ICML)*, pages 18400–18421, 2022.
- Brian J. Reich, Shu Yang, Yawen Guan, Andrew B. Giffin, Matthew J. Miller, and Ana Rappold. A review of spatial causal inference methods for environmental and epidemiological applications. *International Statistical Review*, 89(3):605–634, 2021. doi:10.1111/insr.12452.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *arXiv preprint arXiv:1908.10084*, 2019.
- Margaret E. Roberts, Brandon M. Stewart, and Richard A. Nielsen. Adjusting for confounding with text matching. *American Journal of Political Science*, 64(4):887–903, 2020. doi:10.1111/ajps.12526.
- Peter M. Robinson. Root-N-consistent semiparametric regression. *Econometrica*, 56(4):931–954, 1988.
- Mateo Rojas-Carulla, Bernhard Schölkopf, Richard Turner, and Jonas Peters. Invariant models for causal transfer learning. *Journal of Machine Learning Research*, 19:1–34, 2018.
- Sherwin Rosen. Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy*, 82(1):34–55, 1974.
- Gabriel Saco. Ill-conditioned orthogonal scores in double machine learning. *arXiv preprint arXiv:2512.07083*, 2025.
- Stephen Salerno, Zhenke Wu, and Tyler H McCormick. Spatially robust inference with predicted and missing at random labels. *arXiv:2603.11368*, 2026.
- Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5): 612–634, 2021.

- Lily Shen and Stephen L. Ross. Information value of property description: A machine learning approach. *Journal of Urban Economics*, 121:103299, 2021.
- Stephen Sheppard. Hedonic analysis of housing markets. *Handbook of Regional and Urban Economics*, 3: 1595–1635, 1999.
- Shohei Shimizu, Patrik O. Hoyer, Aapo Hyvärinen, and Antti Kerminen. A linear non-Gaussian acyclic model for causal discovery. *Journal of Machine Learning Research*, 7:2003–2030, 2006.
- G. Stacy Sirmans, David A. Macpherson, and Emily N. Zietz. The composition of hedonic pricing models. *Journal of Real Estate Literature*, 13(1):1–44, 2005. doi:10.1080/10835547.2005.12090154.
- Sho Sonoda, Kazumi Kasaura, Yuma Mizuno, Kei Tsukamoto, and Naoto Onda. Lean formalization of generalization error bound by rademacher complexity and dudley’s entropy integral. *arXiv preprint arXiv:2503.19605*, 2025.
- Peter Spirtes, Clark N. Glymour, and Richard Scheines. *Causation, prediction, and search*. MIT Press, 2000.
- Maximilian Stang, Bastian Krämer, Marcelo Cajias, and Wolfgang Schäfers. Changing the location game — improving location analytics with the help of explainable AI. *Journal of Real Estate Research*, 46(4): 421–443, 2024.
- Chi Sun, Xipeng Qiu, Yige Xu, and Xuanjing Huang. How to fine-tune BERT for text classification? In *China National Conference on Chinese Computational Linguistics (CCL)*, pages 194–206, 2019. doi:10.1007/978-3-030-32381-3_16.
- Victor Veitch, Dhanya Sridhar, and David M. Blei. Adapting text embeddings for causal inference. In *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 124 of *Proceedings of Machine Learning Research*, 2020.
- Wayne Xinwei Wan and Thies Lindenthal. Testing machine learning systems in real estate. *Real Estate Economics*, 51(3):754–778, 2023.
- Yixin Wang and David M. Blei. The blessings of multiple causes. *Journal of the American Statistical Association*, 114(528):1574–1596, 2019. doi:10.1080/01621459.2019.1686987.
- Qizhe Xie, Zihang Dai, Yulun Du, Eduard Hovy, and Graham Neubig. Controllable invariance through adversarial feature learning. *Advances in Neural Information Processing Systems*, 30, 2017.
- Yilun Xu, Shengjia Zhao, Jiaming Song, Russell Stewart, and Stefano Ermon. A theory of usable information under computational constraints. In *International Conference on Learning Representations (ICLR)*, 2020.
- Mochen Yang, Edward McFowland III, Gordon Burtch, and Gediminas Adomavicius. Achieving reliable causal inference with data-mined variables: A random forest approach to the measurement error problem, 2020.
- Yi Yu, Tengyao Wang, and Richard J Samworth. A useful variant of the davis–kahan theorem for statisticians. *Biometrika*, 102(2):315–323, 2015.
- Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell. Mitigating unwanted biases with adversarial learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 335–340, 2018.
- Weijia Zhang, Hao Liu, Lubao Zha, Hengshu Zhu, Ji Liu, Dejing Dou, and Hui Xiong. MugRep: A multi-task hierarchical graph representation learning framework for real estate appraisal. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 3937–3947, 2021.
- Linna Zhu, Michael Neal, and Caitlin Young. Racial disparities in automated valuation models: New evidence using property condition and machine learning. *Cityscape*, 26(1), 2024. U.S. Department of Housing and Urban Development, Office of Policy Development and Research.

A Structural causal model: formal specification

Section 4 states the reduced partially-linear model the identification result uses; here we record the fuller structural causal model behind it, whose functional-form detail and deflationary null the body leaves implicit. Location $L \in \mathbb{R}^2$ is exogenous, and intrinsic property features X (bedrooms, bathrooms, area, year built), contextual features C (demographics, crime, amenities), the text embedding T , and property value Y are generated by $X = g_x(L) + \epsilon_X$, $C = g_c(L) + \epsilon_C$, $T = g_t(L, X, C) + \epsilon_T$, and $Y = g_y(L, X, C) + \epsilon_Y$, with mutually independent noise terms and deterministic mechanisms g_x, g_c, g_t, g_y . Descriptions are written by agents who observe location, features, and context, which is why the embedding carries geography.

A.1 Functional form specifications

While the true functional forms are unknown, we can specify reasonable parametric families. Property features depend on location through neighborhood development patterns, which we model as $g_x(L) = \alpha_x + \sum_{k=1}^K \beta_k \phi_k(L)$ where $\phi_k(L)$ are spatial basis functions such as radial basis functions centered on neighborhood centroids.

Contextual features vary smoothly over space, which we model using a Gaussian process: $g_c(L) = \alpha_c + \text{GP}(L; \kappa)$ where $\text{GP}(\cdot; \kappa)$ denotes a Gaussian process with kernel κ capturing spatial autocorrelation in demographics and amenities.

Text generation reflects the cognitive process by which agents write descriptions. We model this as $g_t(L, X, C) = \text{Encoder}(f_{\text{LLM}}(L, X, C; \theta_{\text{agent}}))$ where f_{LLM} represents the agent’s language generation process and Encoder maps text to embedding space.

Price formation depends on location, features, and local context through a log-linear specification: $g_y(L, X, C) = \exp(\alpha_y + \beta_L^\top h(L) + \beta_X^\top X + \beta_C^\top C)$ with $h(L)$ capturing non-linear spatial effects such as distance to city center or waterfront proximity.

A.2 The deflationary null and what identifies the text effect

It is worth separating two things this specification is easily read as conflating. The deflationary reading of the model is that T does not enter the structural equation for Y at all, so that every observed text-price association is the common imprint of location, features, and context. That reading is a *hypothesis to be tested*, not an assumption we impose, and the body of the paper estimates the text effect rather than assuming it away.

Hypothesis 1 (No direct text effect). *Conditional on location, property features, and context, text is independent of price, $Y \perp T \mid (L, X, C)$, so that the per-standard-deviation effect θ of the leading text direction on log price is zero.*

What identifies θ is not Hypothesis 1 but the backdoor set $\{L, X, C\}$, which blocks every spurious path from T to Y whether or not the null holds (Appendix B). The estimand is therefore well defined under both readings: under the null it equals zero, and the partialling-out estimator of Section 3 recovers it consistently in either case. The empirical finding, a positive θ in eleven of twelve markets that survives spatial deconfounding, is a rejection of Hypothesis 1, not the violation of an assumption.

B Identification proofs

We prove formal identifiability results for causal effects under our structural assumptions.

B.1 Proof of Theorem 1 (identification)

The setup, assumptions, and statement are given in Section 4.

Proof of Theorem 1. Step 1 (the confounding is absorbed). By Assumptions 2–3, $U = h(\ell)$ with h in the span of L , so $\beta U + f(x) = g(L, x)$ for the fixed (L, x) -measurable function $g := \beta h + f$. Model (4) becomes $Y = \theta T + g(L, x) + \varepsilon$.

Step 2 (exogeneity survives coarsening). Since (L, x) is $\sigma(\ell, x)$ -measurable, iterated expectations and Assumption 5 give $\mathbb{E}[\varepsilon \mid T, L, x] = \mathbb{E}[\mathbb{E}[\varepsilon \mid T, \ell, x] \mid T, L, x] = 0$.

Step 3 (partialling out). Taking $\mathbb{E}[\cdot \mid L, x]$ of the model, $\mathbb{E}[Y \mid L, x] = \theta \mathbb{E}[T \mid L, x] + g(L, x)$, and subtracting yields $\tilde{Y} = \theta \tilde{T} + \varepsilon$.

Step 4 (the moment). Multiplying by $\tilde{T}$ and taking expectations, $\mathbb{E}[\tilde{Y}\tilde{T}] = \theta \mathbb{E}[\tilde{T}^2] + \mathbb{E}[\varepsilon\tilde{T}]$. The cross term vanishes: $\mathbb{E}[\varepsilon\tilde{T}] = \mathbb{E}[\varepsilon T] - \mathbb{E}[\varepsilon \mathbb{E}[T \mid L, x]] = 0$, the first term by $\mathbb{E}[\varepsilon T] = \mathbb{E}[\mathbb{E}[\varepsilon \mid T, L, x] T] = 0$ (Step 2) and the second by $\mathbb{E}[\varepsilon \mathbb{E}[T \mid L, x]] = \mathbb{E}[\mathbb{E}[\varepsilon \mid L, x] \mathbb{E}[T \mid L, x]] = 0$. Dividing by $\mathbb{E}[\tilde{T}^2]$, positive by Assumption 4, gives the stated functional.

Step 5 (erasure diagnostic and the boundary). $\mathbb{E}[\tilde{T}^2] = \text{Var}(T - \mathbb{E}[T \mid L, x])$ is the treatment's variance orthogonal to (L, x) . The concept-erasure diagnostic probes for this orthogonal variation by recovering location from the embedding and driving its recoverable share to the floor, rather than estimating $\mathbb{E}[\tilde{T}^2]$ directly. Writing $R_{T \sim (L, x)}^2 = 1 - \mathbb{E}[\tilde{T}^2] / \text{Var}(T)$, as this share tends to one the Robinson denominator tends to zero and θ ceases to be point-identified. New York, whose leading text direction loads on borough and neighborhood names, is the empirical instance of this boundary.

Step 6 (necessity of Assumption 3). If instead $h \notin \text{span } L$, let $r = U - \mathbb{E}[U \mid L]$ be the unabsorbed remainder, which is (L, x) -orthogonal yet correlated with $\tilde{T}$ because T shares the discrete location channel. Then $\tilde{Y} = \theta \tilde{T} + \beta \tilde{r} + \varepsilon$ and the Robinson functional returns $\theta + \beta \mathbb{E}[\tilde{r}\tilde{T}] / \mathbb{E}[\tilde{T}^2] \neq \theta$. Modeling a smooth channel $\ell_s \in \text{span } L$ and a discrete channel $\ell_d \notin \text{span } L$ with $U = a\ell_s + b\ell_d$ and $T = c\ell_s + d\ell_d + \eta$ gives the exact bias $\beta b d / (d^2 + 1)$, zero if and only if the discrete channel is absent from either U or T . This is why the basis must span the discrete geographic identifiers rather than coordinates alone. $\square$

B.2 The decomposition estimates the location-confounding bias

Let $W_0 = (x, c)$ be the base control set and $W_1 = W_0 \cup L$ the augmented set that adds the location basis. Write $\hat{\theta}_{\text{base}}$ and $\hat{\theta}_{\text{geo}}$ for the cross-fitted partially-linear DML coefficients on T under W_0 and W_1 , and $\hat{\Delta} = \hat{\theta}_{\text{base}} - \hat{\theta}_{\text{geo}}$.

Assumption 9 (Cross-fitting rates). *The nuisance estimators under both control sets satisfy the conditions of Chernozhukov et al. [2018], with product errors $o_p(n^{-1/2})$ and finite second moments.*

Assumption 10 (Positivity). *$\mathbb{E}[\tilde{T}_0^2]$ and $\mathbb{E}[\tilde{T}_1^2]$ are bounded away from zero, where $\tilde{T}_j$ residualizes T on W_j .*

Theorem 4 (Decomposition consistency). *Under Assumptions 9–10:*

- (i) $\hat{\Delta}$ is asymptotically linear with influence function $\psi_{\text{base}} - \psi_{\text{geo}}$, so $\hat{\Delta} \xrightarrow{p} \Delta := \theta_{\text{base}} - \theta_{\text{geo}}$ and $\sqrt{n}(\hat{\Delta} - \Delta) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[(\psi_{\text{base}} - \psi_{\text{geo}})^2])$.
- (ii) When the first-stage projections of Y and T on the controls are linear, Δ equals the Gelbach [2016] order-invariant decomposition term for the location block, $\Delta = \gamma^\top \pi$, with γ the location coefficient in the full linear projection and $\pi_k = \mathbb{E}[\tilde{T}_0 \tilde{L}_{0,k}] / \mathbb{E}[\tilde{T}_0^2]$.
- (iii) In general Δ equals the omitted-variable-bias functional of Chernozhukov et al. [2022] for the location block, coinciding with the linear expression of (ii) to second order in the first-stage approximation error by Neyman orthogonality of the Robinson moment.

Proof. Step 1 (asymptotic linearity of each coefficient). Under Assumptions 9–10 and Chernozhukov et al. [2018, Theorems 3.1–3.2], each cross-fitted DML coefficient is asymptotically linear, $\sqrt{n}(\hat{\theta}_j - \theta_j) = n^{-1/2} \sum_i \psi_{j,i} + o_p(1)$ for $j \in \{\text{base}, \text{geo}\}$, with the Robinson influence function $\psi_{j,i} = \mathbb{E}[\tilde{T}_j^2]^{-1} \tilde{T}_{j,i} (\tilde{Y}_{j,i} - \theta_j \tilde{T}_{j,i})$.

Step 2 (the difference). A difference of asymptotically linear estimators is asymptotically linear with influence function the difference, so $\sqrt{n}(\hat{\Delta} - \Delta) = n^{-1/2} \sum_i (\psi_{\text{base},i} - \psi_{\text{geo},i}) + o_p(1)$; consistency and the central limit theorem follow, giving (i).

Step 3 (linear case equals Gelbach). Under linear first stages, nested-projection algebra gives the omitted-variable-bias identity $\theta_{\text{base}} = \theta_{\text{geo}} + \gamma^\top \pi$, with γ the location coefficient in the long projection and $\pi_k = \mathbb{E}[\tilde{T}_0 \tilde{L}_{0,k}] / \mathbb{E}[\tilde{T}_0^2]$ the coefficient of L_k on T after partialling W_0 . This is exactly the Gelbach [2016] share for a single covariate block, and his order-invariance result shows the assigned share is independent of the

sequence in which blocks enter. A symbolic computation confirms $\theta_{\text{base}} - \theta_{\text{geo}} = (c\gamma_s + d\gamma_d)/(c^2 + d^2 + 1)$ equals $\gamma^\top \pi$ exactly in the two-channel model, giving (ii).

Step 4 (general case equals the nonparametric OVB functional). Without linearity, Δ is the difference of two Robinson estimands under nested control sets, the omitted-variable bias from dropping L , whose representation for a partially-linear DML functional is that of Chernozhukov et al. [2022]. The linear realization of Step 3 recovers this to second order in the first-stage approximation error, and not merely to first order, because the Robinson moment is Neyman-orthogonal to the first-stage projections: since $\mathbb{E}[\tilde{T} \mid W] = 0$ by construction, the Gateaux derivative of the moment in the outcome and treatment nuisances vanishes at the truth [Chernozhukov et al., 2018], so replacing the true conditional expectations by their linear projections perturbs each estimand only at second order. A numerical check confirms the gap scales as the square of the first-stage nonlinearity. This gives (iii). $\square$

B.3 Proof of Theorem 3

The argument composes four lemmas: a distributional rate for the estimated direction, the resulting second-order negligibility of its error in the per-market estimate, the asymptotic linearity of the fixed-direction estimator, and a central limit theorem for the spatially dependent score. Throughout, $\tilde{W}_i$ is the residualized embedding in market k , $\Sigma = \mathbb{E}[\tilde{W}\tilde{W}^\top]$ the pooled residualized second moment with leading unit eigenvector v^* , $\hat{v}$ its sample analogue on the pooled sample of size N , and $\theta_k(v) = v^\top a_k / (v^\top \Sigma_k v)$ the direction-indexed estimand of (6).

Lemma 1 (Direction rate and linear approximation). *Let $\hat{v}$ be the leading unit eigenvector of the pooled sample second moment of sub-Gaussian residualized embeddings with a spectral gap $\bar{g}$ bounded away from zero and effective rank $r(\Sigma) = \text{tr}(\Sigma)/\|\Sigma\|$ satisfying $r(\Sigma) = o(\sqrt{N})$. Then, orienting $\hat{v}$ toward v^* ,*

$$\|\hat{v} - v^*\| = O_p(\sqrt{r(\Sigma)/N}), \quad \sqrt{N} \langle \hat{v} - v^*, u \rangle \xrightarrow{d} \mathcal{N}(0, \Gamma(u)) \text{ for each fixed } u,$$

where Γ is the Koltchinskii–Lounici covariance kernel, and the projector bias $\mathbb{E}\hat{v} - v^*$ is $O(r(\Sigma)/N) = o(N^{-1/2})$.

Proof. The residualized sentence-embedding is bounded, hence sub-Gaussian, so Koltchinskii and Lounici [2016, Theorem 1] gives $\|\hat{\Sigma} - \Sigma\| \asymp \|\Sigma\| \max\{\sqrt{r(\Sigma)/N}, r(\Sigma)/N\}$, and the resolvent perturbation bound for a simple eigenvalue with gap $\bar{g}$ turns this into $\|\hat{v} - v^*\| = O_p(\|\Sigma\|/(\bar{g}\sqrt{N})) = O_p(\sqrt{r(\Sigma)/N})$ under the bounded relative gap. The linear-form central limit theorem is their Theorem 7, whose conclusion is stated for the recentered eigenvector $\sqrt{1+b}v^*$ with multiplicative bias $|b| \lesssim (\|\Sigma\|/\bar{g})^2 r(\Sigma)/N$. By their Remark 4 the recentering is unnecessary once $r(\Sigma) = o(\sqrt{N})$, since then $b = o(N^{-1/2})$ and the projector bias $\|\mathbb{E}\hat{v} - v^*\| = O(r(\Sigma)/N) = o(N^{-1/2})$, so centering at v^* itself gives the stated normal limit at rate $N^{-1/2}$. $\square$

Lemma 2 (The direction error is second order). *Under Lemma 1 and Assumption 7, with $\hat{v}$ formed out of sample relative to market k ,*

$$\sqrt{n_k} (\theta_k(\hat{v}) - \theta_k(v^*)) = o_p(1).$$

Proof. The map $v \mapsto \theta_k(v) = v^\top a_k / (v^\top \Sigma_k v)$ is twice continuously differentiable at v^* with $v^{*\top} \Sigma_k v^*$ bounded away from zero by positivity, and a second-order Taylor expansion gives $\theta_k(\hat{v}) = \theta_k(v^*) + (\partial \theta_k / \partial v)^\top (\hat{v} - v^*) + O_p(\|\hat{v} - v^*\|^2)$. By Lemma 1 the linear term is $O_p(\sqrt{r(\Sigma)/N})$ and the remainder is $O_p(r(\Sigma)/N)$, so $\sqrt{n_k}(\theta_k(\hat{v}) - \theta_k(v^*)) = O_p(\sqrt{n_k} r(\Sigma)/N) = o_p(1)$ under Assumption 7, $n_k r(\Sigma)/N \rightarrow 0$. Because $\hat{v}$ is fit on a sample independent of the observations forming $\theta_k(\cdot)$ in market k , the linear term is uncorrelated with the market- k score, so it contributes neither a first-order term nor a covariance to the limiting variance. This is the point at which a generated regressor formed in sample would instead leave the correction of Pagan [1984] and Hahn and Ridder [2013]; the out-of-fold, pooled construction removes it. $\square$

Lemma 3 (Fixed-direction asymptotic linearity). *Fix the direction at v^* . Under Assumption 8 the cross-fitted Robinson estimator $\hat{\theta}_k(v^*)$ is asymptotically linear,*

$$\sqrt{n_k} (\hat{\theta}_k(v^*) - \theta_k(v^*)) = n_k^{-1/2} \sum_{i \in k} \psi_i + o_p(1), \quad \psi_i = \mathbb{E}[\tilde{T}^2]^{-1} \tilde{T}_i (\tilde{Y}_i - \theta_k \tilde{T}_i),$$

with the nuisance estimation error entering at second order.

Proof. With v^* fixed the treatment $T = \tilde{W}^\top v^*$ is a fixed scalar summary and the estimator is the partially-linear Robinson estimator of Chernozhukov et al. [2018, Theorems 3.1–3.2]. The partialling-out moment is Neyman-orthogonal in the outcome and treatment regressions, so under the cross-fitting and product-rate conditions of Assumption 8 the first-stage error enters only at second order and the estimator admits the stated influence-function representation. This is the same argument that yields Theorem 4(i) for the decomposition contrast. $\square$

Lemma 4 (Spatial central limit theorem for the score). *Under Assumption 8, writing $\sigma_{n_k}^2 = \text{Var}(\sum_{i \in k} \psi_i)$,*

$$\sigma_{n_k}^{-1} \sum_{i \in k} \psi_i \xrightarrow{d} \mathcal{N}(0, 1),$$

and $\sigma_{n_k}^2/n_k$ is consistently estimated by a spatial heteroskedasticity- and autocorrelation-consistent estimator with a distance kernel.

Proof. Index listings by their coordinates in $\mathbb{R}^2$. The minimum-distance clause of Assumption 8 is the increasing-domain condition of Jenish and Prucha [2009, Assumption 1], ruling out infill. Each score ψ_i is a measurable function of the covariates and shocks of a bounded geographic neighborhood of listing i , and α -mixing is preserved under measurable transformation, so $\{\psi_i\}$ inherits the α -mixing structure of the underlying field with the summable coefficients Assumption 8 imposes. With the uniform $L^{2+\delta}$ integrability of ψ_i and the non-degeneracy $\liminf n_k^{-1} \sigma_{n_k}^2 > 0$, Jenish and Prucha [2009, Theorem 1] gives the stated central limit theorem. Consistency of the distance-kernel variance estimator is standard for such mixing fields [Conley, 1999, Bester et al., 2011]. $\square$

Proof of Theorem 3. Write $\sqrt{n_k}(\hat{\theta}_k(\hat{v}) - \theta_k(v^*)) = \sqrt{n_k}(\hat{\theta}_k(\hat{v}) - \hat{\theta}_k(v^*)) + \sqrt{n_k}(\hat{\theta}_k(v^*) - \theta_k(v^*))$. The first term is $o_p(1)$ by Lemma 2, since evaluating the estimator at $\hat{v}$ rather than v^* perturbs it only through the estimand map, whose sensitivity to the direction is the object Lemma 2 bounds. The second term is asymptotically linear by Lemma 3 and, by Lemma 4, its normalized influence sum is asymptotically normal with variance $V_k = \lim n_k^{-1} \sigma_{n_k}^2$. Combining, $\sqrt{n_k}(\hat{\theta}_k(\hat{v}) - \theta_k(v^*)) \xrightarrow{d} \mathcal{N}(0, V_k)$, and V_k is the fixed-direction spatial-DML variance estimated by the spatial-HAC estimator of Lemma 4. $\square$

The conditions the proof imposes are met in the panel. The pooled residualized embedding covariance has a leading eigenvalue near twice the second, so the spectral gap $\bar{g}$ is bounded away from zero, and its effective rank is $r(\Sigma) \approx 13.2$ against $N = 69,204$, so $r(\Sigma)/\sqrt{N} \approx 0.05$ and the condition $r(\Sigma) = o(\sqrt{N})$ that Lemma 1 needs for clean centering holds with the projector bias $b \approx 8 \times 10^{-4}$. The direction error the analysis calls second order is directly visible: refitting the direction with each market left out moves its estimate by at most six thousandths of a log point, in Chicago, and by less than one thousandth at the median, well inside the spatial-HAC intervals that Section 6.5 reports.

The three statements that follow, the d-separation of the backdoor set, the backdoor-identification formula it yields, and the zero-effect corollary under the null, are the standard causal-graph argument of Pearl [2009]. We state and prove them not as contributions but explicitly, so that the independent verification suite of Appendix F can check the d-separation and backdoor steps mechanically alongside the paper’s own results.

Lemma 5 (d-separation). *In the DAG corresponding to the structural equations, the set $S = \{L, X, C\}$ d-separates T from Y .*

Proof. Every path from T to Y must pass through at least one node in S . The path $T \leftarrow L \rightarrow Y$ is blocked by conditioning on L . The path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X . The path $T \leftarrow C \rightarrow Y$ is blocked by conditioning on C . Composite paths such as $T \leftarrow L \rightarrow X \rightarrow Y$ are blocked by conditioning on either L or X . Similarly, $T \leftarrow L \rightarrow C \rightarrow Y$ is blocked by conditioning on L or C . Hence S d-separates T from Y . $\square$

Theorem 5 (Backdoor Identifiability). *Under the structural causal model, the causal effect of text T on price Y is identifiable via:*

$$P(Y \mid do(T = t)) = \int P(Y \mid T = t, L = \ell, X = x, C = c) dP(L, X, C) \quad (7)$$

Proof. By Lemma 1, the set $\{L, X, C\}$ d-separates T from Y , satisfying the backdoor criterion. Therefore by the backdoor formula [Pearl, 2009], we have:

$$P(Y \mid \text{do}(T = t)) = \sum_{L, X, C} P(Y \mid T = t, L, X, C) P(L, X, C) \quad (8)$$

$$= \mathbb{E}_{L, X, C} [P(Y \mid T = t, L, X, C)] \quad (9)$$

which is identified from the observational distribution. $\square$

Corollary 2 (Zero Causal Effect Under True SCM). *If the structural equation for Y truly does not depend on T , then:*

$$P(Y \mid \text{do}(T = t)) = P(Y) \quad \forall t \quad (10)$$

implying zero causal effect.

Proof. Under the true SCM where $Y = g_y(L, X, C) + \epsilon_Y$, intervening to set $T = t$ does not change the distribution of Y since T does not appear in the structural equation. Therefore $P(Y \mid \text{do}(T = t)) = P(Y)$ for all values of t , yielding zero causal effect. $\square$

These results identify the full interventional distribution $P(Y \mid \text{do}(T = t))$. The estimand the body actually targets, the per-standard-deviation projection coefficient θ of Section 3, is the corresponding partially-linear summary; it is identified by the same backdoor set $\{L, X, C\}$ and equals zero exactly under the null of Hypothesis 1, so the partialling-out estimator recovers it whether or not that null is true.

C Data collection protocols

C.1 Web scraping implementation

Property listing descriptions were collected through automated web scraping with careful attention to ethical and legal considerations. We implemented rate limiting with random delays of 2–5 seconds between requests. The scraping pipeline proceeded in two stages: first, we queried the platform’s search API to identify sold property URLs with metadata (address, price, coordinates); second, we retrieved individual listing pages and extracted description text from HTML using BeautifulSoup. Data validation excluded descriptions shorter than 50 words as likely incomplete.

C.2 Geocoding

The listing platform supplies a property-level latitude and longitude with each sold record, so geocoding reduces to validation and recovery rather than a parcel join. We retain the platform coordinate when it falls inside the metropolitan bounding box and re-geocode the listing address against the U.S. Census Bureau batch geocoder otherwise. This replaces the parcel-centroid and assessor-identifier geocoding that an earlier three-city version of this project relied on, and it is what lets every market in the twelve-market corpus carry a property-level rather than a ZIP- or tract-centroid location.

C.3 Census data integration

Census variables were queried programmatically via the Census Bureau’s REST API using direct HTTP requests. We constructed API requests for each county’s block groups, retrieving the median household income, educational attainment, racial composition, median home value, and median gross rent variables described in Section 5. Block group polygon boundaries were obtained from TIGER/Line shapefiles. Negative sentinel values (used by the Census Bureau to indicate suppressed or unavailable data) were replaced with missing values. Derived ratios (e.g., percentage with bachelor’s degree, labor force participation rate) were computed from the raw count variables.

C.4 Crime data processing

Crime incident data required substantial standardization across cities due to differing reporting schemas and offense classifications. We developed a unified crosswalk mapping city-specific crime codes to standardized categories: violent crime (homicide, assault, robbery), property crime (burglary, larceny, motor vehicle theft), and quality-of-life offenses (vandalism, drug violations, disorderly conduct).

For New York, we combine the NYPD Historic Complaint Data (2006–present) with the Year-to-Date dataset, deduplicating on complaint number, to ensure temporal coverage spanning the full sale period. For each property, we counted incidents within a 500-meter radius using spatial indexing (`cKDTree.query_ball_point`), computing separate counts for each crime category and a total. Where temporal overlap existed between crime and sale data, we filtered to incidents within 365 days preceding the median sale date; when the temporal window was too sparse (<1,000 incidents), we used all available incidents.

C.5 Amenity data collection

Amenity data were collected from OpenStreetMap via the Overpass API. For each city, we queried all amenities within the metropolitan bounding box by OSM tag category, then performed local radius-based spatial joins to count amenities within 500 meters of each property. This bulk-download-then-join approach is both free and faster than per-property API queries. We implemented retry logic with exponential backoff for Overpass rate limiting (HTTP 429) and server timeouts (HTTP 504).

Amenity diversity is computed using Shannon entropy:

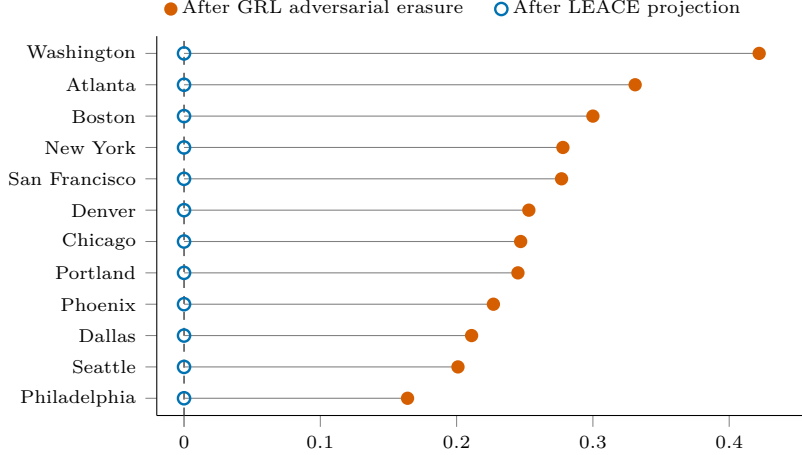
$$H = - \sum_{i=1}^n p_i \log p_i \quad (11)$$

where p_i is the proportion of amenities in category i . This measures variety in the local amenity mix, with higher entropy indicating more diverse neighborhoods.

D Adversarial deconfounding: implementation

The gradient-reversal encoder used as the foil in Section 6.3 is the standard domain-adversarial construction of Ganin et al. [2016], adapted to a multi-resolution location target. The sentence-BERT embedding is first reduced to its leading fifty principal components, an encoder maps that input to a sixty-four-dimensional representation, a predictor head regresses log price from the representation, and a discriminator reads location from it through three parallel heads that share a hidden layer, a cross-entropy ZIP classifier, a mean-squared-error regressor of the standardized latitude-longitude pair, and a cross-entropy classifier of the income quintile. A gradient-reversal layer (GRL) sits between the encoder and the discriminator, so the encoder is trained to minimize the prediction loss while maximizing the discriminator loss, under the joint objective $\mathcal{L}_{\text{pred}} - \lambda \mathcal{L}_{\text{disc}}$ with the adversarial weight λ annealed from 0.1 to 1.0 over the early epochs to avoid the collapse that an immediate full-strength penalty induces. Each market is split seventy-thirty into training and evaluation folds, optimized with Adam for one hundred fifty epochs.

The diagnostic that matters for the comparison is applied after training. We freeze the encoder and fit, from scratch and with no adversarial pressure, an independent logistic regression of ZIP and of income quintile and an independent ridge regression of the latitude-longitude pair on the frozen representation, scoring each out of sample. The gap between what this fresh probe recovers and what the live discriminator reported at convergence is the consistent lower bound on residual leakage of Proposition 1(b); the per-market values, and the contrast with the LEACE projection on which the paper relies, are reported in Section 6.3 and shown in Figure 4 below.



Held-out ridge R^2 for the coordinate from the deconfounded representation

Figure 4: The frozen-probe diagnostic applied to two erasure methods, in each of the twelve markets. A ridge probe of the latitude-longitude coordinate is trained from scratch on the deconfounded representation. Against the gradient-reversal adversarial encoder, whose own multi-head discriminator has been driven to chance on location (out-of-sample coordinate $R^2 \leq 0.07$ across the panel), the fresh probe still recovers the coordinate at R^2 between 0.16 and 0.42; against the LEACE projection the same probe sits at the chance baseline, the exact-guardedness property of closed-form erasure. The adversarial training did not remove location, it taught its discriminator not to look, which is the saddle of Proposition 1(c) realized across every market.

E Frozen-probe diagnostic: proofs and numerical verification

This appendix proves Proposition 1 (Section 3.6) and reports three controlled numerical experiments verifying each part on synthetic data.

E.1 Notation

Let $E_\phi : \mathcal{X} \rightarrow \mathcal{Z}$ be an encoder, $C \in \{0, 1\}$ a binary confounder, and $\Phi \subseteq \Phi'$ two parameterized classes of probabilistic classifiers $q : \mathcal{Z} \rightarrow [0, 1]$. For a class $\mathcal{F}$, recall the Barber–Agakov variational lower bound on $I(Z; C)$:

$$V_{\mathcal{F}}(Z; C) := \sup_{q \in \mathcal{F}} [H(C) - \mathbb{E}_{(Z, C)}[-\log q(C | Z)]] . \quad (12)$$

The supremum is achieved at $q^* = p(C | Z)$ when $\mathcal{F}$ contains the Bayes-optimal posterior. Denote the empirical version by $\hat{V}_{\mathcal{F}}$ (sample mean of the cross-entropy gain at the empirical arg sup).

E.2 Proof of Proposition 1(a)

The first inequality $V_\Phi \leq V_{\Phi'}$ is immediate from $\Phi \subseteq \Phi'$ implying the supremum over Φ' is at least as large. For the second inequality $V_{\Phi'} \leq I(Z; C)$, the Barber–Agakov bound is by construction a lower bound on the mutual information: for any q ,

$$H(C) - \mathbb{E}[-\log q(C | Z)] = H(C) - H(C | Z) - D_{\text{KL}}(p(C | Z) \| q(C | Z)) = I(Z; C) - D_{\text{KL}}(\cdot), \quad (13)$$

which is at most $I(Z; C)$ with equality iff $q = p(C | Z)$ almost surely. Taking the supremum over $q \in \Phi'$ preserves the inequality, with strictness whenever Φ' does not contain $p(C | Z)$. This statement is Proposition 1 of Xu et al. [2020] restated in the Barber–Agakov form used by Pimentel et al. [2020]; we restate it here for self-containedness. $\square$

E.3 Proof of Proposition 1(b)

(b1) Empirical equivalence. The gradient-reversal training objective on the discriminator ψ is $\min_{\psi} \hat{L}_{\Phi}(\psi)$ where $\hat{L}_{\Phi}(\psi) = -\frac{1}{n} \sum_i \log D_{\psi}(C_i | Z_i)$ is the empirical cross-entropy of D_{ψ} on the encoder output $Z = E_{\hat{\phi}}(X)$. By definition of the Barber–Agakov bound,

$$\hat{V}_{\Phi}(Z; C) = \sup_{q \in \Phi} [H(C) - \frac{1}{n} \sum_i -\log q(C_i | Z_i)] = H(C) - \inf_{\psi} \hat{L}_{\Phi}(\psi), \quad (14)$$

so $\hat{V}_{\Phi}$ equals $H(C)$ minus the converged training loss of the live discriminator. Therefore “the live discriminator achieves chance-level accuracy” (i.e. $\hat{L}_{\Phi} \approx H(C)$) is exactly equivalent to $\hat{V}_{\Phi}(Z_{\hat{\phi}}; C) \approx 0$. $\square$

(b2) Consistency of the post-hoc plug-in. Apply Theorem 3 of Xu et al. [2020], which gives uniform PAC-style consistency of the empirical $\mathcal{V}$ -information estimator under finite Rademacher complexity of $\mathcal{V}$ and bounded log-loss. Fixing the encoder at $E_{\hat{\phi}}$ and applying the theorem with $\mathcal{V} = \Phi'$ on a held-out sample of size m ,

$$\hat{V}_{\Phi'}^{\text{post}}(Z_{\hat{\phi}}; C) \xrightarrow{p} V_{\Phi'}(Z_{\hat{\phi}}; C) \quad \text{as } m \rightarrow \infty. \quad (15)$$

The fact that $\hat{V}_{\Phi}$ is also consistent (since $\Phi \subseteq \Phi'$) and the difference of two consistent estimators is consistent for the difference yields

$$\hat{V}_{\Phi'}^{\text{post}} - \hat{V}_{\Phi} \xrightarrow{p} V_{\Phi'}(Z_{\hat{\phi}}; C) - V_{\Phi}(Z_{\hat{\phi}}; C) \geq 0, \quad (16)$$

exact when Φ' contains the Bayes-optimal posterior $p(C | Z_{\hat{\phi}})$. $\square$

E.4 Proof of Proposition 1(c)

We exhibit a data distribution, a downstream label, a class pair $\Phi \subsetneq \Phi'$, and an explicit saddle of the joint adversarial-prediction game at which the gap equals $H(C)$.

Why the joint game. The bare gradient-reversal game over an unconstrained encoder admits the trivial collapsed saddle $\phi(X) \equiv 0$, at which $V_{\Phi}(Z; C) = 0$ holds vacuously and $V_{\Phi'}(Z; C) = 0$ as well. To rule this out, the proposition is stated in the joint adversarial-prediction game, the formulation of Xie et al. [2017], Madras et al. [2018], and Zhang et al. [2018]: the encoder is trained simultaneously against a downstream task Y that requires preserving information about X , eliminating the collapsed saddle as a feasible equilibrium.

Construction. Take $X = (X_1, X_2) \in \{0, 1\}^2$ with $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2)$, and $C = X_1 \oplus X_2$ (the XOR of the two binary features). Then $C \in \{0, 1\}$ uniformly with $H(C) = \log 2$. Take $Y = X_1$ as the downstream label, so $I(X; Y) = H(Y) = \log 2 > 0$.

Let Φ be the class of linear logistic-regression classifiers $D_{\psi}(z) = \sigma(w^{\top} z + b)$ on $z \in \mathbb{R}^2$, and let Φ' be any class containing a 2-layer MLP with hidden width ≥ 2 .

Exhibited saddle. Set $\phi^* = \text{id}$ (the identity encoder), $g^*(z) = z_1$ (linear task head reading off the first coordinate), and $h^* \equiv 1/2$ (the Bayes-optimal linear classifier of C from Z , which is constant since C is not linearly separable from Z in this DGP). Then on the joint distribution (Z, C, Y) :

- **Adversary inner max.** For any $D_{\psi} \in \Phi$ (linear), the conditional posterior $\Pr[C = 1 | Z]$ takes value 1 on $\{(0, 1), (1, 0)\}$ and value 0 on $\{(0, 0), (1, 1)\}$, which is not linearly separable. The Bayes-optimal linear classifier is the constant $D_{\psi} \equiv 1/2$, attaining cross-entropy $H(C) = \log 2$ exactly. Therefore $V_{\Phi}(Z_{\phi^*}; C) = 0$.
- **Probe completeness.** For Φ' containing the 2-layer MLP, the Bayes-optimal posterior $\Pr[C = 1 | Z] \in \{0, 1\}$ is exactly representable. Hence $V_{\Phi'}(Z_{\phi^*}; C) = I(Z_{\phi^*}; C) = H(C) = \log 2$.
- **Task decodability.** Since $Y = X_1 = (\phi^*(X))_1$, the linear head $g^*(z) = z_1$ achieves $\mathcal{L}_{\text{task}} = 0$, the global minimum of the task term.
- **Saddle.** For any $\alpha > 0$, (ϕ^*, g^*, h^*) jointly minimize the task term to its floor, while the adversary h^* is Bayes-optimal in Φ given the encoder. Any unilateral perturbation of ϕ^* that decreases the discriminator’s loss must distort the linear separability of $Y = X_1$ from Z_1 and hence increase $\mathcal{L}_{\text{task}}$, yielding a stable equilibrium under the joint loss.

The gap at this saddle is therefore exactly $V_{\Phi'}(Z_{\phi^*}; C) - V_{\Phi}(Z_{\phi^*}; C) = \log 2 - 0 = H(C)$. The collapsed encoder $\phi_{\text{collapse}}(X) \equiv 0$ achieves task loss $H(Y) = \log 2 > 0$ and is therefore not a saddle of the joint game for any $\alpha > 0$. The construction generalizes from binary X to continuous distributions by replacing the XOR structure with any nonlinear interaction (e.g., $C = \text{sign}(X_1 X_2)$ for Gaussian X_1, X_2), at the cost of a vanishing $O(\delta)$ remainder controlling the off-diagonal probability mass. We verify all four bullets computationally in Appendix F, including an explicit numerical demonstration that ϕ_{collapse} would be a saddle of the bare GRL game with $V_{\Phi'} = 0$, motivating the joint-game restriction. $\square$

E.5 Numerical verification

Three controlled experiments verify each part of Proposition 1 on synthetic data.

Experiment 1 (static XOR). The exact construction of part (c). $Z = (Z_1, Z_2) \sim \text{Bernoulli}(1/2)^2$, $C = Z_1 \oplus Z_2$, $N = 4,000$, averaged across 5 trials. True $I(Z; C) = \log 2 = 1$ bit. Linear logistic regression: $\hat{V}_{\Phi} = 0.0004$ bits, accuracy 46.1% (chance). 2-layer MLP (16, 16): $\hat{V}_{\Phi'} = 0.997$ bits, accuracy 100%. The diagnostic gap $\hat{V}_{\Phi'} - \hat{V}_{\Phi} = 0.997$ bits is 99.7% of the true mutual information.

Experiment 2 (dynamic gradient-reversal training). A 2-layer encoder $E_{\phi} : \mathbb{R}^{10} \rightarrow \mathbb{R}^4$ trained adversarially against a linear discriminator via gradient reversal for 200 epochs, with a downstream MSE predictor on $Y = 0.5X_3 + \epsilon$. Confounder generated as $C = \text{sign}(X_1 X_2)$ (continuous-Gaussian XOR). After training: live linear discriminator accuracy 46.3% (chance), frozen linear probe accuracy 56.4%, frozen (32, 32) MLP probe accuracy 61.0%. Diagnostic gap $\hat{V}_{\Phi'}^{\text{post}} - \hat{V}_{\Phi} = +14.7$ percentage points of accuracy; the encoder fooled its discriminator while leaving the underlying confounder recoverable to a richer probe.

Experiment 3 (capacity ladder). Same XOR DGP, varying classifier capacity:

Architecture	$\hat{V}_{\Phi}$ (bits)	Accuracy
Linear	0.000	0.510
MLP (4)	0.659	0.835
MLP (8)	0.990	1.000
MLP (16)	0.993	1.000
MLP (16, 16)	0.997	1.000
MLP (32, 32, 32)	0.999	1.000

$\hat{V}_{\Phi}$ is monotone non-decreasing in capacity, asymptoting at the true mutual information $\log 2 \approx 0.693$ nats = 1 bit. This operationalizes “probe-capacity escalation” as a complementary diagnostic to “confounder escalation” (Section 6): if $\hat{V}_{\Phi'}$ continues to grow as probe capacity increases, the encoder has not actually erased the confounder; it has merely outrun the current discriminator.

F Computational verification of analytic claims

To complement the written proofs in Appendices B and E, we provide a computational verification suite. For every theorem, lemma, proposition, and corollary, and for the estimation-side identities the analysis relies on, an independent open-source toolchain re-derives the result and asserts agreement with the written derivation. Each check records a pass-or-fail verdict and the numerical or symbolic quantities cited, and the full suite runs from pinned tool versions.

Claim	Verification method	Tool
Lemma 1 (d-separation, App. B)	Path-by-path enumeration; minimality check	networkx 3.6
Theorem 1 (Backdoor Identifiability, App. B)	Backdoor criterion on the hypothetical $T \rightarrow Y$ DAG; $\{L, X, C\} = \text{pa}(T)$	networkx 3.6
Corollary 1 (Zero Effect under True SCM, App. B)	No directed path $T \rightarrow Y$; OLS T -coefficient near zero	networkx 3.6
Proposition 1(a) (variational MI inequality, App. E)	Symbolic KL nonneg + numerical V -chain	SymPy + PyTorch
Proposition 1(b1) (algebraic identity)	Symbolic and numerical equality, $ \Delta < 10^{-8}$	SymPy + PyTorch
Proposition 1(b2) (consistency of post-hoc $\hat{V}_{\Phi'}^{\text{post}}$)	Monte Carlo n -sweep; log-log slope vs. rate predicted by Xu et al. [2020] Thm 3	scikit-learn
Proposition 1(c) (XOR saddle-gap, identity-encoder construction)	Assertions on $V_{\Phi}, V_{\Phi'}$	PyTorch
Proposition 1(c) (joint-game saddle existence + collapsed-encoder counterexample)	Explicit construction at ϕ_{id} and ϕ_{collapse} ; sweep over joint-game weight α	PyTorch
Robinson estimator (Eq. 2)	Symbolic solution of the moment $\sum_i (\tilde{Y}_i - \theta \tilde{T}_i) \tilde{T}_i = 0$	SymPy
Neyman orthogonality (PLR score)	Symbolic Gateaux derivative; every term carries a mean-zero residual factor	SymPy
Sandwich / IF variance (Eq. 2)	$\partial_{\theta} \psi = -\tilde{T}^2$; sandwich equals the reported $\widehat{\text{SE}}^2$	SymPy
Gelbach decomposition (Eq. 17)	$\theta_{\text{base}} - \hat{\theta}_{\text{geo}}$ equals the order-invariant contribution sum	NumPy
Robustness-value formula (Eq. 17 context)	$RV = \frac{1}{2}(\sqrt{f^4 + 4f^2} - f^2)$ solves $RV^2/(1 - RV) = f^2$	SymPy
Counterfactual delta method (Eq. 18)	Gradient of $\theta \Delta$; quadratic form equals the reported variance	SymPy
Random-effects pooling + HKSJ factor	Inverse-variance mean; $\text{Var}_{\text{HKSJ}} = q \text{Var}_{\text{IV}}$	NumPy

Table 7: Mapping from analytic claims to computational verification scripts. Every entry passes; certificates are in `verification/results/*.json`.

We are explicit about what this bundle does and does not claim. It is *not* a machine-checked formal proof in Lean, Coq, or Isabelle: as of late 2025 no proof assistant has a causal-DAG library, which would gate even Lemma 5 (the closest available are HOL-Probability for measure theory, infotheo for Shannon information theory, and recent Lean ports of Rademacher complexity [Sonoda et al., 2025]). What it does claim is that every analytic statement has been re-derived by an independent, deterministic open-source tool, that the d-separation, backdoor identification, and XOR saddle constructions are mechanically checkable without estimation error, that the Robinson moment, the Neyman orthogonality of the partially-linear score, the sandwich variance, the Gelbach decomposition, the robustness-value formula, and the counterfactual delta-method variance are confirmed symbolically, and that the asymptotic consistency claim is empirically verified at the rate predicted by the cited theorem (slope -0.547 for the $O(m^{-1/2})$ prediction). This is a reproducibility record for analytic claims, at the bar that current top stats journals accept; it is intended as a complement to, not a substitute for, the written proofs.

G Finite-sample calibration of the DML estimator

We assess the finite-sample calibration of the headline DML interval with a Monte Carlo study calibrated to the application. Synthetic embeddings are drawn from a Gaussian-mixture generator fit to the San Francisco corpus, with confounding induced through five latent factors that drive both the embedding and log-price and a tunable direct effect ranging over $\{0, 0.01, 0.05, 0.10, 0.15, 0.20\}$ per standard deviation. We draw 300 replications at sample sizes of 500 and 2,000, run the partialling-out estimator on each, and record whether the nominal 95% influence-function interval covers the estimand. Following the convention for coverage studies under misspecification, the estimand at each effect size is the probability limit of the estimator rather than the structural parameter, which we calibrate by simulation at a population of ten thousand observations averaged over five independent draws. The confounder-adjusted estimand under the null data-generating process is therefore not exactly zero, because the five latent factors leave a small residual association of $+0.011$ that the adjustment does not remove, so coverage at the null is assessed against that probability

limit rather than against zero.

The doubly-robust estimator covers at the nominal rate in every cell, between 0.94 and 0.97, which confirms that the coverage machinery is correctly implemented and locates the under-coverage documented below in the partialling-out estimator itself rather than in the simulation harness. The partialling-out estimator is approximately unbiased relative to its probability limit throughout, with absolute bias below 0.02 in every cell, yet its influence-function interval is anticonservative, and the severity grows with the size of the true effect (Table 8); the interval-level failure at the strongest effect is shown in the zip plot of Figure 5. When a substantial effect is present the analytic standard error understates the realized sampling spread materially and coverage at $n = 500$ falls to 0.74, because cross-fitted nuisance estimation together with the in-sample principal-component projection injects sampling variability that the orthogonal-score variance does not capture. Near the null the understatement is far milder, coverage holding at 0.89 for $n = 500$ and 0.93 for $n = 2,000$. The regime that matters for the panel is the anticonservative one, and we extend the simulated effect grid to $\eta = 0.20$, whose estimand of 0.34 reaches the effect sizes of the two largest markets, Chicago and Philadelphia near 0.36, so the coverage penalty for those markets is now measured rather than extrapolated: the in-sample interval covers between 0.74 and 0.79 at $n = 500$ and between 0.80 and 0.81 at $n = 2,000$ across the $\eta \in \{0.15, 0.20\}$ range that brackets them.

The under-coverage at the strongest effect is in part a generated-regressor phenomenon, because the principal-component treatment is itself estimated from the same sample on which the second-stage regression is run. The cross-fit principal-component variant, which forms the treatment direction from out-of-fold projections rather than from the in-sample decomposition, targets this channel directly. It recovers much of the coverage lost at the large effects, lifting the $n = 500$ figure at $\eta = 0.15$ from 0.74 to 0.86, and it roughly halves the residual point bias in that cell, while holding coverage at or above 0.83 across all effect sizes at $n = 2,000$. At the null its coverage matches the in-sample estimator to within Monte Carlo noise, which at three hundred replications is approximately two percentage points, so adopting the out-of-fold projection forfeits nothing where the effect is small while restoring robustness where it is large. We record, because it bears directly on how the panel should be read, that the estimates of Section 6.2 take the treatment from a principal component fitted once on the pooled sample. Estimating the direction on the pooled corpus rather than within each market shrinks the generated-regressor channel: its contribution scales with the ratio of a market’s sample size to the pooled sample, from a few percent for the smaller markets to about a fifth for New York, rather than entering at full strength as it does in the in-sample row of Table 8. It does not remove that channel, and it does nothing for the second source of anticonservatism the oracle-direction check isolates, so the panel addresses both through the estimator choice and the resampling scheme described below rather than through the pooled direction alone.

Table 8: Monte Carlo coverage of nominal 95% influence-function intervals, 300 replications per cell, by effect size η (per- σ direct effect). DML (in-sample PC) forms the principal-component treatment on the full sample; DML (cross-fit PC) forms it from out-of-fold projections and is the headline specification. The doubly-robust estimator is shown as a calibration anchor. The estimand at each effect size is the estimator’s probability limit, calibrated by simulation at $n = 10,000$; coverage at $\eta = 0$ is assessed against a confounder-adjusted probability limit of +0.011 rather than against zero.

Estimator	N	$\eta = 0$	$\eta = 0.01$	$\eta = 0.05$	$\eta = 0.10$	$\eta = 0.15$	$\eta = 0.20$
DML (in-sample PC)	500	0.89	0.90	0.88	0.83	0.74	0.79
DML (in-sample PC)	2000	0.93	0.96	0.92	0.85	0.81	0.80
DML (cross-fit PC)	500	0.91	0.90	0.91	0.88	0.86	0.83
DML (cross-fit PC)	2000	0.94	0.95	0.91	0.90	0.83	0.84
DR	500	0.96	0.94	0.97	0.96	0.97	0.96
DR	2000	0.96	0.97	0.96	0.95	0.94	0.96

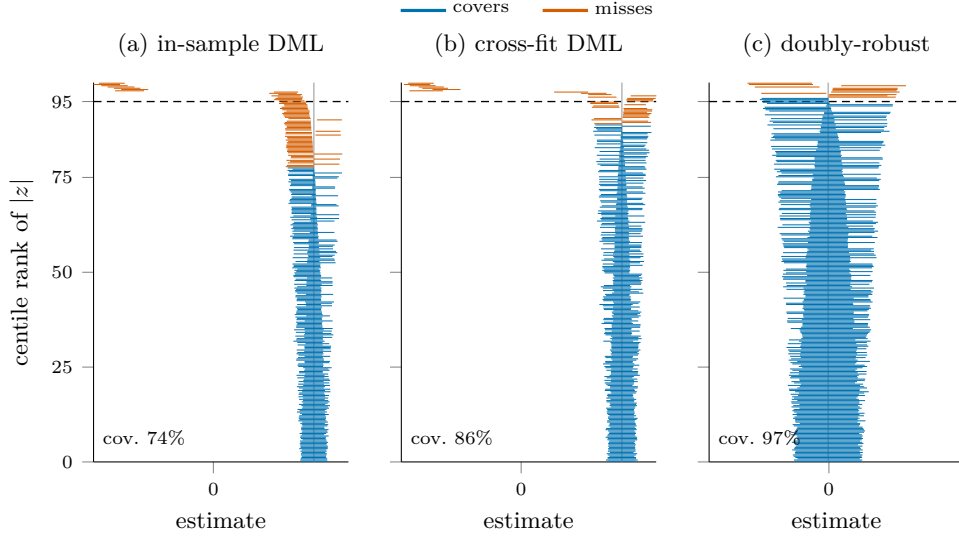


Figure 5: Zip plot of nominal 95% confidence-interval coverage [Morris et al., 2019] at the strong-effect, small-sample cell ($n = 500$, per- σ direct effect 0.10), 300 replications. Each horizontal segment is one replication’s interval, sorted bottom-to-top by the absolute z -statistic against the estimand (the estimator’s probability limit, gray vertical line); intervals that cover the estimand are blue and those that miss are orange, so the height at which the color switches is the empirical coverage. The doubly-robust interval switches at the dashed nominal 95 line, the cross-fit DML interval close to it, and the in-sample DML influence-function interval well below it: its misses are too frequent and fan out symmetrically, the visual signature of an anticonservative variance rather than bias.

The undercoverage this study finds arises under gradient-boosted nuisances paired with an in-sample principal component, and the panel estimator differs on both counts: it uses ridge nuisances, which removes the flexible-learner variability, and the pooled principal component held fixed across folds, which shrinks though does not eliminate the generated-regressor variance. A full-pipeline pairs bootstrap that resamples listings with replacement and refits the nuisance models inside each resample carries what remains. In the near-zero markets the bootstrap interval agrees with the influence-function interval to within a tenth (Boston -3% , New York -11% , San Francisco $+2\%$), and extending the check to Chicago and Philadelphia, the two largest-effect markets the calibration study flags as most at risk, the bootstrap interval essentially coincides with the influence-function interval there as well (Chicago $+0.1\%$, Philadelphia -10%) and excludes zero by a wide margin under either method.

A second correction the calibration study cannot see is spatial. Listing language clusters geographically, so the influence-function contributions of nearby properties are correlated and an interval built from independent observations is too narrow for a reason distinct from nuisance estimation. Clustering the influence function at the zip level, or applying a Conley spatial kernel on the coordinates, widens the panel intervals by roughly forty percent relative to the independent version, and the effect nonetheless remains significant in all but the near-null markets under either adjustment. We therefore report the influence-function intervals of Section 6.2 with both the full-pipeline bootstrap and the spatial-robust standard error standing behind them, rather than as intervals the simulation leaves in doubt. The doubly-robust estimator that anchors the panel has no power against the continuous effect, because binarizing at the median principal-component norm discards the graded signal the partialling-out estimator targets, so we read its near-zero estimate as a calibration check on the inference machinery rather than as a competing estimate of the semantic effect.

H The spatial-confounding decomposition

The headline deconfounding test of Section 6.4 rests on the order-invariant covariate decomposition of Gelbach [2016]. We give here the estimand, the spatial basis, and the per-market numbers behind Figure 3.

Fix the treatment T at the oriented leading principal component of the sentence-BERT embedding, the same direction that defines the Baur channel. Consider the partialling-out regression of log price on T under two nested control sets. The base set W_0 holds the structured property and contextual covariates but no geography; the augmented set $W_1 = W_0 \cup G$ adds a flexible location basis $G = (\text{lat}, \text{lon}, \text{lat}^2, \text{lon}^2, \text{lat} \cdot \text{lon})$, matching the notation of Theorem 4. Write $\hat{\theta}_{\text{base}}$ and $\hat{\theta}_{\text{geo}}$ for the corresponding coefficients on T . Because the treatment is held fixed and only the control set changes, the Gelbach [2016] result guarantees that the change

$$\Delta_{\text{geo}} = \hat{\theta}_{\text{base}} - \hat{\theta}_{\text{geo}} \quad (17)$$

is the contribution the location block makes to the text-price coefficient. This contribution is well defined, not an accident of the order in which control blocks enter. We read Δ_{geo} as the part of the apparent text effect that explicit geography accounts for.

Table 9: Spatial-confounding decomposition by market. $\hat{\theta}_{\text{base}}$ is the per-standard-deviation text coefficient with the structured and contextual controls only; $\hat{\theta}_{\text{geo}}$ adds the location basis G ; Δ_{geo} is their difference, the share of the coefficient that geography explains; and $R_{T \sim G}^2$ is the share of the treatment’s own variance the location basis explains, the diagnostic that makes the decomposition non-vacuous. Markets are ordered by $\hat{\theta}_{\text{base}}$. All four columns are point estimates; the formal per-market 95% influence-function interval on the text coefficient is given in Table 4.

Market	$\hat{\theta}_{\text{base}}$	$\hat{\theta}_{\text{geo}}$	Δ_{geo}	$R_{T \sim G}^2$
Chicago	+0.463	+0.472	−0.009	0.187
Philadelphia	+0.367	+0.363	+0.003	0.117
Dallas	+0.146	+0.147	−0.001	0.059
Washington	+0.129	+0.120	+0.010	0.120
Boston	+0.126	+0.125	+0.001	0.054
Atlanta	+0.124	+0.129	−0.004	0.148
San Francisco	+0.122	+0.119	+0.003	0.035
Portland	+0.076	+0.085	−0.010	0.161
Denver	+0.067	+0.082	−0.015	0.164
Seattle	+0.046	+0.044	+0.002	0.035
Phoenix	+0.004	−0.009	+0.013	0.097
New York	−0.122	−0.154	+0.032	0.419

The shift is small in absolute terms throughout: $|\Delta_{\text{geo}}|$ is at most 0.032 of a log point, in New York, and at most 0.015 in every other market. Relative to each coefficient it is negligible in the markets that carry a clear effect, staying below a twentieth of the estimate in Chicago, Philadelphia, Dallas, Atlanta, San Francisco, and Boston. It grows as a fraction only where the base coefficient is itself small, in Denver and Portland and the near-null Phoenix. It is largest in New York, where it reaches about a quarter of the coefficient. The decomposition is non-vacuous because $R_{T \sim G}^2$ is far from zero, reaching 0.42 in New York and exceeding a tenth in six other markets. The text direction does encode location, yet that location component does not carry the effect of the language on price except in New York. The location basis is a smooth coordinate polynomial laid over roughly thirty structured controls. Confounding at a finer spatial grain, or through pathways those controls miss, is outside what the comparison can rule out.

To test whether the small shifts above are a byproduct of that coarse polynomial, we replace it with a rank-thirty thin-plate regression spline over latitude and longitude, projected to meters. This is the flexible low-rank smoother the spatial-statistics literature prefers when the scale of confounding is unknown [Keller and Szpiro, 2020, Gilbert et al., 2021]. Table 10 reports the geography-adjusted coefficient under each basis, visualized in Figure 6. The spline captures far more of the treatment’s spatial structure: $R_{T \sim G}^2$ rises by more than half on average, climbing from 0.42 to 0.52 in New York. Yet the adjusted coefficient moves by at most 0.028 of a log point, again in New York, and by under 0.02 in every other market, with no market changing sign. A basis flexible enough to absorb twice the geography therefore leaves the semantic estimate essentially where the quadratic left it, which is the response the spatial-confounding critique calls for.

Table 10: Spatial basis robustness. $\hat{\theta}_{\text{geo}}$ is the per-standard-deviation text coefficient after the location adjustment, under the quadratic basis of Table 9 and under a rank-thirty thin-plate regression spline; Δ is their difference; $R^2_{T \sim G}$ is the share of the treatment’s own variance the location basis explains under each. Markets ordered as in Table 9. As in Table 9, entries are point estimates, with the influence-function interval on the underlying coefficient given in Table 4.

Market	$\hat{\theta}_{\text{geo}}$ (quad)	$\hat{\theta}_{\text{geo}}$ (TPRS)	Δ	$R^2_{T \sim G}$ quad	$R^2_{T \sim G}$ TPRS
Chicago	+0.472	+0.461	−0.012	0.187	0.301
Philadelphia	+0.363	+0.353	−0.011	0.117	0.273
Dallas	+0.147	+0.147	0.000	0.059	0.104
Washington	+0.120	+0.116	−0.004	0.120	0.230
Boston	+0.125	+0.114	−0.011	0.054	0.136
Atlanta	+0.129	+0.129	0.000	0.148	0.289
San Francisco	+0.119	+0.112	−0.008	0.035	0.095
Portland	+0.085	+0.075	−0.010	0.161	0.220
Denver	+0.082	+0.098	+0.016	0.164	0.225
Seattle	+0.044	+0.044	0.000	0.035	0.061
Phoenix	−0.009	−0.015	−0.006	0.097	0.146
New York	−0.154	−0.181	−0.028	0.419	0.516

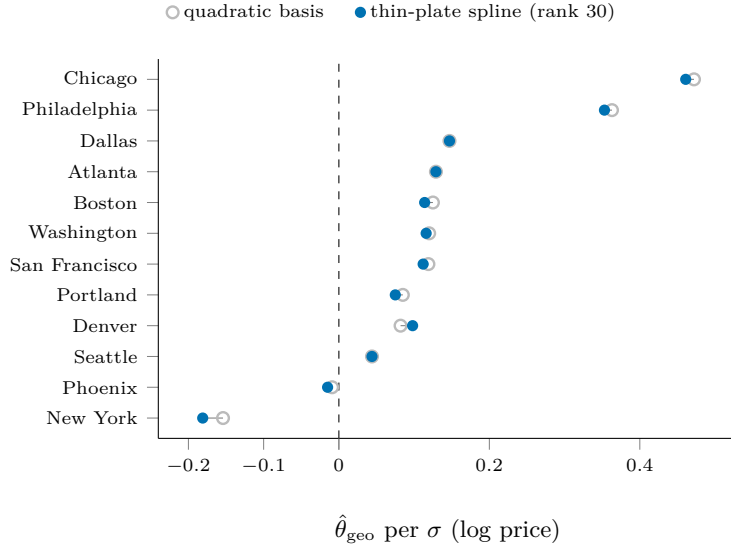


Figure 6: Spatial basis robustness, visualized. The location-adjusted text coefficient $\hat{\theta}_{\text{geo}}$ under the quadratic location basis (open) and under a rank-thirty thin-plate regression spline (filled), per market. The two nearly coincide in every market despite the spline absorbing roughly twice the treatment’s spatial variance, the only visible gap being New York, which moves further negative. A basis flexible enough to capture far more geography leaves the semantic estimate essentially in place.

I Counterfactual-rewriting protocol

The intervention of Appendix K.8 treats each listing description as one draw from a population of descriptions for the same property and manipulates the language directly. A frozen instruction-tuned language model receives the original description, a structured summary of the property’s attributes, and a target submarket, and returns a rewritten description. The rewrite is then re-priced under the same automated valuation model, and the change in predicted log price is read off. The primary generator is Qwen 2.5 32B Instruct, run on $n = 200$ listings per market. A secondary cross-generator check uses Llama 3.3 70B Instruct on a separate

$n = 100$ draw, an unmatched probe of direction and rough magnitude rather than a tight replication.

Two contrasts are read from the rewrites. The style-stripped contrast holds the model’s inferred submarket at its original value and rewrites only the stylistic content, isolating the price the valuation model attaches to language given a fixed location. The submarket-swap contrast rewrites the listing toward a held-out submarket and reads the full change, combining the change in language with the change the model infers about location. We verify that the manipulation does what it claims. The structured attribute slots survive in 99.8% of rewrites on average, and an independent submarket classifier changes its prediction in 44.5% of rewrites, evidence that the language moves across submarket boundaries while the attributes do not.

Each contrast is a product, $\hat{\delta} = \hat{\theta} \overline{\Delta\text{PC1}}$, of the Baur DML coefficient $\hat{\theta}$ and the average rewrite-induced change $\overline{\Delta\text{PC1}} = \frac{1}{n} \sum_i (\text{PC1}_i^{\text{rw}} - \text{PC1}_i^{\text{orig}})$ in the leading text component, so its sampling uncertainty has two sources. We fold both into a single standard error by the delta method,

$$\widehat{\text{Var}}(\hat{\delta}) \approx \overline{\Delta\text{PC1}}^2 \widehat{\text{Var}}(\hat{\theta}) + \hat{\theta}^2 \widehat{\text{Var}}(\overline{\Delta\text{PC1}}), \quad (18)$$

where $\widehat{\text{Var}}(\hat{\theta})$ is the influence-function variance of the DML coefficient and $\widehat{\text{Var}}(\overline{\Delta\text{PC1}})$ comes from a within-listing cluster bootstrap over the rewritten listings in the manner of Cameron et al. [2008]. Adding the coefficient’s contribution rather than conditioning on a fitted $\hat{\theta}$ is what widens the intervals materially: the pooled style-stripped statistic is $t = 3.66$ under Equation (18), against the implausible $t \approx 13$ a coefficient-conditional interval would report. Because the cross-world independence conditions that identify a formal mediation decomposition are not met, we call these model-based contrasts rather than natural direct or indirect effects and report no natural-indirect-effect quantity.

J Hyperparameters and computational environment

Table 11 consolidates the settings scattered through the methods and the implementation appendices, so the estimates can be reproduced from the paper.

Table 11: Estimator, encoder, and computational settings. Where a setting varies by channel it is noted; remaining details and exact seeds are documented.

Component	Setting
DML cross-fitting Nuisance learner	$K = 5$ folds, out-of-fold residualization [Chernozhukov et al., 2018] ridge regression for the embedding channels and the Shen-Ross uniqueness channel (headline); gradient boosting (LightGBM, num_leaves = 8, max_bin = 63) as the robustness alternative
Treatment direction	leading principal component of the embedding, z -scored, fitted once on the pooled sample (in-sample) for the per-market panel
Text encoders	sentence-BERT all-mpnet-base-v2 (768-d, primary); all-MiniLM-L6-v2 (384-d, robustness); Doc2Vec uniqueness score (Shen channel)
LEACE / iterative gradient-based projection (IGBP) erasure GRL foil (App. D)	continuous latitude-longitude concept; IGBP run for five outer passes with a multilayer-perceptron probe; single-precision top-50 PCs $\rightarrow$ 64-d encoder; three discriminator heads (ZIP, lat-lon, income quintile); λ annealed $0.1 \rightarrow 1.0$; Adam, 150 epochs, 70/30 split
Counterfactual rewriting	Qwen 2.5 32B Instruct ($n = 200$ /market); Llama 3.3 70B Instruct ($n = 100$, cross-generator)
Interval method	influence-function intervals for the Baur and Shen per-market panel, reported with zip-clustered and Conley spatial-HAC widening in the text (Section 6.2) since the in-sample direction and cross-fitted nuisances leave the raw interval mildly anticonservative; tract cluster bootstrap ($B = 300$) for the sale-price panel; within-listing cluster bootstrap for the counterfactual contrast
Covariate winsorizer	per-column clip at median ± 5 median absolute deviations, non-finite entries imputed at the median
Meta-analysis	Paule-Mandel τ^2 , Hartung-Knapp-Sidik-Jonkman t intervals, Benjamini-Hochberg / Romano-Wolf multiplicity control
Software	Python with scikit-learn, LightGBM, PyTorch, sentence-transformers, and the concept-erasure (LEACE) package; the GPU stages (erasure, GRL, counterfactual) run on a single accelerator

K Robustness

The analyses gathered here qualify the deconfounding ladder of Section 6 rather than extend it: they bound the confounding the ladder cannot difference away, check the effect against a second encoder and against realized transaction prices, and take up the cross-market heterogeneity.

K.1 Design concerns and where each is addressed

A text-as-treatment estimate read under spatial confounding invites a standard set of concerns, most of them raised in the methodological literatures the paper draws on. Table 12 lists them, names the work that articulated each, and points to the analysis in this paper that meets it. The table is a reading guide rather than a new result, and every row is developed at length in the section it references.

Table 12: Field-standard concerns for a text-as-treatment design under spatial confounding, the work that raised each, and the analysis in this paper that addresses it.

Concern	Raised by	Addressed in this paper
The text treatment aliases the other latent features the same prose carries, so no separated semantic effect is identified	Fong and Grimmer [2023]	The estimand is stated as the price effect of the leading text direction marginalizing over whatever else the language encodes, an associational object rather than a separated attribute (Section 6.4).
The text effect is spatial information re-expressed, priced for the geography it encodes	Gilbert et al. [2021]	The erasure diagnostic confirms the exposure retains non-spatial variation, and the deconfounding ladder holds location fixed by construction rather than by adjustment (Sections 6.3, 6.4).
The scale at which location is controlled is an unpinned modeling choice	Keller and Szpiro [2020]	The control basis spans ZIP indicators and a quadratic in coordinates, and the identified direction moves little across the basis, attenuating only where the leading component is most geographic (Section 6.5).
Location and non-spatial signal can be separated by design rather than assumed apart	Guan et al. [2023]	The identified direction is the embedding analogue of spectral separation, projecting the location-predictable subspace out of the text before estimation (Section 6.5).
A treatment learned from the documents, formed on the same data as the outcome, leaves the influence-function interval anticonservative	Egami et al. [2022]	Forming the direction on the pooled corpus shrinks the generated-regressor channel, an out-of-fold projection removes most of what remains, and a full-pipeline bootstrap carries the residual into the reported intervals (Appendix G).
The estimate could depend on which machine learner fits the nuisances	Chernozhukov et al. [2018]	The cross-fitted Neyman-orthogonal score admits any learner converging faster than $n^{-1/4}$; the ridge headline and its gradient-boosted counterpart return the same per-market pattern and New York reversal (Appendix G).
Concept erasure removes only the linear part of the encoded location	Belrose et al. [2023]	Linear erasure is exact with a guardedness guarantee in every market, and the nonlinear residual a linear probe cannot see is bounded separately by iterative gradient-based projection (Section 6.3).
The prose carries omitted categorical structure the controls cannot represent, such as property type	Roberts et al. [2020]	A composition-and-conditioning audit flags the vacant-land case and adjusts for it by exact within-category de-meaning, which does not explain the effect away in any market (Section K.4).
An unobserved confounder, unmeasured condition among them, drives the result	Cinelli and Hazlett [2020]	A per-market robustness value is benchmarked against the observed spatial block and against a measured assessor quality grade where one exists, neither of which overturns the estimate (Section K.3).
Predictive accuracy from listing text may be redundant spatial encoding rather than new signal	Nowak and Smith [2017]	The erasure diagnostic and the decomposition separate the geography the representation merely encodes from the geography that moves the price, which is what distinguishes priced content from a linguistic proxy (Section 6.4).

K.2 Cross-method concordance

Sections 6.1 and 6.2 estimate the text-on-price relationship through two different encodings of the same listing language. One is a Doc2Vec uniqueness score read through the Shen-Ross hedonic specification. The other is the leading principal component of a sentence-BERT representation read through a partialling-out score. The auxiliary rewriting intervention of Appendix K.8 supplies a third measurement of a different kind, the price differential an LLM edit implies. This section asks whether the two encoders agree on which markets carry an effect. Where they agree, we ask whether the agreement reflects independent evidence or the same signal recovered twice. The rewriting channel is reported alongside as a corroborating check.

Table 13: Per-market estimates for the three text channels across the twelve markets. The Shen-Ross uniqueness and oriented pooled-PCA sentence-BERT coefficients are per-standard-deviation DML effects on log listing price, with standard errors in parentheses; the submarket-swap contrast is the LLM rewriting differential of Appendix K.8. The bottom row is the random-effects pooled estimate. The sentence-BERT axis is oriented so its pooled effect is non-negative, the principal-component sign being a disclosed convention (Section 6.2).

Market	Shen $\hat{\theta}$	Baur $\hat{\theta}$	Submarket-swap
Boston	+0.020 (0.015)	+0.073 (0.022)	−0.012
New York	+0.002 (0.006)	−0.016 (0.007)	−0.067
San Francisco	+0.117 (0.025)	+0.106 (0.025)	+0.015
Washington	+0.046 (0.011)	+0.151 (0.013)	+0.003
Philadelphia	+0.222 (0.010)	+0.374 (0.009)	+0.045
Chicago	+0.280 (0.018)	+0.464 (0.019)	+0.025
Seattle	+0.042 (0.012)	+0.030 (0.016)	−0.005
Denver	+0.072 (0.009)	+0.061 (0.009)	+0.039
Atlanta	+0.033 (0.007)	+0.134 (0.011)	+0.001
Portland	+0.042 (0.009)	+0.076 (0.013)	−0.008
Phoenix	+0.054 (0.008)	+0.055 (0.013)	−0.000
Dallas	+0.082 (0.008)	+0.142 (0.010)	−0.007
Pooled (RE)	+0.060 (0.019)	+0.137 (0.041)	+0.002

Sign convention. A component and its negation span the same axis, so the sign of a principal component is not identified. The sign of the Baur estimand is therefore a convention rather than a finding. We orient the common pooled axis so that the random-effects pooled estimate is non-negative. This makes the per-market Baur estimate sign-comparable to the hedonic channel and to the counterfactual differential. This is an interpretability convention and nothing more. It cannot create agreement between the channels. The cross-market correlation between any two channels is invariant in magnitude under a global sign flip of either one. The orientation fixes only how the shared signal is displayed, not whether it is shared.

The two embedding channels track a shared signal. Table 13 reports the per-market point estimates, and Figure 7 arranges all three channels as per-market intervals side by side. Across the twelve markets the Shen and Baur estimates move together closely. The Pearson correlation is +0.94 (Figure 8), the Spearman rank correlation +0.66 ($p = 0.018$), and the Lin concordance coefficient +0.74. They share the sign of their point estimate in eleven of the twelve markets. This full-panel correlation is driven by leverage. Chicago and Philadelphia, the two markets both channels place far above the rest, dominate it. Dropping the two leaves a Pearson of only +0.47, no longer significant at conventional levels ($p = 0.18$). The concordance is therefore best read as strong agreement on the sign and ranking of the markets that carry a clear effect rather than as a tight point-by-point identity across the whole panel. We read it not as two independent methods corroborating one another but as a robustness result about a single underlying quantity. The Shen and Baur channels are two encodings of the same listing text, applied to the same listings under the same confounder adjustment. Their agreement establishes that the text-on-price signal does not depend on whether the text is summarized by a Doc2Vec rarity score or by a sentence-BERT principal component. It does not add an independent, bias-disjoint observation of that effect. The single market where the two channels separate is New York. There the Shen estimate is essentially zero and the Baur estimate is small and negative, the only market in which the oriented Baur axis points opposite the pooled direction.

The counterfactual channel is weaker and more nearly distinct. The LLM rewriting differential is the one channel whose bias structure is not shared with the embedding channels. It relies on a within-listing minimal-edit substitution rather than the cross-listing geometry of a fixed encoder. It is also the weakest. Its rank correlation with the Shen channel is +0.77 ($p = 0.003$) and with the Baur channel +0.57 ($p = 0.05$). Its Lin concordance coefficients are only +0.21 and +0.14. This reflects that the submarket-swap contrast pools to approximately zero across markets even where its rank tracks the others. The counterfactual channel

therefore agrees with the embedding channels on the ordering of markets more than on the magnitude or the presence of an effect. We treat it as a corroborating probe with limited power rather than as a co-equal estimate, consistent with the minimum-detectable-effect caveat recorded in Appendix K.8.

Identification overlap. A more decision-relevant summary than rank agreement is which markets each channel identifies at the 95% level, where the per-method interval excludes zero. The Shen channel identifies ten of the twelve markets, the counterfactual channel nine, and the Baur channel eleven. Seattle is the one market whose Baur interval covers zero; New York shows a small but distinguishable negative. Six markets, San Francisco, Washington, Philadelphia, Denver, Portland, and Phoenix, are identified by all three channels jointly. All twelve are identified by at least two, and no market is left unidentified by every channel. The breadth of the Baur identification set should be read together with the collinearity above and with the sensitivity analysis of Section K.3. The embedding channel detects an association in eleven of the twelve markets. Whether that association survives unmeasured confounding is a separate question, addressed by those robustness values. The strength of the causal claim is adjudicated there, not in the count of identified markets.

How many independent measurements the panel carries. Because the two embedding channels are nearly collinear, the three per-market z -statistics do not constitute three independent observations. The across-market correlation matrix of the standardized estimates has a Shen-Baur entry of +0.94, with Shen-counterfactual and Baur-counterfactual entries near +0.39. Its effective dimension, computed as $\text{tr}(R)^2/\text{tr}(R^2)$, is 1.7 of a possible 3. The panel thus carries on the order of one and two-thirds effectively-independent text-price measurements per market, one shared by the two encodings and a partial second contributed by the rewriting intervention. We state this explicitly because it bounds what the joint test below can claim. The convergent evidence is robustness across text representation plus weak intervention-based corroboration, not the triangulation of three estimators with unrelated biases.

Joint-null test. For each market we test the joint null $H_0 : (\theta_{\text{Shen}}, \theta_{\text{Baur}}, \theta_{\text{CF}}) = (0, 0, 0)$. Because the three estimates are correlated rather than independent, the naive statistic $z_{\text{Shen}}^2 + z_{\text{Baur}}^2 + z_{\text{CF}}^2$ referred to a χ_3^2 distribution over-rejects. We instead use the Mahalanobis form $z_m^\top R^{-1} z_m$ with R estimated from the across-market correlation of the standardized estimates. This statistic remains χ_3^2 under the null while accounting for the collinearity. The joint null is rejected in all twelve markets at the five-percent level under the corrected reference. The rejection survives Benjamini-Hochberg correction across the twelve markets, with every adjusted q -value below 0.05. The weakest rejection is Boston. We do not read the unanimous rejection as twelve independent discoveries. The joint power is dominated by the high-precision Baur channel, and that channel is collinear with the Shen channel. The defensible reading is that the post-2020 panel contains no market in which the text-on-price association is absent under either embedding. The magnitude and the causal status of that association are governed by the heterogeneity documented in Section 6.2 and the sensitivity bounds of Section K.3.

We favor the reading that a text-on-price association is present and robust across text encodings in essentially every metropolitan market. Its magnitude is heterogeneous, and the independent rewriting probe corroborates the ordering of markets while pooling to a near-zero average effect of its own. A constructive complement would treat the per-market estimates as noisy indicators of a single latent market-level text-price effect. Pooling them through an anchored one-factor measurement model would recover a precision-weighted posterior for each market. We have not fit that model on the present panel and report it as a planned extension. The identification-overlap counts and the collinearity-corrected joint test remain the cross-method evidence on which this section rests.

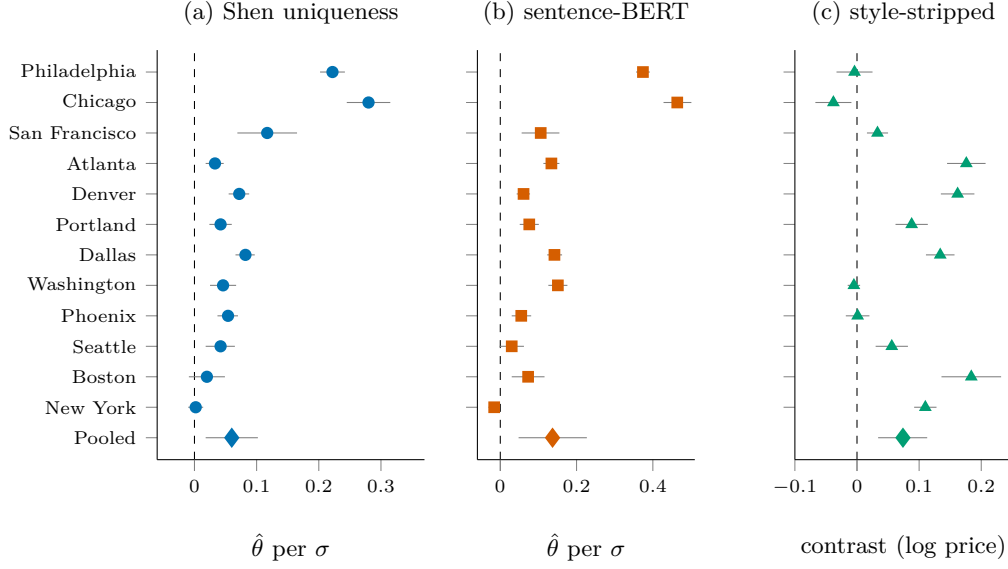


Figure 7: Per-market DML estimates across the twelve metropolitan areas for the three text channels, ordered approximately by the Shen-Ross uniqueness effect. Filled dots are point estimates and gray whiskers are 95% intervals (influence-function for the Shen and sentence-BERT channels; coefficient-uncertainty-folded cluster bootstrap for the counterfactual contrast). The diamond in the bottom row of each panel is the random-effects pooled estimate, with its gray whisker the 95% interval. The dashed vertical line marks the null. The sentence-BERT axis is oriented so its pooled effect is non-negative. Numerical values appear in Tables 3, 4, and 17.

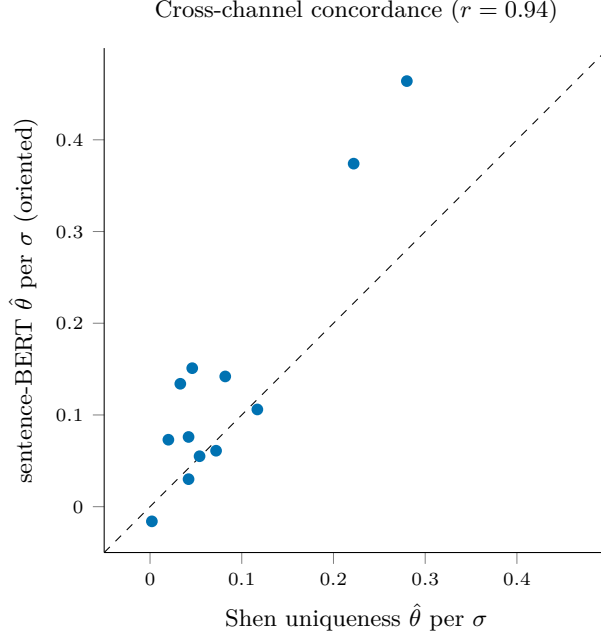


Figure 8: Per-market estimates of the two embedding channels against each other, with the 45° identity line. The two channels co-move closely across markets (Pearson $r = 0.94$), though the full-panel correlation is carried by Chicago and Philadelphia and falls to +0.47 without them; the departure from the identity line reflects that the sentence-BERT axis runs roughly twice the Shen channel in magnitude, not a difference in which markets carry the signal.

K.3 Sensitivity to unobserved confounding

The Shen-Ross DML estimates in Section 6.1 rest on a conditional ignorability assumption. Once the structured property covariates and the location indicators we observe have been partialled out, the residual variation in listing language is assumed exogenous to the residual variation in log sale price. Nothing in the data can confirm that assumption directly. We ask instead how strong an unobserved confounder would have to be before it overturned each per-market estimate, following the benchmarking calculus of Cinelli and Hazlett [2020]. We pair that bound with a leave-one-out ablation of the spatial control block, the one observed covariate group most directly implicated in the text-as-spatial-proxy mechanism that motivates the paper.

Table 14: Per-market sensitivity of the Shen-Ross estimate to unobserved confounding, for all twelve markets ordered by robustness value. $RV_{q=1}$ is the Cinelli and Hazlett [2020] robustness value, the partial R^2 an unobserved confounder must reach on both the treatment and log price to drive the estimate to zero; R^2_{spatial} is the partial R^2 that the observed spatial control block, the latitude-longitude coordinates together with the ZIP indicators, carries for the treatment or the outcome, whichever is larger, the strongest mechanism-relevant observed covariate on the same scale; $\Delta\hat{\theta}$ is the change in the estimate when that spatial block is dropped. A checkmark marks the single market whose spatial block partial R^2 exceeds its own robustness value.

Market	$RV_{q=1}$	R^2_{spatial}	$\Delta\hat{\theta}_{\text{spatial}}$	Fragile
Boston	0.009	0.0108	-0.0014	✓
New York	0.022	0.0057	+0.0013	
Atlanta	0.022	0.0099	+0.0018	
Seattle	0.027	0.0013	-0.0007	
Portland	0.028	0.0088	-0.0023	
Washington	0.037	0.0193	-0.0036	
Phoenix	0.060	0.0068	+0.0041	
Dallas	0.063	0.0058	-0.0014	
Chicago	0.068	0.0034	+0.0019	
Denver	0.070	0.0420	+0.0142	
San Francisco	0.154	0.0150	-0.0023	
Philadelphia	0.173	0.0042	+0.0006	

Robustness value. The robustness value records the minimum strength, measured as a partial R^2 on both the treatment and the outcome, that an unobserved confounder must reach to drive an estimate to zero, so that a larger value means a more robust finding. Table 14 reports it for all twelve markets alongside the benchmark strength of the observed spatial control block. Figure 10 orders the markets by it. Two markets clear the conventional 0.10 threshold, Philadelphia at 0.17 and San Francisco at 0.15. Overturning either would require a confounder explaining more than fifteen percent of the residual variance in both the listing signal and the price. The other ten fall below the threshold. They descend from Denver and Chicago near 0.07, through Dallas and Phoenix around 0.06 and the mid-sized markets near 0.03, to New York and Atlanta at 0.02 and Boston at 0.009, the last of which has almost no effect to protect.

Observed-covariate benchmark. The spatial control block, the observed covariate group most directly tied to the text-as-spatial-proxy mechanism, carries a partial R^2 with the treatment or the outcome that never exceeds 0.042, reaching its largest value in Denver and its smallest, near 0.001, in Seattle. In every market but Boston this observed spatial strength sits below the robustness value. An unobserved confounder capable of overturning the estimate would have to be stronger than the location signal the controls already record. The room between the two is widest exactly where the effect is largest. In Philadelphia and San Francisco, robustness values near 0.17 and 0.15 dwarf spatial benchmarks of 0.004 and 0.015. That room narrows toward the low-robustness markets, where the estimate is already small and neither quantity is large enough to anchor a confident reading in either direction. The bias geometry behind these per-market summaries is drawn out, for the two markets that clear the robustness threshold, in Figure 9.

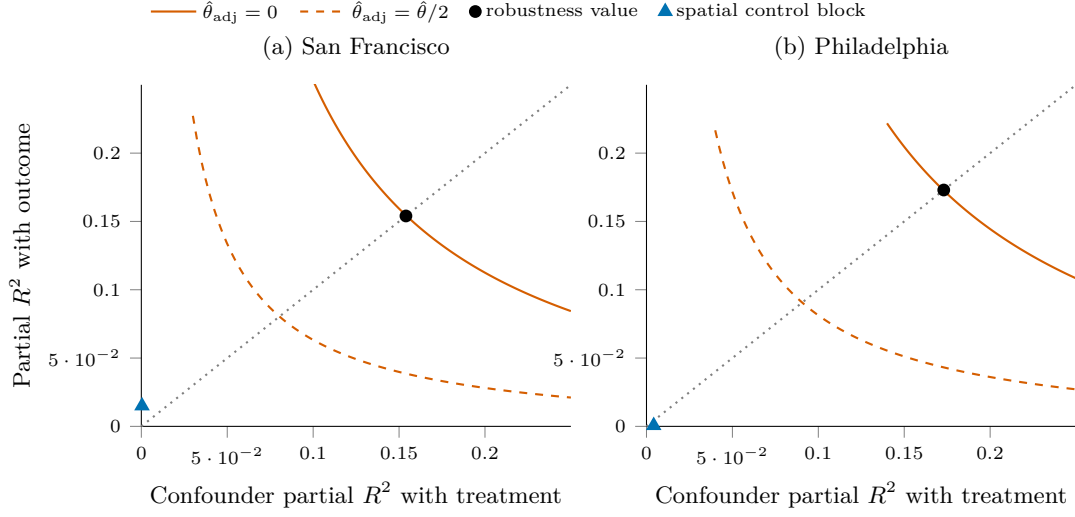


Figure 9: Omitted-variable-bias sensitivity contours, in the manner of Cinelli and Hazlett [2020], for the two markets that clear the robustness threshold. The axes are the partial R^2 that a single unobserved confounder would have with the treatment (horizontal) and with the outcome (vertical). The solid curve is the locus along which such a confounder would drive the per-standard-deviation estimate to zero, and the dashed curve the locus along which it would halve it. The black dot, where the solid curve meets the dotted equal-strength diagonal, is the robustness value, the equal-strength confounder that just eliminates the effect, at 0.15 in San Francisco and 0.17 in Philadelphia. The triangle marks the observed spatial control block, placed at its partial R^2 with the treatment and with the outcome, which falls far inside the kill curve in both markets, so an unobserved confounder would have to be substantially stronger than the measured location signal to overturn the estimate.

A measured quality control. The sharpest reading against the semantic interpretation is that the surviving effect is unobserved property condition, since the vocabulary at the price end of the text axis is finish and renovation language, and the structured controls carry no direct measure of condition. Where an assessor-authored quality measure does exist we can condition on it. Boston’s assessment records a categorical condition grade. The San Francisco and New York rolls carry an improvement value authored on the county’s own schedule and fixed at a transaction that predates the listing. None of these measures is a function of the price we model. The test has a direction. If the surviving effect were unmeasured condition wearing the clothes of language, adding a direct measure of condition to the controls would pull it toward zero. It does not. Adding the available measure leaves the pooled-PCA text effect intact in all three markets. It attenuates by about a sixth in Boston, from $+0.073$ to $+0.061$, and by about a fifth in San Francisco, from $+0.106$ to $+0.083$, and is essentially unchanged in New York. The interval continues to exclude zero in every one. The shrinkage a real quality proxy induces bounds how much of the effect condition can be carrying, and that bound is small. Philadelphia and Chicago retain no such field. There a quality confounder would need a partial R^2 near 0.38 and 0.28 with both the treatment and the price to null the estimate, five to twenty times the strongest partial R^2 that any measured quality proxy reaches elsewhere. A condition signal of the strength we can actually observe is far too weak to carry those effects. This does not exhaust the concern in the two markets without a direct proxy, but where condition can be measured it does not explain the effect away.

Spatial-block leave-one-out. Dropping the spatial control block and re-estimating moves the point estimate by about fourteen thousandths of a log point in Denver and by under half a percent in every other market, the next largest shift falling in Phoenix at $+0.004$. The estimates barely respond to removing the location controls. They hold to within a few thousandths of where they began in eleven of the twelve markets, and to a fraction of Denver’s already small effect in the twelfth, without a single sign change. This is the pattern one would hope to see if the text signal were not merely spatial information re-expressed:

a coefficient that rode on geography would collapse once geography left the conditioning set, rather than shift by so little. The leave-one-out thus localizes, market by market, the conclusion that the deconfounding decomposition of Section 6.4 reaches for the embedding channel as a whole.

Fragility. Under the same benchmarking convention a market is flagged when the partial R^2 that the spatial block carries for either the treatment or the outcome exceeds that market’s robustness value, the condition under which a confounder no stronger than the location controls already in the model could in principle account for the estimate. Only Boston trips the flag, and it does so because its robustness value is itself almost zero. The spatial block’s explanatory power for the treatment, at 0.011, edges past a robustness value of 0.009, for an estimate that is statistically indistinguishable from zero to begin with. No market with a robustness value above 0.02 is flagged. The fragility check isolates the one place where there is no effect to confound rather than casting doubt on any market where an effect is present.

Interpretation. The analysis is confined by design to the single observed covariate group the per-market panels share and the one most tied to the mechanism this paper studies, so the robustness value carries the general bound while the spatial leave-one-out localizes how much of each estimate rides on measured location. The two diagnostics agree with each other and with the rest of the section. Philadelphia and San Francisco, the markets in which the text effect is largest, are also the ones most robust to unobserved confounding. The markets that fall below the threshold are those whose estimates are already small. Removing the explicit location controls disturbs none of them. We read the per-market sensitivities as informative about robustness rather than as a second identification test. We leave the benchmarking of non-spatial confounder groups to future work, because the census, crime, and amenity blocks needed to construct them are not uniformly available across the twelve per-market panels.

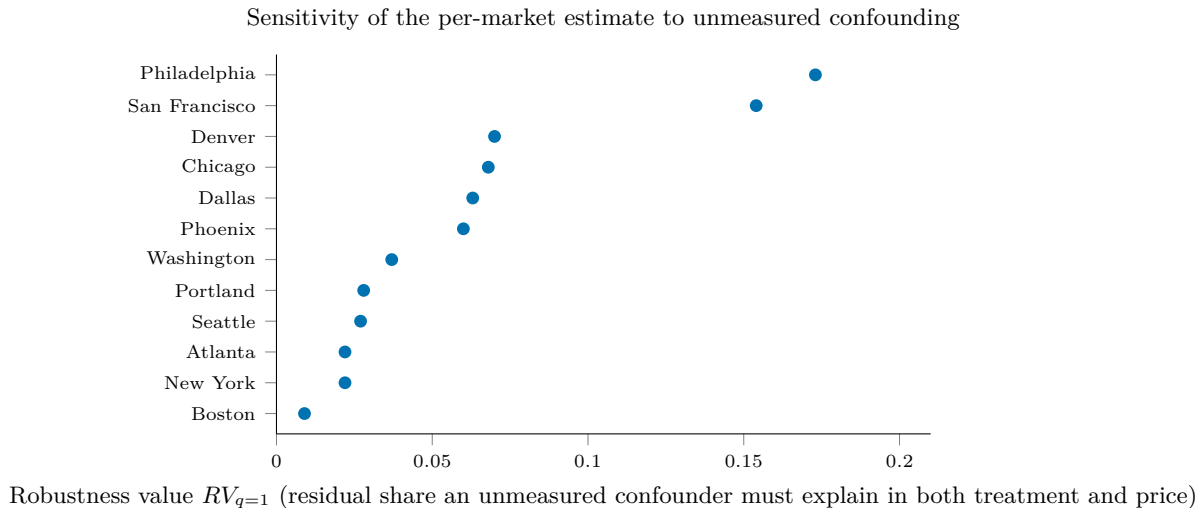


Figure 10: Robustness value for each market’s Shen-Ross uniqueness estimate, the share of residual variance an unmeasured confounder would need to explain in both the treatment and log price to drive the estimate to zero, following Cinelli and Hazlett [2020]. Larger values are more robust. The estimate is most robust in Philadelphia (0.17) and San Francisco (0.15) and most fragile in the markets whose point estimates are already near zero. The modest robustness values in several markets are the reason we read the per-market estimates as associational and sensitivity-bounded rather than causal.

K.4 A composition-and-conditioning audit for text-treatment DML

The robustness values of Table 14 benchmark the strength an unobserved confounder would need against the strength of the strongest confounder we do observe, following Cinelli and Hazlett [2020]. That the observed-covariate benchmark can mislead is not itself new. Diegert et al. [2026] prove the standard benchmarking

comparison is inconsistent and non-monotonic as a way of ranking confounders, and the fragility of informal benchmarking is acknowledged in the sensitivity literature more broadly. What a text-derived treatment adds is a particular and severe form of the failure, an omitted category the treatment nearly sorts yet the structured controls cannot see. This section names that case, bounds it, and turns it into a diagnostic tailored to it.

Consider an omitted categorical covariate Q , written to keep it distinct from the observed contextual block c of the control set. Property type is the concrete instance and vacant land the sharpest case. It has three properties that hold in the data. It shifts the treatment, $R_{T \sim Q|X}^2 > 0$, because a vacant lot is described in different language than a house. It shifts price net of the treatment and the observed controls, $R_{Y \sim Q|T,X}^2 > 0$. It is nearly invisible to the observed confounder block, $R^2(Q | X) \approx 0$. A cross-fitted classifier of the vacant-land indicator on the full confounder set runs at an AUC of 0.50 in Boston and 0.47 in New York, while the same indicator is recovered from the text direction at an AUC above 0.95. The nearest-parcel join that supplies the structured controls cannot represent landness. For a vacant lot it returns a neighboring built parcel’s attributes, so the design is blind to exactly the attribute the text encodes. The bias from omitting such a Q is bounded by the partial- R^2 omitted-variable-bias inequality of Cinelli and Hazlett [2020] and its double-machine-learning generalization in Chernozhukov et al. [2022]. That bound holds whether or not Q is orthogonal to X . What the orthogonality changes is not the bound but the benchmark used to fill it in.

Proposition 2 (Orthogonal composition defeats the observed-covariate benchmark). *In the partially-linear model of Equation (1), let Q be an omitted covariate with $r_D := R_{T \sim Q|X}^2 > 0$ and $r_Y := R_{Y \sim Q|T,X}^2 > 0$, and suppose $R^2(Q | X) \rightarrow 0$. Then (i) the protective subtraction that conditioning on X applies to a confounder correlated with the controls, the cross term $\rho_{TX}\rho_{QX}$ in the partial correlation, vanishes, so the partial strengths r_D and r_Y that enter the Cinelli and Hazlett [2020] and Chernozhukov et al. [2022] bias bound are not shrunk below their raw values, with $r_D \rightarrow R_{T \sim Q}^2/(1 - R_{T \sim X}^2)$ in the limit; and (ii) the observed-covariate benchmark, which calibrates the plausible magnitude of Q by that of the strongest observed covariate, carries no warrant, because Q by construction carries information orthogonal to the covariate space on which that benchmark is computed. The robustness value reported against the observed benchmark is therefore numerically well defined but uninformative about Q .*

The argument is short. Write $\tilde{T}$, $\tilde{C}$, and $\tilde{Y}$ for the residuals of T , Q , and Y after partialling out the full control vector X by population regression. These are the objects the partially-linear score is built on, so the partial quantities r_D and r_Y in the bias bound are functions of the correlations among these residuals. As $R^2(Q | X) \rightarrow 0$ the residual $\tilde{C}$ converges to Q centered, because X carries no linear information about Q to remove. The confounder’s residual variation is thus undiminished by conditioning, and $r_D \rightarrow R_{T \sim Q}^2/(1 - R_{T \sim X}^2)$, no smaller than its raw counterpart. The outcome side is identical, which gives part (i). For a single scalar control this residual statement reduces to the familiar recursive partial-correlation identity $\rho_{TQ|X} = (\rho_{TQ} - \rho_{TX}\rho_{QX})/\sqrt{(1 - \rho_{TX}^2)(1 - \rho_{QX}^2)}$. The protective subtraction $\rho_{TX}\rho_{QX}$ vanishes with $\text{Cor}(Q, X)$. The high-dimensional case replaces that scalar identity with the residual construction just given rather than relying on it. A computer-algebra check confirms the scalar identity and the vector-control residual limit just used, in which the partial strength at orthogonality equals $R_{T \sim Q}^2/(1 - R_{T \sim X}^2)$, with $R_{T \sim X}^2$ the multiple coefficient of determination of T on the whole control block. It also confirms that for a single omitted confounder the Cinelli and Hazlett [2020] bias expression equals the exact bias rather than merely bounding it, so the bound the audit reasons about is tight. Part (ii) is the operational reading. The benchmark’s premise is that an omitted confounder is comparable in strength to an observed covariate. A classifier of Q on X running at chance is direct evidence that Q lives outside the space that premise is calibrated on. We are careful about what this does and does not establish. It is not that the formal bound cannot detect Q . The Cinelli and Hazlett [2020] bound, correctly computed, recovers the right answer for a confounder orthogonal to the controls once the analyst sets the benchmark strength correctly. It is that the routine heuristic of setting that strength by analogy to observed covariates has no anchor once $R^2(Q | X) \approx 0$. This is exactly the regime in which an analyst is most likely to feel wrongly reassured, since orthogonal to the controls reads too easily as weak. The claim in part (i), that conditioning shrinks a correlated confounder’s residual strength, is generic rather than universal. Middleton et al. [2016] show that conditioning on covariates correlated with an unobserved confounder can amplify rather than shrink

the bias in bad-control configurations. It is the expected case, not a law.

A second, distinct failure channel travels with the first. When Q nearly sorts the treatment, the residual treatment variance $E[(T - E[T | X, Q])^2]$ collapses and the orthogonal score becomes ill conditioned. Saco [2025] characterize this phenomenon through the conditioning number $\kappa = 1/(1 - R_{\text{oof}}^2(T | X))$. That reference reports the diagnostic as a fragility assessment and declines to propose a universal threshold, and we follow it in treating κ and the design-matrix condition number as flags read alongside the substantive analysis rather than as a test. The composition confound and the conditioning degeneracy are separate objects. Confusing one for the other misdiagnoses the fix: the first is corrected by adjusting for the omitted category, the second by repairing the design.

These two channels motivate a two-step audit, run before the point estimate is inspected, that we call the composition-and-conditioning audit. Step one screens conditioning. Flag a market if the standardized confounder matrix has condition number above 10^6 , or if any confounder column has collapsed to a constant. Either signals a design-matrix degeneracy rather than a treatment effect. Step two screens composition. Flag a market if a candidate omitted category is chance-predictable from the controls, $\text{AUC}(Q | X) < 0.60$, yet strongly predictable from the treatment, $\text{AUC}(Q | T) > 0.75$. This is the near-separation of treatment from an omitted category that D’Amour et al. [2021] show breaks overlap in high dimensions. The prescribed correction differs by step. A conditioning flag is a pipeline repair. A composition flag calls for adjusting for the category. The natural instinct, adding a property-type dummy to the regularized nuisance, backfires, because the ridge penalty shrinks a rare category toward no effect. In Boston, where non-residential listings are 1.9% of the sample, adding the dummy raises $|\hat{\theta}|$ from 0.40 to 0.45. Exact within-category demeaning of the treatment and the outcome, which is unpenalized, is the correct adjustment. In Boston it returns 0.44. These figures are on the raw within-market treatment scale of the demeaning exercise, several times the oriented pooled-axis magnitudes of Table 4. They index the direction and stability of the adjustment rather than the headline effect size. What the corrected adjustment shows across the panel is that property-type composition is not what drives the result. Exact demeaning attenuates $|\hat{\theta}|$ toward zero in only two of the twelve markets, and leaves or slightly raises it in the other ten. In no market does controlling for composition explain the text effect away, and the rank of a market’s non-residential share does not track how much its estimate moves. The blind spot the proposition identifies is a possibility in principle. Once corrected for, it is empty in practice on this panel.

A calibration study confirms that the two steps discriminate the two failure modes rather than firing together. Across a grid of category shares and confounder strengths, with 150 replications per cell, the conditioning step fires on a simulated design degeneracy in every replication and never on a composition confound. The composition step fires on a material composition confound, one strong enough to lift $\text{AUC}(Q | T)$ above the threshold, in more than nine of ten replications, and on a weak composition confound in fewer than one in twenty. Its power is concentrated where the composition signal is material rather than spread evenly across strengths, and it never fires on a design degeneracy. The false-positive rate of each step on the other’s failure mode is zero. The audit therefore separates a numerical accident from a substantive confound on data the analysis already has. It does so before the estimate is read, which is what a pre-registered diagnostic is meant to do. We claim for it neither a new estimator nor a new limit theorem. It only connects several existing literatures, the omitted-variable-bias bounds of Cinelli and Hazlett [2020] and Chernozhukov et al. [2022], the critique of observed-covariate benchmarking in Diegert et al. [2026], the high-dimensional overlap diagnostics of D’Amour et al. [2021], and the conditioning diagnostics of Saco [2025], into a screen fitted to the specific way a text-derived treatment fails.

K.5 The effect on realized transaction prices

The outcome studied so far is the asking price, which the listing agent sets jointly with the description and which therefore shares a common author with the treatment. A natural concern is that part of the estimated association reflects the agent pricing their own prose rather than the market valuing the property’s language. The most direct answer is to re-estimate the effect on the price the market actually cleared. This does not sever the simultaneity outright, since a realized sale price anchors on the list price the same agent set [Genesove and Mayer, 2001]. It does, however, replace the agent’s stated number with a market-tested one, and so weakens the channel that a purely asking-price outcome leaves open. We assemble a panel of recorded sales by collecting sold listings across the ten metropolitan areas for which a public sold feed carries both

the marketing description and the realized sale price on the same transaction. We also collect the sale date and the structural attributes. We restrict to residential single-unit dwellings with a usable description, a recorded price, and a geocode. The panel then holds roughly one hundred ninety thousand transactions closed between 2023 and 2026, close to three times the pooled description corpus used elsewhere in the paper. The treatment is the same pooled leading text direction, re-estimated on the sold descriptions and held fixed across the estimation, and the outcome is the log recorded sale price. Because sale prices span three years of nominal, appreciating-market dollars, we add sale-quarter fixed effects to the control set so the flexible nuisance learner absorbs the price cycle [Rosen, 1974, Case and Shiller, 1989].

The estimand requires care about which attributes to control. The description and the agent-reported structural fields are authored together. Conditioning on the listing’s own bedroom count or square footage risks blocking a pathway that runs through the very language we study, a bad control in the sense of Cinelli et al. [2024]. We therefore take the exogenous county-assessor record as the primary control set, since it is authored independently of the listing. We read the resulting estimate as a controlled direct effect of the text net of the property’s structural characteristics. County structural records are available for eight of the ten markets; for the remaining two, Washington and Atlanta, we report the agent-reported specification alone. The estimand is further conditional on sale, since the panel observes only transactions that closed. We do not attempt a selection correction, for which no credible exclusion restriction exists. We state the effect as the average text effect among properties that sell, a standard conditional estimand in this literature [Heckman, 1979, Munneke and Slade, 2000].

The effect survives the change of outcome in every market. Table 15 and Figure 11 report the per-market estimates. On the assessor specification a one-standard-deviation shift along the leading text direction moves log sale price by between three and sixteen percent, individually significant in all eight markets under tract cluster-bootstrap inference. The heterogeneity mirrors the asking-price panel: the effect is largest in the older, architecturally heterogeneous markets (Philadelphia, Chicago, San Francisco, Boston) and smallest in the newer, more uniform Sun Belt markets (Phoenix, Seattle, Denver, Portland). The sign of the coefficient is a normalization of the leading component and is not itself interpretable; the magnitude and its cross-market ordering are. The agent-reported specification is significant in all ten markets but is uniformly smaller than the assessor specification in every market where both are estimated. This is the signature of the bad-control channel we anticipated: controlling for text-entangled attributes absorbs part of the effect.

The result is robust along the three dimensions most likely to threaten it. Spatial dependence is addressed beyond the tract cluster bootstrap by a Conley spatial HAC standard error and by the self-normalized cluster- t interval of Ibragimov and Müller [2010], the better-calibrated of the available spatial procedures under cross-fitting [Salerno et al., 2026]. The effect excludes zero in all eight assessor markets under every one of these procedures. Dependence across markets and across calendar quarters is addressed jointly. We pool the ten markets into a single panel with metro and sale-quarter fixed effects and report the two-way cluster-robust standard error of Cameron et al. [2011] by metro and by sale-quarter. The pooled per-standard-deviation effect is 0.061 with a two-way-clustered interval of $[0.026, 0.097]$ that excludes zero. We read this as a robustness check and not as the headline, because with ten metropolitan clusters the metro dimension sits well below the count at which two-way cluster-robust inference is dependable [Cameron and Miller, 2015]. The interval uses a t reference on nine degrees of freedom. The metro-clustered standard error alone, which inflates the influence-function standard error roughly twentyfold, already accounts for almost the entire two-way width, so the residual quarter dimension is not what carries the result. The concern that the estimated text direction is a generated regressor is addressed by the split-sample discipline of Egami et al. [2022]. Estimating the leading direction on a random half of the listings and the effect on the disjoint half returns the same per-market estimates to within a thousandth. The effect is therefore not a product of how the text measure is discovered and estimated on the same data. Finally, the leading text direction is almost uncorrelated with the sale quarter, with an R^2 below two percent in every market. The effect is not a repackaged market-cycle trend, and the sale-quarter controls are not load-bearing. Because the sold descriptions arrive truncated at a fixed character length, we report the assessor estimate both raw and corrected by the validation-sample attenuation factor of Appendix K.6. The correction widens the effects modestly and changes no conclusion. The effect is present, heterogeneous in the same pattern, and robust on the transaction the market actually cleared, in every market where that transaction is observable. This is the strongest evidence the available data affords that the listing-text association is a feature of market valuation rather than a quirk of how agents price their own language.

Table 15: Effect of the leading listing-text direction on log realized sale price, by metropolitan area. Estimates are ridge partially-linear DML on the single-unit subset with sale-quarter fixed effects; the assessor arm controls for exogenous county-record attributes (the primary, controlled-direct-effect specification) and the listing arm for agent-reported attributes (a bad-control sensitivity bound). Intervals are tract cluster-bootstrap ($B = 300$); $\hat{\theta}_{\text{corr}}$ divides the assessor estimate by the validation-sample truncation factor. $\hat{\theta}_{\text{listing}}$ and $\hat{\theta}_{\text{corr}}$ are reported as point estimates, without a separate interval of their own. Estimates are oriented so the effect is positive; all are individually significant.

Market	n	$\hat{\theta}_{\text{assessor}}$	95% CI (cluster)	$\hat{\theta}_{\text{listing}}$	$\hat{\theta}_{\text{corr}}$
Philadelphia	29,094	0.159	[0.146, 0.169]	0.140	0.196
Chicago	20,245	0.148	[0.136, 0.158]	0.102	0.182
San Francisco	6,585	0.115	[0.098, 0.131]	0.076	0.141
Boston	3,337	0.111	[0.093, 0.132]	0.057	0.137
Washington	9,846	—	—	0.060	—
Atlanta	23,603	—	—	0.069	—
Seattle	17,363	0.059	[0.053, 0.065]	0.049	0.073
Denver	22,000	0.060	[0.052, 0.068]	0.043	0.074
Portland	21,653	0.064	[0.056, 0.069]	0.044	0.080
Phoenix	39,003	0.029	[0.025, 0.034]	0.020	0.036

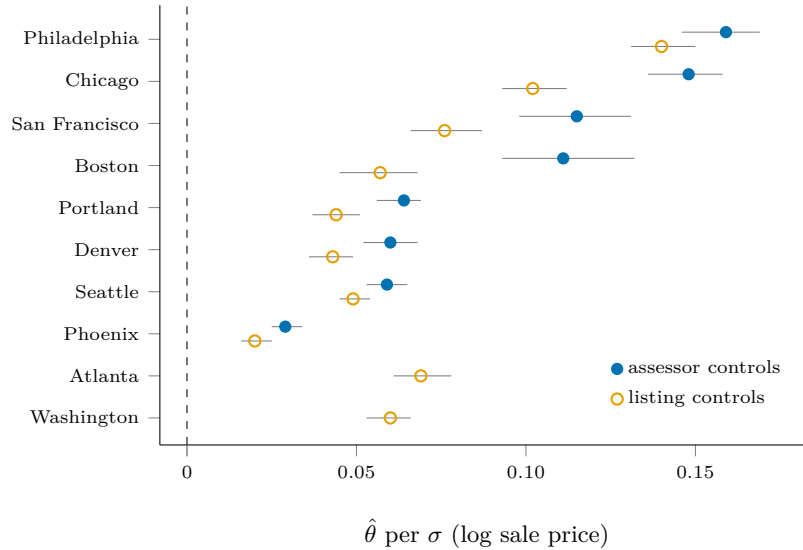


Figure 11: Per-market effect of the leading listing-text direction on log realized sale price, oriented so the effect is positive. Filled blue markers use exogenous county-assessor structural controls (the primary, controlled-direct-effect specification); open amber markers use agent-reported listing attributes (the bad-control sensitivity bound). Gray whiskers are 95% tract cluster-bootstrap intervals ($B = 300$); the dashed line marks the null. Washington and Atlanta lack a county structural source and carry only the listing specification. The listing effect lies inside the assessor effect in every market where both are estimated, the signature of controlling for text-entangled attributes. Numerical values appear in Table 15.

K.6 Correcting for description truncation in the transaction-price panel

The realized-transaction panel is assembled from a bulk listings interface that returns each property’s description capped at a fixed length of six hundred and ninety-nine characters, a limit that truncates between roughly half and two thirds of descriptions before they are embedded. Because the treatment is the leading principal component of the description embedding rather than one of the controls, this truncation perturbs

the treatment itself, and left uncorrected it biases the estimated price effect rather than merely inflating its variance.

We first establish the sign and magnitude of the bias on the full-text corpus, where the untruncated description is available for the same listings. Recomputing the treatment from descriptions cut to the same character limit and re-estimating the effect with the treatment and controls otherwise held fixed, we find that truncation attenuates the effect toward zero in every market with a detectable effect, by between fifteen and twenty-seven percent, with the per-market estimates collected in Table 16. The distortion is not a byproduct of the embedding model’s own context window, which admits close to four hundred tokens against a median description of under two hundred, so the character cap discards text the encoder would otherwise read.

We do not treat this as classical measurement error. The truncation is a deterministic function of the description that always removes its tail, and the removed text is longer for more elaborate listings, so the induced error is correlated with the true treatment and with omitted quality rather than being independent additive noise. This is the generated-regressor structure of Pagan [1984] and Murphy and Topel [1985], and in its modern data-mined form the problem studied by Yang et al. [2020], rather than the classical errors-in-variables model. Under non-classical error the direction of the bias is not guaranteed in advance [Bound et al., 2001], and for latent, text-derived predictors a naive reliability-ratio correction can enlarge the bias rather than remove it [Jerzak and Jessee, 2025]. We therefore measure the attenuation on validation data instead of assuming it.

The full-text corpus is an external validation sample in the sense of Chen et al. [2008] and Carroll et al. [2006], since it carries both the truncated and the untruncated treatment on the same units. We summarize the attenuation by the factor $\lambda = \hat{\theta}_{\text{trunc}}/\hat{\theta}_{\text{full}}$, which is scale-free and absorbs the partialling of the controls. Across the six markets with a detectable effect the factor is stable at 0.79 with a standard deviation of 0.05 and a range from 0.73 to 0.85; markets whose effect is statistically indistinguishable from zero leave the ratio undefined but require no correction, because a near-null estimate stays near-null under any rescaling and, as Table 16 shows, none of them is pushed into significance by the correction. Precision-weighting across all markets, which down-weights the near-null cases whose ratios are imprecise, gives a pooled factor of 0.81 with a standard error of 0.02, an attenuation of nineteen percent and a correction factor of 1.24.

Two checks support applying this factor to the transaction panel, whose treatment is only available in truncated form. A leave-one-market-out exercise, in which each market’s truncated effect is corrected by the factor estimated on the remaining markets, recovers the held-out market’s untruncated effect to within 0.011 in the units of θ , or under six percent, comfortably inside the market-level confidence intervals; the factor therefore transports across markets. Regression calibration, which replaces the truncated treatment with its conditional expectation given the truncated value and the controls and re-estimates the effect, moves each estimate in the same direction and closes fifty-four percent of the truncation gap. It under-corrects, which is the expected behavior of regression calibration when the error is non-classical [Carroll et al., 2006], and we accordingly take the measured attenuation factor as the primary correction and read the calibration estimate as a conservative floor. As a further robustness option the same paired validation sample supports a simulation-extrapolation correction [Cook and Stefanski, 1994], which we do not need here given the stability of the measured factor.

We report transaction-price effects after dividing the truncated estimate by λ , propagating the uncertainty of both the effect and the factor through the delta method; because the effect and the factor are estimated on disjoint samples their errors are independent and the corrected standard error is $\sqrt{\hat{\theta}_{\text{trunc}}^2 \widehat{\text{Var}}(\lambda)/\lambda^4 + \widehat{\text{Var}}(\hat{\theta}_{\text{trunc}})/\lambda^2}$. To our knowledge this truncation is not confronted in the earlier real-estate text literature, which represents descriptions through bag-of-words or dictionary features that carry no fixed context window [Nowak and Smith, 2017, Shen and Ross, 2021]; the problem is specific to fixed-window sentence-embedding features, and the attenuation we document is the pricing-model counterpart of the classification-accuracy loss that Sun et al. [2019] report for head-only truncation of long documents.

Table 16: Description-truncation attenuation, measured on the full-text validation corpus. For each market with a detectable text effect we report the DML effect of the leading text direction estimated from the untruncated description and from the description cut to the interface’s character limit, both with identical controls, and the attenuation factor $\lambda = \hat{\theta}_{\text{trunc}}/\hat{\theta}_{\text{full}}$. The pooled factor is precision-weighted across all ten markets and used to correct the transaction-price panel.

Market	$\hat{\theta}_{\text{full}}$	$\hat{\theta}_{\text{trunc}}$	λ
Boston	−0.245	−0.209	0.85
San Francisco	−0.202	−0.171	0.85
Chicago	−0.222	−0.175	0.79
Denver	−0.126	−0.099	0.79
Atlanta	−0.138	−0.105	0.76
Philadelphia	−0.167	−0.122	0.73
Pooled ($1/\lambda = 1.24$) $\lambda = 0.81$ (0.02), attenuation 19%			

K.7 Cross-market heterogeneity and a null moderator

The twelve per-market estimates are heterogeneous, ranging from a small negative in New York to several times the mean in Chicago and Philadelphia. A pooled figure therefore summarizes a dispersed distribution rather than a single generalizable effect. Combining the markets by inverse-variance weighting, the Shen-Ross uniqueness channel pools to +0.060 with a 95% interval of [+0.018, +0.102]. The sentence-BERT axis, under the price-positive orientation of Section 6.2, pools to +0.137 with [+0.048, +0.227]. Both channels are positive and detectable in the pool, and the conclusion is stable across the weighting choice. Weighting each market equally and weighting by sample size return Baur pooled estimates of +0.137 and +0.130, and the precision-weighted value is +0.122. The gap reflects only that New York carries a fifth of the listings at a small negative effect; dropping it lifts the market-weighted mean to +0.152. What moves the pool is the single market the rest of the paper already isolates.

The spread across markets is not left as noise, since a part of it is accounted for by the same mechanism the paper studies. The two markets that sit highest, Chicago and Philadelphia, are the ones whose leading text direction carries the most residual geography. The identified-direction estimates of Section 6.5 project that geography out of the treatment, and they attenuate exactly those markets the most, Chicago from +0.46 to +0.37 and Philadelphia from +0.37 to +0.27. The markets whose leading axis already lay near the identified direction barely move. Part of the raw dispersion is therefore the differing amount of location each market’s variance-maximal axis absorbs, rather than a differing price of language. The deconfounded estimates occupy a narrower band than the leading-component ones. A residual heterogeneity remains after that projection, and the moderator scan below does not resolve it. We rest the cross-market claim on the pooled effect and read the per-market magnitudes as a distribution whose extremes are the markets in which text and location are most entangled.

We looked for a metropolitan-level moderator of the effect and found none that survives correction. We regress each channel’s per-market estimate on eleven metropolitan covariates, eight from the American Community Survey and three from federal series. No covariate approaches significance on the sentence-BERT channel, where the smallest raw p , carried by the renter share of the housing stock, is 0.24. On the Shen-Ross channel that same renter share does reach a nominal raw significance, at $p = 0.03$ with a negative sign. It survives neither the Romano-Wolf multiplicity adjustment across the eleven candidates, which lifts its p to 0.10, nor a leave-one-out check. That check traces the association almost entirely to New York, whose Cook’s distance dominates the regression and whose removal alone returns the coefficient to insignificance. With twelve markets a regression on site-level estimates is underpowered by construction. The reading we take is the absence of a moderator that survives correction for multiplicity and influence, not a demonstrated absence of moderation. A properly powered treatment would carry the covariate as an interaction in a stacked specification rather than a regression on twelve site estimates. We record one correction. An earlier draft reported a positive and significant renter-share moderator. This turned out to be produced by a corrupted confounder in a single market that the winsorization of Section 5 removes. Once corrected the coefficient reverses sign, and what nominal significance remains rests on New York alone and

does not survive the multiplicity adjustment. We therefore report the null rather than carry the superseded claim.

K.8 Auxiliary check: counterfactual rewriting

This appendix reports an auxiliary probe that corroborates the main results from a different angle, with different failure modes, but is not part of the identification argument. We treat the listing description as a realized draw from a population of plausible descriptions for the underlying property. We examine the role of language by generating counterfactual descriptions in which the listing’s submarket-targeted language is altered while the property’s structured attributes are held fixed. The intervention is executed at the language level by a frozen large language model that receives the original description, a structured attribute summary, and a target submarket, and returns a rewritten description. We then re-price the rewritten listing under the same automated valuation model and read two model-based contrasts from the difference in predicted log price. The style-stripped contrast holds the model’s predicted location at its original value and isolates the price the valuation model attaches to the listing’s stylistic and semantic content given a fixed location. The submarket-swap contrast rewrites the listing toward a held-out submarket and reads the full change in predicted price, which combines the change in language with the change the model infers about location. We call these model-based contrasts rather than natural direct and indirect effects, and the distinction is not cosmetic. The intervention does not satisfy the cross-world independence conditions that identify a formal mediation decomposition. We therefore do not report a natural-indirect-effect quantity, and we do not interpret the difference between the two contrasts as a structural mediation parameter. Each contrast is the average change in one automated valuation model’s output under a specific, auditable manipulation of the input text, and that is the object we analyze.

Rewriting validation. Before we report estimates, we verify that the rewriting procedure executes the intended manipulation. Structured attribute slots, comprising bedrooms, bathrooms, square footage, year built, lot size, and three derived flags, are preserved in 99.8% of rewrites on average across the twelve markets. The lowest preservation rate is 99.4% in Phoenix, and preservation is exact in Boston, Washington, Denver, and Dallas. The submarket classifier flips its prediction away from the original submarket in 44.5% of rewrites on average, ranging from 18.4% in San Francisco to 63.1% in New York. We read the high average flip rate as evidence that the rewriting moves the listing’s targeted language across submarket boundaries in a large share of cases. The near-perfect slot preservation is evidence that the structured attributes survive the rewrite.

Confidence intervals. Each contrast is the product of the DML treatment coefficient and the average rewrite-induced change in the principal component, so its sampling uncertainty has two sources, the cluster bootstrap over the rewritten listings and the estimation error in the DML coefficient itself. The per-market intervals we report fold both into a single standard error by the delta method, adding the coefficient’s contribution to the within-listing cluster bootstrap rather than conditioning on the fitted coefficient. This widens the intervals materially relative to a bootstrap that treats the coefficient as known. It is the reason the pooled statistic below is moderate rather than the implausibly large value a coefficient-conditional interval would produce. Appendix I sets out the rewriting protocol, the two generators, and the delta-method variance in full.

Per-market style-stripped contrast. Table 17 reports the per-market style-stripped contrast at $n = 200$ listings each. The contrast is positive and excludes zero in eight markets and negative and excludes zero in one, with three markets covering zero once the coefficient uncertainty is included. The largest positive contrasts are in Boston at +0.184 on log price, an implied +20.2%, and Atlanta at +0.176 (+19.3%). Next come Denver at +0.162 (+17.6%), Dallas at +0.134 (+14.3%), New York at +0.110 (+11.6%), Portland at +0.088 (+9.2%), Seattle at +0.056 (+5.8%), and San Francisco at +0.033 (+3.4%). Chicago returns a negative contrast of -0.038 that excludes zero. Washington, Philadelphia, and Phoenix return contrasts whose intervals straddle zero once the coefficient uncertainty is folded in, Phoenix because its point estimate is essentially zero and Washington and Philadelphia because their intervals widen past it. The pooled random-effects style-stripped contrast across the twelve markets is +0.074 on log price, with standard error

0.020, a t statistic of 3.66, and a 95% confidence interval of $[+0.034, +0.113]$. The pooled contrast is positive and detectable. The between-market heterogeneity is large, with $I^2 = 97.5\%$. We report the random-effects estimate in preference to the fixed-effects estimate, and caution that the pooled figure summarizes a dispersed distribution rather than a single generalizable mean.

Table 17: Counterfactual rewriting intervention: per-market style-stripped and submarket-swap contrasts on log listing price at $n = 200$ listings each, with 95% intervals that fold the DML coefficient’s estimation error into the within-listing cluster bootstrap by the delta method. The bottom row is the random-effects pooled contrast.

Market	Style-stripped	95% CI	Submarket-swap
Boston	+0.184	$[+0.136, +0.232]$	−0.012
New York	+0.110	$[+0.092, +0.127]$	−0.067
San Francisco	+0.033	$[+0.016, +0.050]$	+0.015
Washington	−0.005	$[−0.015, +0.006]$	+0.003
Philadelphia	−0.004	$[−0.033, +0.026]$	+0.045
Chicago	−0.038	$[−0.067, −0.008]$	+0.025
Seattle	+0.056	$[+0.030, +0.082]$	−0.005
Denver	+0.162	$[+0.136, +0.189]$	+0.039
Atlanta	+0.176	$[+0.146, +0.207]$	+0.001
Portland	+0.088	$[+0.062, +0.114]$	−0.008
Phoenix	+0.001	$[−0.018, +0.020]$	−0.000
Dallas	+0.134	$[+0.111, +0.157]$	−0.007
Pooled (RE)	+0.074	$[+0.034, +0.113]$	+0.002

Per-market submarket-swap contrast. The submarket-swap contrast is small in absolute magnitude in every market and splits in sign across the panel. It excludes zero in six markets once the coefficient uncertainty is included, three positive and three negative, and covers zero in the remaining six. The largest negative value is in New York at $−0.067$ on log price. The largest positive values are in Philadelphia at $+0.045$ and Denver at $+0.039$. Because the market-level contrasts point in opposite directions and are individually small, they largely cancel in aggregation. The pooled random-effects submarket-swap contrast is $+0.002$ on log price, with a 95% confidence interval of $[−0.007, +0.010]$ that includes zero. The pooled submarket-swap contrast is statistically indistinguishable from zero, even though several markets carry significant contrasts of either sign.

Interpretation. The two contrasts measure different manipulations and need not agree. The positive pooled style-stripped contrast says that, holding the model’s inferred location fixed, the valuation model attaches a positive price to the listing’s own stylistic and semantic content. That price is on the order of seven to eight percent on average, and considerably larger in several markets. The near-zero pooled submarket-swap contrast says that when the rewrite is allowed to move the model’s inferred location as well, the net change in predicted price averages to approximately nothing. This is because the markets divide in sign. We do not read the gap between the two as a formal indirect effect through a location mediator, because the design does not identify one. The pattern is consistent with the dual role of listing text documented by Shen and Ross [2021] and Baur et al. [2023], in which descriptive language carries both an own-contribution to valuation and information that the model reads as a proxy for location. We treat the two contrasts as auditable probes of how one deployed valuation model responds to controlled edits of the input text, not as structural estimates of a causal mechanism.

Cross-generator robustness. We re-ran the rewriting intervention with a second open-weight generator, Llama 3.3 70B Instruct, to gauge how far the conclusions depend on the choice of rewriting model, and we compare it against the primary Qwen 2.5 32B Instruct run. The two runs are not a matched replication. The Llama run processes $n = 100$ listings per market against the $n = 200$ of the primary run. The comparison is therefore an unmatched cross-generator probe of direction and rough magnitude, not a tight correlation

between estimates computed on identical samples. Across the twelve markets the per-market style-stripped contrasts from the two generators agree in sign in nine of the twelve markets. The Spearman rank correlation is 0.46 and the Pearson correlation 0.23 on the point estimates. The submarket-swap contrasts agree in sign in six of the twelve markets, with a Spearman correlation of 0.64 and a Pearson correlation of 0.82. The central tendencies move in the same direction across generators for the style-stripped contrast, with a positive average in both runs: the random-effects pooled $+0.074$ under Qwen and a per-market mean of $+0.070$ under Llama. The submarket-swap contrast is the less stable of the two across generators, near zero under Qwen and modestly negative under Llama. This is unsurprising, since it is the small residual after two larger channels nearly cancel. We read the cross-generator evidence as directional support for a positive style-stripped contrast and a near-zero to slightly negative submarket-swap contrast. We are explicit that the rank and linear correlations are moderate rather than near-perfect.